| Responsible division : | Responsible unit : | Document type : | Confidentiality status : | |
|---|---|---|---|---|
| PGR | MLN/ENG | Requirement specification | Internal | ALSTOM |
| Prepared : 2022-02-14 A Memoli | | Title : **Functional Vehicle Req Specification (FVRS) VZ ES (ILSA)** | Document state : Released | |
| Checked : 2022-02-15 M Ciabatti1 | | | **3ECP413839-0001** | |
| Approved : 2022-02-18 R Niklass | | File name : 3ECP413839-0001_Ben.docx | Revision : _B | Language : en / Pages : 1/89 |
| | | External Revision Date: External Approver: | Customer / External Identity: | |
| | | Saved: 2022-02-18 11:00 | Sequence index : 1 | |

# Functional Vehicle Requirement Specification

# VZ ES (ILSA)

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**3 Revision** **87**

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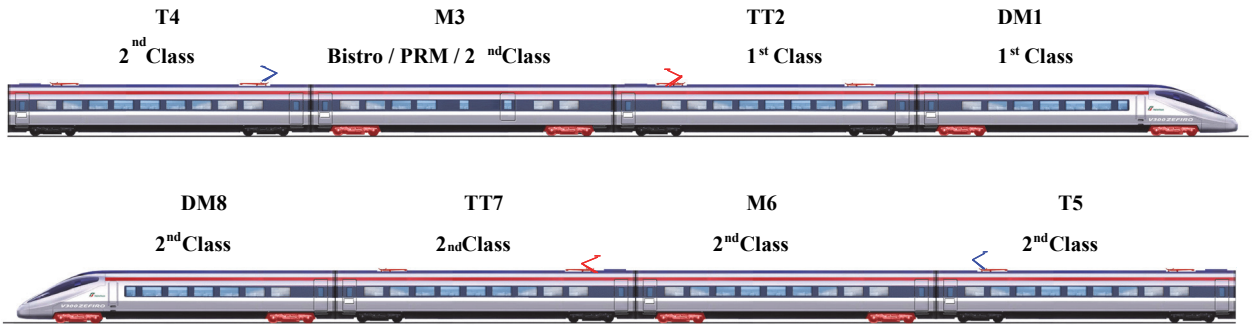
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# 1 Introduction

This document is a Function Vehicle Requirement Specification for the V300ZEFIRO Spain for customer Trenitalia.

## 1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:




- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

## 1.2 Definitions and abbreviations

| | |
|---|---|
| FVRS | Function Vehicle Requirement Specification |
| OTC | Ordinary Train Control (logic by circuit diagram) |
| TCMS | Train Control and Management System |


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# 2 2F Vehicle-Functional Integration Areas

## 2.1 2F_02 Exterior Fittings-VF

### 2.1.1 2F_02.01 Sanding System-VF

Included in Brakes, see section 2.4.2

### 2.1.2 2F_02.02 Wash Wipe-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 Activation of Windscreen Wiper + Washer** | |
| 2F_02.02.Zefiro-Europe.TRS.12 | Requirement | There shall be a control on the drivers desk and when operated fluid shall come over the windscreen. | N/A |
| 2F_02.02.Zefiro-Europe.TRS.11 | Requirement | The washer and wiper functions shall be operated with a control that have interval operation and continuously operation modes. | N/A |
| 2F_02.02.Zefiro-Europe.TRS.22 | Requirement | To increase availability of the wiping system there shall be an emergency button in the driver's desk. | N/A |

### 2.1.3 2F_02.03 Windscreen-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 Activation of Windscreen Heating** | |
| | Information | The front window is heated by foil heating. | |
| 2F_02.03.Zefiro-Europe.TRS.21 | Requirement | There shall be one operation modes of front window heating:<br>– Normal Heating | N/A |
| 2F_02.03.Zefiro-Europe.TRS.11 | Requirement | Front window heating shall be activated by the driver with a push-button or switch in the active cab when the ambient temperature is below threshold. The driver shall also be able to de-activate front-window heating manually. | N/A |
| 2F_02.03.Zefiro-Europe.TRS.12 | Requirement | If not de-activated manually by the driver the front window heating shall be de-activated automatically:<br>– when ambient temperature is above threshold or<br>– heating has been active for time period. | N/A |

### 2.1.4 2F_02.04 Coupler-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 Coupling** | |
| 2F_02.04.Zefiro-Europe.TRS.82 | Requirement | It must be possible to couple the Trainsets in multiple compositions, without any loss of command and control system functionality or other functionalities as compared to those of the single Trainset and without any restriction or limitation on circulation. | N/A |
| 2F_02.04.Zefiro-Europe.TRS.83 | Requirement | It must be possible to couple the Trainset, for emergency rescue, mechanically and pneumatically with other rolling stock in the client's fleet, fitted with automatic coupler with "type 10" head. | N/A |
| | Information | It will only be possible to connect up to two train together. | |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | Information | The coupling operation shall be a controlled process with a predetermined speed when the driver selects the coupling function. | |
| | Information | During coupling the minimum speed shall be 0.5 km/h and the maximum speed shall be 3 km/h. | |
| 2F_02.04.Zefiro-Europe.TRS.46 | Requirement | The coupler shall automatically connect two trains together mechanically, pneumatically and electrically and make it possible to operate the whole train from the front-end cab. | N/A |
| | | **2 Decoupling** | |
| 2F_02.04.Zefiro-Europe.TRS.77 | Requirement | The driver shall be able to order the two trainsets to decouple by a control in the cab. | SIL 0 |
| 2F_02.04.Zefiro-Europe.TRS.78 | Requirement | Decoupling of two trainsets shall only be possible from the active cab. It shall be possible to disconnect the two trainsets in front of the active cab (if a cab in the middle of the 16 cars train is activated) or on the opposite side in remote (if a cab at the end of the 16 cars is activated and the disconnection is done in the middle). | SIL 0 |
| 2F_02.04.Zefiro-Europe.TRS.79 | Requirement | Decoupling shall only be possible if the train is standing still. | SIL 0 |
| 2F_02.04.Zefiro-Europe.TRS.49 | Requirement | Two trainsets shall be decoupled when a decoupling order from the cab is received. | N/A |
| 2F_02.04.Zefiro-Europe.TRS.80 | Requirement | After uncoupling, the decoupled train part shall apply emergency brake. In the part with active cab the new configuration of the security loops shall be acknowledged by the driver. | SIL 0 |
| 2F_02.04.Zefiro-Europe.TRS.66 | Requirement | The driver shall be able to order the coupler cover to close by a control in the drivers desk after the train has been decoupled. | N/A |
| 2F_02.04.Zefiro-Europe.TRS.81 | Requirement | Failure of the decoupling sequence shall be indicated to the driver. | SIL 0 |
| 2F_02.04.Zefiro-Europe.TRS.84 | Requirement | Driver shall be able to electrically uncoupling the Multiple Unit. | N/A |
| | | **3 Control of coupler covers** | |
| 2F_02.04.Zefiro-Europe.TRS.47 | Requirement | The driver shall be able to order the coupler covers to open by a control in the drivers desk. | SIL 0 |
| 2F_02.04.Zefiro-Europe.TRS.63 | Requirement | The opening of the coupler covers shall be interlocked at speeds >10 km/h. | SIL 0 |
| 2F_02.04.Zefiro-Europe.TRS.51 | Requirement | Failure of opening or closing of the coupler cover shall be indicated to the driver. | N/A |
| | | **4 Detection of unintented decoupling** | |
| 2F_02.04.Zefiro-Europe.TRS.42 | Requirement | Emergency brake shall automatically be applied in both trainparts in case of unintended decoupling. | SIL 3 |
| | | **5 Coupler heating** | |
| | Information | Heating to the coupler system is available to ensure that the contacts in the coupler and other mechanics do not freeze. | |
| 2F_02.04.Zefiro-Europe.TRS.68 | Requirement | Heating cartridge in coupler shall be activated by TCMS if the outside temperature is < 1º C. When heating is already activated it shall be deactivated if the outside temperature is > 8º C. | N/A |

### 2.1.5 2F_02.05 De-Icing-VF

N/A


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### 2.1.6 2F_02.06 Exterior Lighting-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 Front and rear lamps** | |
| | Information | Headlamp<br>A white light on the front end of a train, intended to provide visual warning of an approaching train, and to illuminate lineside signs. | |
| | Information | Marker lamp<br>A white light on the front end of a train, intended to indicate the presence of a train. | |
| | Information | Tail lamp<br>A red light on the rear end of a train, intended to indicate the presence of a train. | |
| | Information | Combined lamp<br>Combined lamps (e.g. lamps capable of different functions) are permitted only where the requirements for the individual lamp functions are achieved. | |
| | Information | Headlights<br>Two white headlamps are provided at the front end of the train, arranged on the horizontal axis at the same height above the rail level, symmetrical about the centre line.<br>Marker lights<br>Three white marker lamps are provided at the front end of the train. Two marker lamps are arranged on the horizontal axis at the same height above the rail level, symmetrical about the centre line. The third marker lamp is arranged centrally above the two lower lamps.<br>Tail lights<br>Two red tail lamps are provided at the rear end of the train, arranged on the horizontal axis at the same height above the rail level, symmetrical about the centre line. | |
| | | **1.1 Front and rear lamps control** | |
| | Information | Dimmed marker lamps are used in daytime and night-time when there is bad weather conditions.<br>Full marker lamps are used in daytime and night-time in normal weather conditions.<br>Dimmed headlamps are used in daytime and night-time by driver's choice and when passing trains, intersecting road and passing stations.<br>Full-beam headlamps are used in daytime and night-time by driver's choice. | |
| 2F_02.06.Zefiro-Europe.TRS.12 | Requirement | The driver shall be able to control the head and marker lamps from the normal driving position. | N/A |
| 2F_02.06.Zefiro-Europe.TRS.14 | Requirement | The driver shall be able to turn all lamps off | SIL 1 |
| 2F_02.06.Zefiro-Europe.TRS.11 | Requirement | The driver shall be able to turn dimmed marker lamps on | N/A |
| 2F_02.06.Zefiro-Europe.TRS.19 | Requirement | The driver shall be able to turn full marker lamps on | SIL 1 |
| 2F_02.06.Zefiro-Europe.TRS.16 | Requirement | The driver shall be able to turn dimmed headlamps on | N/A |
| 2F_02.06.Zefiro-Europe.TRS.22 | Requirement | The driver shall be able to turn full-beam headlamps on | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_02.06.Zefiro-Europe.TRS.33 | Requirement | The tail lamps shall be automatically switched on when:<br>the dimmed marker lamps are turned on<br>or<br>the full marker lamps are turned on<br>or<br>the dimmed headlamps are turned on<br>or<br>the full-beam headlamps are turned on.<br>This requirement shall not be applicable for train ends that are coupled. | SIL 0 |
| 2F_02.06.Zefiro-Europe.TRS.23 | Requirement | In addition to their traditional role as front and rear lamps, the lamps may be used in specific ways and arrangements in emergency. | N/A |
| 2F_02.06.Zefiro-Europe.TRS.28 | Requirement | The external lamps located at intermediate points within the train shall be switched off. | N/A |

### 2.1.7 2F_02.07 Horn-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 Horn** | |
| | Information | The trains are fitted with pneumatically operated horns with two distinct tones. The notes of the audible warning horns are intended to be recognisable as being from a train and not be similar to warning devices used in road transport or as factory or other common warning devices | |
| | Information | The second horn may be used independently in the event of failure of the first, even if it has a different tone. | |
| 2F_02.07.Zefiro-Europe.TRS.13 | Requirement | The horn shall be pneumatically operated. | N/A |
| 2F_02.07.Zefiro-Europe.TRS.11 | Requirement | The horn shall be activated by a control in the driver´s desk. | SIL 0 |
| 2F_02.07.Zefiro-Europe.TRS.15 | Requirement | There shall be two controls in the cab to operate the horn. One control shall be for the driver to operate the horn and one control shall be for the second driver to operate the horn.ko | SIL 0 |

## 2.2 2F_03 Bogies-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 Flange Lubrication** | |
| 2F_03.Zefiro-Europe.TRS.14 | Requirement | Wheel flange lubrication shall be installed on at least two axle of the leading bogie. It shall operate curve dependently. | N/A |
| 2F_03.Zefiro-Europe.TRS.11 | Requirement | The lubrication shall be enabled if<br>– speed v > 3 km/h<br>– sanding not active<br>– train speed v <= 160km/h | N/A |
| 2F_03.Zefiro-Europe.TRS.16 | Requirement | Spraying of the grease shall be triggered by a pulse from the lateral accelerometer if a curve is detected. The wheel/rail contact area shall not be contaminated. | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_03.Zefiro-Europe.TRS.20 | Requirement | Test of the lubrication system shall be possible in maintenance mode by using:<br>– a push button on the lubrication control unit | N/A |
| | | **2 Hotbox Detection** | |
| 2F_03.Zefiro-Europe.TRS.67 | Requirement | The temperature of all axle bearings shall be supervised by hot box detectors installed on board the train. | SIL 2 |
| 2F_03.Zefiro-Europe.TRS.107 | Requirement | A thermal fuse shall be installed in every axle box. | SIL 2 |
| 2F_03.Zefiro-Europe.TRS.56 | Requirement | If any hotbox detector reports "over temperature" then the driver shall be notified. | SIL 2 |
| 2F_03.Zefiro-Europe.TRS.102 | Requirement | If any hotbox detector reports "over temperature" then an event shall be set in the diagnostic system. | N/A |
| 2F_03.Zefiro-Europe.TRS.109 | Requirement | If any gearbox sensor detects "over temperature" then an event shall be set in the diagnostic system and the driver shall be notified | N/A |
| 2F_03.Zefiro-Europe.TRS.108 | Requirement | A test function shall be implemented for the hot box detection | N/A |
| | | **3 Instability Monitoring** | |
| 2F_03.Zefiro-Europe.TRS.82 | Requirement | An on board instability monitoring system shall be fitted. The detection of instability shall be based on acceleration measurements taken on the bogie frame. An instability warning or alarm shall advise the driver to reduce the speed. The criteria for activation of this alarm shall be as defined in <<HS RST TSI 2008>> clause 4.2.3.4.2 d2. | SIL 2 |
| 2F_03.Zefiro-Europe.TRS.91 | Requirement | For IMS, the detection threshold levels and number of consecutive cycles shall be adaptable. | N/A |
| 2F_03.Zefiro-Europe.TRS.92 | Requirement | The safety relays of all IMS units of the train shall be connected in serial connection to respective train-lines to communicate the instability warning to a lamp (and/or buzzer) in the cab. The trainline shall be of fail safe type i.e. the trainline is de-energised by the safety relays breaking the loop. | SIL 2 |
| 2F_03.Zefiro-Europe.TRS.93 | Requirement | For availability reasons there shall be two redundant train-lines and two IMS units per bogie. The two IMS units per bogie shall be connected to different train lines. | N/A |
| 2F_03.Zefiro-Europe.TRS.95 | Requirement | Two digital inputs shall be provided to indicate 4 states to TCMS:<br>– System OK<br>– High Level Detection (level 2)<br>– Instability Detection (level 0 and level 1)<br>– System Error | SIL 0 |
| | | **3.1 Level 0 Warning** | |
| 2F_03.Zefiro-Europe.TRS.68 | Requirement | Instability warning level 0 shall be reported via diagnostics (TCMS) if a lateral acceleration of low level is detected (Fe => 8m/s² is detected between 3Hz and 9Hz for 10 consecutive cycles). The driver shall have the possibility to decide upon actions considering the track status. | SIL 0 |
| | | **3.2 Level 1 Warning** | |
| 2F_03.Zefiro-Europe.TRS.69 | Requirement | Instability warning level 1 shall be indicated if the timer for an active level 0 warning exceeds 1s. This warning - now escalating into an alarm - shall be reported via the IMS trainlines. | SIL 2 |

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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_03.Zefiro-Europe.TRS.103 | Requirement | Instability warning level 1 shall be indicated if the timer for an active level 0 warning exceeds 1s. This warning - now escalating into an alarm - shall be reported via the diagnostic system (TCMS). | N/A |
| | | **3.3 Level 2 Alarm** | |
| 2F_03.Zefiro-Europe.TRS.77 | Requirement | If in the mean time a peak acceleration is detected by IMS in excess of 15 m/s² (high level detected), then the alarm shall be reported via:<br>– the IMS trainline being deenergised | SIL 2 |
| 2F_03.Zefiro-Europe.TRS.104 | Requirement | If in the mean time a peak acceleration is detected by IMS in excess of 15 m/s² (high level detected), then the alarm shall be reported via the IMS delivering the signal "high level" (level 2) to TCMS | SIL 0 |
| 2F_03.Zefiro-Europe.TRS.76 | Requirement | If the instability level 2 alarm is triggered, train speed shall be limited. If speed is higher than the speed limit when the alarm is triggered, | SIL 0 |
| | | **3.6 Test** | |
| 2F_03.Zefiro-Europe.TRS.44 | Requirement | The IMS shall perform an automatic self-test after Power-up | SIL 2 |
| 2F_03.Zefiro-Europe.TRS.48 | Requirement | A IMS self test shall be triggered by TCMS if a cab is activated and the train is standstill. | SIL 0 |
| 2F_03.Zefiro-Europe.TRS.52 | Requirement | The IMS train line during shall be open during test of the system. | SIL 2 |
| 2F_03.Zefiro-Europe.TRS.94 | Requirement | Every part shall be tested simultaneously on all IMS systems of the train. If the test is not successful, the driver is informed via the IMS lamp associated with the loop. | SIL 2 |
| 2F_03.Zefiro-Europe.TRS.101 | Requirement | If the IMS test is not successful, the driver shall be informed via the diagnostic system (TCMS). | N/A |
| | | **4 Active Lateral Suspension (ALS)** | |
| 2F_03.Zefiro-Europe.TRS.88 | Requirement | An ALS shall be provided on all bogies to ensure a good running behaviour and comfort. | SIL 3 |
| 2F_03.Zefiro-Europe.TRS.96 | Requirement | The ALS is integrated in the bogie controller which shall be connected to TCMS Ethernet to receive train data (e.g. speed) and to provide diagnostics and status reporting. | SIL 3 |
| | | **5 Diagnostics & CBM** | |
| 2F_03.Zefiro-Europe.TRS.79 | Requirement | The conditions of the bogie components shall be monitored continuously by a CBM system to optimise the behaviour in maintenance by early detecting degradations of e.g axle bearings, wheels and axles. | N/A |
| 2F_03.Zefiro-Europe.TRS.60 | Requirement | Periodical measurement analysis shall be done for the following components: bearings, wheels and axles. Notifications shall be given to the maintenance regime and if necessary also to the driver. | SIL 0 |


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## 2.3 2F_04 Power Supply-VF

### 2.3.1 2F_04.01 Line Voltage-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 Line Voltage** | |
| 2F_04.01.Zefiro-Europe.TRS.670 | Requirement | The architecture of the power supply system shall support the following four different line voltage systems: AC 25 kV 50 Hz, AC 15 kV 16.7 Hz, 3 kV DC to 1.5 kV DC. | N/A |
| 2F_04.01.Zefiro-Europe.TRS.771 | Requirement | The following power supply system shall be possible to be selected by the driver at driver's desk:<br>AC 25 kV 50 Hz,<br>AC 15 kV 16.7 Hz,<br>3 kV DC,<br>1.5 kV DC. | N/A |
| | | **1.1 Pantograph Control** | |
| | | **1.1.1 Pantograph Control** | |
| | Information | There will be two stable switches for pantograph control. One for raising/lowering the front pantograph and one for raising/lowering the rear pantograph.<br>In Annex 19 only one control for pantograph control is foreseen but agreements with TI ended up with two pantograph switches. | |
| | | **1.1.2 Pantograph - Lifting** | |
| 2F_04.01.Zefiro-Europe.TRS.767 | Requirement | If the "Pantograph 1" switch is turned to "up" position the front pantograph in the train shall be raised. | N/A |
| 2F_04.01.Zefiro-Europe.TRS.768 | Requirement | If the "Pantograph 2" switch is turned to "up" position the rear pantograph in the train shall be raised. | N/A |
| 2F_04.01.Zefiro-Europe.TRS.619 | Requirement | The following sequence shall be implemented:<br>* Driver selects the Country where the train is operated from the IDU.<br>* Driver selects the line voltage system<br>* The (not blocked or cut out) pantograph (one per 8 car train unit) for the chosen line voltage system shall be lifted when requested by the driver<br>* Line voltage shall be detected and the chosen line voltage system verified<br>If the detection and verification are successful, the closing of line circuit breakers shall be enabled. ELSE a diagnostic event shall be generated. | N/A |
| 2F_04.01.Zefiro-Europe.TRS.304 | Requirement | To free the catenary line from ice it shall be possible to lift two pantographs per 8car train, however only one line circuit breaker is closed to supply the train with energy.<br>In this case it is necessary to foresee a speed limitation. | SIL 0 |
| 2F_04.01.Zefiro-Europe.TRS.685 | Requirement | For speeds greater than or equal to 300 km/h, the maximum number of pantographs raised shall be limited to two per train. | N/A |
| | | **1.1.3 Pantograph - Lowering** | |
| 2F_04.01.Zefiro-Europe.TRS.769 | Requirement | If the "Pantograph 1" switch is turned to "down" position the front pantograph in the train shall be lowered. | N/A |
| 2F_04.01.Zefiro-Europe.TRS.770 | Requirement | If the "Pantograph 2" switch is turned to "down" position the rear pantograph in the train shall be lowered. | N/A |

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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_04.01.Zefiro-Europe.TRS.266 | Requirement | Pantograph shall be lowered when line voltage system shall be de-activated<br>– by switching the selector switch to the down position OR<br>– by drivers emergency push button OR<br>– in case of failures.<br>All line circuit breakers shall be open before lowering the pantograph except in emergency cases. | SIL 0 |
| 2F_04.01.Zefiro-Europe.TRS.686 | Requirement | In separation sections (neutral sections where supply voltage changes) the pantograph shall be lowered. | N/A |
| 2F_04.01.Zefiro-Europe.TRS.687 | Requirement | When multiple pantographs are lifted and spacing allowed in coming neutral section is exceeded, the pantographs shall be lowered before entering the neutral section. | SIL 0 |
| | | **1.1.4 Automatic Dropping Device (ADD)** | |
| 2F_04.01.Zefiro-Europe.TRS.445 | Requirement | The ADD shall detect damages of contact strips (rupture, abnormal wear) and insulated end horns. | SIL 2 |
| 2F_04.01.Zefiro-Europe.TRS.270 | Requirement | In the event of collector head failure all pantographs of the train shall be lowered immediately. After activation of ADD it must be possible to lift faultless pantographs. | SIL 2 |
| 2F_04.01.Zefiro-Europe.TRS.794 | Requirement | In case of unavailability of both pantographs of the available system due to ADD intervention (or failure), through a sealed switch in driver's cab the driver, under his responsibility, can allow to raise the pantograph and move the train at reduced speed in order to reach a siding track to clear the main line. | SIL 2 |
| | | **1.1.5 Pantograph - Interlock** | |
| 2F_04.01.Zefiro-Europe.TRS.275 | Requirement | The driver shall be able to manually block a pantograph. | N/A |
| 2F_04.01.Zefiro-Europe.TRS.621 | Requirement | A faulty pantograph shall automatically be blocked. | SIL 0 |
| 2F_04.01.Zefiro-Europe.TRS.286 | Requirement | The 4 pantographs of one half-train (4 cars), which are used for AC voltage of 25 kV 50 Hz or 15 kV 16.7 Hz and DC voltage of 3 kV or 1,5 kV, shall be interlocked against each other. Switch over between the systems shall be performed automatically onboard the train set while the train is in motion. | SIL 0 |
| 2F_04.01.Zefiro-Europe.TRS.700 | Requirement | It shall be ensured that the DC measurement device is not damaged if a DC panto is raised at an AC catenary by mistake. | SIL 1 |
| 2F_04.01.Zefiro-Europe.TRS.702 | Requirement | It shall be ensured that the main trafo is not damaged if a AC panto is raised at a DC catenary by mistake. | SIL 1 |
| | | **1.1.6 Pantograph Selection** | |
| 2F_04.01.Zefiro-Europe.TRS.325 | Requirement | It shall be possible, through appropriate forcing procedures, to collect energy from any type of catenary by using the other AC pantograph of the trainset for AC lines or the other DC pantograph of the trainset for DC lines. This is limited for special purpose (like testing) under specific and supervised conditions. | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_04.01.Zefiro-Europe.TRS.772 | Requirement | Vehicle shall be configured by TCMS to operate in one of the following counties:<br>– Italy, Austria, Belgium, France, Germany, Netherlands, Spain and Switzerland.<br>The country selection shall be used for country specific configurations such as:<br>– selection of wide or narrow pantograph<br>– levels of line voltage supervision<br>– levels of line current harmonics. | SIL 2 |
| | | **1.1.7 Pantograph - Control of Static Contact Force** | |
| 2F_04.01.Zefiro-Europe.TRS.454 | Requirement | The pressure shall be a function of voltage system, speed, travel direction and position of pantograph | N/A |
| | | **1.2 Supervision of Line Voltage and Line Current** | |
| | | **1.2.1 Line Voltage Supervision** | |
| 2F_04.01.Zefiro-Europe.TRS.287 | Requirement | For AC and for DC it shall be detected which line voltage system is connected. | N/A |
| 2F_04.01.Zefiro-Europe.TRS.458 | Requirement | Line voltage shall be supervised if it is within acceptable range to allow the line circuit breakers to close and to release full or reduced traction | SIL 1 |
| 2F_04.01.Zefiro-Europe.TRS.793 | Requirement | The vehicle shall display the actual line voltage value to the driver. | SIL 0 |
| | | **1.2.2 Line Current Supervision** | |
| 2F_04.01.Zefiro-Europe.TRS.460 | Requirement | At each main transformer there shall be a current transformer to measure the line current. | SIL 1 |
| 2F_04.01.Zefiro-Europe.TRS.251 | Requirement | Current measuring unit is used for line current measurement on high voltage side. The output signal shall be used for train protection (over currents, short circuits, differential currents) and for control purpose (line converter control). | SIL 1 |
| | | **1.2.3 Line Power Limitation** | |
| 2F_04.01.Zefiro-Europe.TRS.773 | Requirement | The maximum allowed power to be extracted from the catenary by the train (also in multiple) shall be possible to be set by the driver at the IDU. | SIL 0 |
| | | **1.2.4 Line Interference Monitor** | |
| 2F_04.01.Zefiro-Europe.TRS.309 | Requirement | One line interference monitor per half-train shall be installed. (e.g. 50 Hz supervision in Italian DC line). | SIL 2 |
| 2F_04.01.Zefiro-Europe.TRS.691 | Requirement | The LIM shall realize the following functions:<br>– Line current measurement<br>– Return current measurement<br>– Line voltage measurement<br>– Supervision of Transformer oil temperatures<br>– Self test | SIL 2 |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_04.01.Zefiro-Europe.TRS.692 | Requirement | The line circuit breaker shall open if<br>– limits for line current exceeded (limits depending on country of operation)<br>– Main transformer differential current exceeded<br>– Short circuit detected<br>– maximum line voltage exceeded | SIL 2 |
| 2F_04.01.Zefiro-Europe.TRS.328 | Requirement | When the maximum line voltage for the nominal 25kV system is exceeded; after 5 minutes operation at line voltage above 29kV or after 1 second at line voltage above 31 kV; the line circuit breaker shall be opened. | SIL 2 |
| | | **1.2.5 LIM 50Hz protection bypass** | |
| 2F_04.01.Zefiro-Europe.TRS.792 | Requirement | A sealed switch to bypass the LIM 50Hz harmonic control (on DC lines only) shall be installed in the driver's cab. | SIL 2 |
| | | **1.2.6 Neutral Section Passing** | |
| | Information | Trains intended to operate on lines fitted with ATP equipment communicating neutral section information shall act on order from ATP before entering neutral section. | |
| 2F_04.01.Zefiro-Europe.TRS.463 | Requirement | The AC phase separation sections and separation section sequences shall be activated by<br>External sensors e.g. ATP or track side magnets. | SIL 0 |
| 2F_04.01.Zefiro-Europe.TRS.708 | Requirement | The AC phase separation or separation section shall be activated via External sensors (e.g. ATP). The External sensors also gives additional information to lower pantograph after the Line circuit breaker has been opened. | SIL 0 |
| 2F_04.01.Zefiro-Europe.TRS.335 | Requirement | Where AC phase separation sections require pantographs to be lowered and subsequently raised, these additional actions are permitted to be automatically initiated by the ATP. | SIL 0 |
| 2F_04.01.Zefiro-Europe.TRS.464 | Requirement | When the AC phase separation section sequence is activated, the Line Converters shall be ramped down and the line circuit breakers shall be opened for the time of passage of the AC phase separation sections. | SIL 0 |
| | | **1.3 Main Circuit Breaker, High Voltage Switches** | |
| | | **1.3.1 Main Circuit Breaker Closing** | |
| | Information | The line circuit breakers for AC and DC are used to connect and disconnect the electrical power equipment of the train set with the 25 kV, 15kV, 3kV or 1,5kV catenary line | |
| 2F_04.01.Zefiro-Europe.TRS.468 | Requirement | Line circuit breaker shall be closed when<br>– 'Line circuit breaker close' command given by driver OR automatically when leaving a neutral section AND<br>– pantograph is already lifted AND<br>– no interlocking for maintenance activities active AND<br>– no line circuit breaker opening command<br><br>The 'Line circuit breaker close' command is done by turning the Line Circuit Breaker switch on driver's desk to position ON. | SIL 0 |
| 2F_04.01.Zefiro-Europe.TRS.703 | Requirement | Line circuit breaker shall only be allowed to close if valid line voltage is detected | SIL 1 |
| 2F_04.01.Zefiro-Europe.TRS.623 | Requirement | The line circuit breakers of the train shall be closed one after another to limit inrush currents. | SIL 0 |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_04.01.Zefiro-Europe.TRS.705 | Requirement | The Line circuit breaker closure shall be synchronised to close at the top of the sinusoidal line voltage of 15kV AC. This shall avoid over magnetisation of the main trafo. | SIL 1 |
| | | **1.3.2 Line circuit breaker Opening** | |
| 2F_04.01.Zefiro-Europe.TRS.234 | Requirement | Line circuit breaker shall be opened when<br>– open command from LIM is received OR<br>– line voltage system detection protection or<br>– interlocking for maintenance activities is active OR<br>– drivers emergency push button was pressed OR<br>– ADD on pantograph or<br>– protective shut down from any DCU. | SIL 2 |
| 2F_04.01.Zefiro-Europe.TRS.712 | Requirement | Line circuit breaker shall be opened fast enough when system change section is entered under speed to avoid damage of electric. equipment | SIL 1 |
| 2F_04.01.Zefiro-Europe.TRS.714 | Requirement | Line circuit breaker shall open in case of risk for overtemperature for trafo | SIL 1 |
| 2F_04.01.Zefiro-Europe.TRS.704 | Requirement | Line circuit breaker shall be opened when<br>– 'Line circuit breaker open' command given by driver OR<br>– entering neutral section | SIL 0 |
| 2F_04.01.Zefiro-Europe.TRS.624 | Requirement | In case of normal opening (no protection situation) the Line Converter Modules shall be ordered to ramp down before opening the line circuit breaker | SIL 0 |
| 2F_04.01.Zefiro-Europe.TRS.326 | Requirement | In case of fire, the power supply to the affected system shall be interrupted. If one traction circuit is concerned, the respective line circuit breaker shall open, the affected system isolated from the traction circuit and the line circuit breaker then close again. | SIL 0 |
| | | **1.3.3 Isolate High Voltage** | |
| | Information | The High voltage system is divided into two sections: AC high voltage section and DC high voltage section. Both parts are built-up redundantly, that means they consist of two pantographs for each catenary system and identical subsystems.<br><br>In order to continue operation in case of a system failure in one half-train, subsystems can be isolated via an isolating switch. | |
| 2F_04.01.Zefiro-Europe.TRS.474 | Requirement | It shall be possible to isolate separate high voltage sections. | SIL 0 |
| | | **1.4 System Selection** | |
| 2F_04.01.Zefiro-Europe.TRS.337 | Requirement | When pantographs are not lowered from the catenary line when passing a neutral section, only those electric circuits on the traction units, which instantaneously conform to the power supply system at the pantograph, may remain connected. | SIL 0 |
| 2F_04.01.Zefiro-Europe.TRS.707 | Requirement | The high voltage system system shall protect itself against false voltage system selection. | SIL 0 |
| | | **1.5 Energy Meter** | |
| 2F_04.01.Zefiro-Europe.TRS.796 | Requirement | The vehicle shall provide the location data and time data to the Energy meter system (DHS) for billing purpose. | SIL 0 |
| 2F_04.01.Zefiro-Europe.TRS.797 | Requirement | The vehicle shall forward the energy data from the Energy meter system (DHS) to the ground station Data Collection System DCS). | SIL 0 |
| 2F_04.01.Zefiro-Europe.TRS.798 | Requirement | The vehicle shall manage data from/to the Energy meter system (DHS). | SIL 0 |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_04.01.Zefiro-Europe.TRS.799 | Requirement | Status of Energy meter system (DHS) and relevant diagnostic events shall be showed on IDU. | SIL 0 |
| | | **1.6 Key interlocking** | |
| 2F_04.01.Zefiro-Europe.TRS.679 | Requirement | Two levels of key interlocking shall be supported:<br>– Operation mode<br>– Maintenance mode | SIL 2 |
| 2F_04.01.Zefiro-Europe.TRS.680 | Requirement | In operation mode, high voltage components and auxiliary power supply shall be locked to prevent unwanted and dangerous switching operations. | SIL 2 |
| 2F_04.01.Zefiro-Europe.TRS.681 | Requirement | Changing from “operation” to “maintenance” mode, the following steps shall be performed:<br>– open line circuit breakers<br>– lowering pantographs<br>– earthing the high voltage equipment, if necessary | N/A |
| 2F_04.01.Zefiro-Europe.TRS.682 | Requirement | The hatches of the converter unit must be locked to prevent access to dangerous voltages. | SIL 2 |
| | | **1.6.1 Earthing of High Voltage Supply** | |
| 2F_04.01.Zefiro-Europe.TRS.291 | Requirement | It shall be possible to earth the high voltage supply with a switch remotely controlled. Earthing of the high voltage AC supply shall be done after the line circuit breaker. With this measure the main transformer shall also be earthed. | SIL 2 |
| | | **1.6.2 Earthing of Converter boxes** | |
| 2F_04.01.Zefiro-Europe.TRS.683 | Requirement | An earth switch shall be installed near each converter unit. The earth switch is operated to earth the internal connections (DC-link) of the converter unit in maintenance mode. In order to access the converter unit hatches, the earth switch shall be operated, which gives access to a key to the hatch. | SIL 2 |
| | | **2 Customer Requirements and Tender Documents** | |
| 2F_04.01.Zefiro-Europe.TRS.778 | Requirement | The key switch may be in the switched on or off position, depending on the investigation from the maintenance staff. In this mode the vehicle control shall disable the ‘pantograph up’ command and the ‘main circuit breaker on’ command. Driving shall also be disabled. | SIL 0 |

### 2.3.2 2F_04.02 Battery System-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 Battery System** | |
| | Information | The batteries provide following functions:<br>– Energy storage device for start of systems onboard at begin of operation<br>– Low inductive buffer for short term high loads (e.g. inrush current from electric DC drives)<br>– Emergency supply of DC consumers in case of missing line voltage - priority loads | |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1.1 Battery Isolation** | |
| 2F_04.02.Zefiro-Europe.TRS.364 | Requirement | It shall be possible to disconnect/isolate manually both poles on each battery completely from the battery bus and from the battery charger. | N/A |
| 2F_04.02.Zefiro-Europe.TRS.650 | Requirement | Each battery pole shall be fused. | N/A |
| | | **1.2 Battery Contactors** | |
| | | **1.2.1 Activation** | |
| 2F_04.02.Zefiro-Europe.TRS.378 | Requirement | The contactor connecting battery (and charger) to the battery bus shall be closed on a manual order. | SIL 0 |
| | | **1.2.2 Deactivation** | |
| 2F_04.02.Zefiro-Europe.TRS.384 | Requirement | The contactor connecting battery (and charger) to the battery bus shall open on manual order or automatic battery protection order, but only if the train is in a safe condition. | SIL 0 |
| 2F_04.02.Zefiro-Europe.TRS.770 | Requirement | A warning to equipment that needs to be shut down before power off shall be sent prior opening of battery contactor | N/A |
| | | **1.3 DC Busline Distribution and Loadshedding Control** | |
| | | **1.3.1 DC Busline Distribution Control** | |
| | | **1.3.1.1 Connection of loads to battery bus** | |
| 2F_04.02.Zefiro-Europe.TRS.320 | Requirement | The distribution of loads shall work according to a hierarchical switch-off management by connecting different groups of consumers to different loadshedding groups. | N/A |
| 2F_04.02.Zefiro-Europe.TRS.370 | Requirement | Safety-relevant DC consumers shall be connected to a permanent battery bus that is not disconnected from the batteries at shut down. | N/A |
| | | **1.3.1.2 Connection of battery and battery charger to battery bus** | |
| | Information | The batteries and the battery chargers shall be connected to the permanent battery bus via a Miniature Circuit Braker | |
| 2F_04.02.Zefiro-Europe.TRS.777 | Requirement | It shall be possible to connect loads in a way that a load is still powered if a failure occurs in one of the two separate battery buses. | N/A |
| | | **1.3.2 Loadshedding Control** | |
| 2F_04.02.Zefiro-Europe.TRS.374 | Requirement | It shall be possible to automatically disconnect battery loads with lower priority using the loadshed contactors. | SIL 0 |
| | | **1.4 Battery Charging** | |
| | | **1.4.1 Battery Monitoring, Battery Management** | |
| 2F_04.02.Zefiro-Europe.TRS.348 | Requirement | A battery management system shall monitor<br>– battery voltage<br>– charging and discharging current<br>– battery temperature<br>– battery network voltage<br>These values shall be available on the IDU. | N/A |
| | | **1.4.2 Start and Operation of Battery Charger** | |
| 2F_04.02.Zefiro-Europe.TRS.411 | Requirement | Battery charger shall start automatically when input voltage is detected | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_04.02.Zefiro-Europe.TRS.639 | Requirement | The battery charger shall at least have following functions:<br>– Charging current and voltage control (acc. charging characteristics)<br>– Battery temperature compensation of charging voltage, for which temperature sensors shall be installed in the battery.<br>– Total output current limitation.<br>– Over temperature protection/indication in charger or battery | SIL 0 |
| 2F_04.02.Zefiro-Europe.TRS.787 | Requirement | The battery charger controls the Batt Charg Contactor-K | N/A |
| | Information | The battery charger shall charge with less power if the auxiliary power is limited | |
| | | **1.4.3 Dead Battery Start** | |
| 2F_04.02.Zefiro-Europe.TRS.415 | Requirement | Start of battery charging with totally discharged battery shall be possible if the train is fed from internal or external three-phase supply | N/A |
| | | **1.4.4 Measuring of battery temperature by battery charger** | |
| 2F_04.02.Zefiro-Europe.TRS.425 | Requirement | The battery temperature shall be measured and that signal shall be hard-wired from the battery to the battery charger. | N/A |
| | | **1.4.5 Loss of battery charging** | |
| 2F_04.02.Zefiro-Europe.TRS.775 | Requirement | If no battery charger is charging and the train has been stationary for a certain time the battery contactors shall open to prevent deep discharge. | SIL 0 |
| | | **1.5 Indicators** | |
| 2F_04.02.Zefiro-Europe.TRS.368 | Requirement | The actual status of each battery circuit breaker ( open / closed ) shall be available on the IDU | SIL 0 |
| 2F_04.02.Zefiro-Europe.TRS.657 | Requirement | The actual value of the battery voltage shall be available on the IDU. | SIL 0 |
| 2F_04.02.Zefiro-Europe.TRS.656 | Requirement | The charging/discharging current of the battery shall be available on the IDU. | SIL 0 |
| 2F_04.02.Zefiro-Europe.TRS.655 | Requirement | The temperature of the batteries shall be available on the IDU. | SIL 0 |
| 2F_04.02.Zefiro-Europe.TRS.658 | Requirement | The battery bus voltage shall be available on the IDU. | SIL 0 |
| 2F_04.02.Zefiro-Europe.TRS.431 | Requirement | The actual status of the battery charger shall be available on the IDU. | SIL 0 |
| | | **1.6 Supervision** | |
| 2F_04.02.Zefiro-Europe.TRS.366 | Requirement | The position of the battery isolator switch shall be supervised | SIL 0 |
| 2F_04.02.Zefiro-Europe.TRS.390 | Requirement | The position of all battery contactors including the contactors for load shedding shall be supervised | SIL 0 |
| 2F_04.02.Zefiro-Europe.TRS.127 | Requirement | At least the following fault messages shall be provided<br>– battery charger active (able to charge and not faulty) | N/A |
| 2F_04.02.Zefiro-Europe.TRS.696 | Requirement | All DC mini circuit breakers shall be supervised. A tripped MCB shall be indicated by a diagnostic message indicating the location of the tripped MCB. | N/A |
| 2F_04.02.Zefiro-Europe.TRS.396 | Requirement | Supervision of low battery voltage shall be used to allow normal shut down before the battery is deep discharged. | SIL 0 |


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## 2.4 2F_05 Drive and Brake Control-VF

### 2.4.1 2F_05.01 Drive Control-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 General functions** | |
| | Information | Given the traction system configuration and its modularity, the availability of 50% traction power of a train unit is guaranteed by the correct functioning of the traction chain of a semi-train.<br>To assure 50% traction in case of fire, separation of the traction systems of the two semi-trains is made possible by means of knife-switches and contactors. | |
| | | **1.1 Train Weight Correction Factor Propulsion** | |
| 2F_05.01.Zefiro-Europe.TRS.120 | Requirement | The applied tractive force shall be compensated with the current weight of the train. | SIL 0 |
| | | **1.2 Anti-slip/slide** | |
| 2F_05.01.Zefiro-Europe.TRS.161 | Requirement | The traction axles shall be equipped with an anti-slip system. | N/A |
| 2F_05.01.Zefiro-Europe.TRS.201 | Requirement | The traction axles shall be equipped with an anti-slide system, this shall interact with the brake systems wheel side protection (WSP). | SIL 0 |
| | | **1.3 Car ID and Vehicle Type** | |
| 2F_05.01.Zefiro-Europe.TRS.104 | Requirement | TCMS shall send a car ID to the Propulsion Control (PCU) | SIL 0 |
| | | **2 MCM Enabling** | |
| 2F_05.01.Zefiro-Europe.TRS.89 | Requirement | MCM shall be enabled when the Master Controller is in traction position or train speed is >0. | N/A |
| 2F_05.01.Zefiro-Europe.TRS.163 | Requirement | The MCM shall be stopped when<br>– traction safe trainline is low OR<br>– Train speed is 0 and Master Controller is not in a traction position<br>– doors are released | N/A |
| | | **3 LCM Interlacing** | |
| 2F_05.01.Zefiro-Europe.TRS.191 | Requirement | Interlacing between line converters shall be provided: the line converters shall switch with shifted pulse pattern between each other. The switching angels shall be distributed evently over the whole train over 360 degrees. | SIL 1 |
| | | **4 Motor rotation supervision** | |
| 2F_05.01.Zefiro-Europe.TRS.184 | Requirement | The torque direction exerted by the traction motor (both, in traction or dynamic braking) shall be monitored and compared to the driving direction requested by the driver. | SIL 1 |
| 2F_05.01.Zefiro-Europe.TRS.186 | Requirement | Driving: In case of discrepancy (traction motor vs. driver request direction), the torque shall immediately be canceled. | SIL 1 |
| | | **5 Brake Backup mode** | |
| 2F_05.01.Zefiro-Europe.TRS.151 | Requirement | The MCM shall work in Brake Backup mode to supply auxiliary power and to regulate DC link voltage in the following cases:<br>– Crossing neutral sections<br>– Loss of line voltage<br>– Towing operation | N/A |

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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **6 Boost Mode** | |
| 2F_05.01.Zefiro-Europe.TRS.74 | Requirement | In case of loss of traction power the remaining traction units shall be used to compensate the reduction as much as possible. If the remaining traction units have sufficient thermal reserves under the current external temperature conditions, the lost traction power shall be compensated. | N/A |
| | | **7 Commissioning and Test** | |
| | | **7.1 Activation of Transient Recorder** | |
| 2F_05.01.Zefiro-Europe.TRS.97 | Requirement | For commissioning and tests it shall be possible to activate all transient recorders within the Propulsion equipment in the train by setting a parameter in the TCMS | N/A |
| | | **7.2 Cut Out of Converter** | |
| 2F_05.01.Zefiro-Europe.TRS.91 | Requirement | For commissioning and tests it shall be possible to cut-out each LCM, ACM and MCM (via IDU) | N/A |
| | | **8 Maintenance Tests** | |
| | | **8.1 Automatic Test** | |
| 2F_05.01.Zefiro-Europe.TRS.99 | Requirement | It shall only be possible to start the Automatic test when<br>– Train speed is zero AND<br>– Maintainer logged in on IDU | N/A |
| 2F_05.01.Zefiro-Europe.TRS.167 | Requirement | When the Automatic test is activated the following sequence shall manage the test:<br>1. Apply full service brake in train (equivalent to Brake Position 7 of Master Controller)<br>2. Set Traction Block<br>3. Send Automatic test command to all propulsion control's in the train<br>4. Collect test status from all propulsion control's<br>5. Reset Traction Block (note: only test condition) | N/A |

### 2.4.2 2F_05.02 Brake Control-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 Brake Control** | |
| | Information | The Brake control function can be divided into:<br>– brake demand generation<br>– Management of brake demand and conversion into a cylinder pressure | |
| | Information | The following types of brake application exist:<br>– service brake<br>– emergency brake<br>– passenger emergency brake<br>– holding brake<br>– parking brake | |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | Information | Brake application is achieved by one or a combination of several of the following means:<br>– Electrodynamic brake (ED) with power recuperation<br>– Electrodynamic brake (ED) without power recuperation, i.e. rheostatic brake using brake resistors<br>– Direct electrodynamical brake (DEP)<br>– Indirect pneumatic brake (IP) using UIC brake pipe control | |
| 2F_05.02.Zefiro-Europe.TRS.161 | Requirement | The train shall be able to run under attack of any fire for 15min, this corresponds to a distance 20km at speed 80 km/h. | N/A |
| 2F_05.02.Zefiro-Europe.TRS.322 | Requirement | Fire shall never cause an immediate automatic brake application. | N/A |
| 2F_05.02.Zefiro-Europe.TRS.181 | Requirement | The status of the brake system and its components shall be supervised continuously. The available brake performance (brake percentage) shall be indicated to the driver and to the train crew on the IDU in the cabs and crew cabinet. Errors, status of cut out switches and limitations in brake performance shall be indicated via the diagnostics system. | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.182 | Requirement | The status (applied, released, cut-out, faulty) shall be indicated for the pneumatic brake on a brake cylinder or axle basis. | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.321 | Requirement | The status (applied, released, cut-out, faulty) shall be indicated for the parking brake on an axle basis. | N/A |
| 2F_05.02.Zefiro-Europe.TRS.202 | Requirement | Dragging brake shall be detected by the brake system and the driver shall be alerted. | SIL 2 |
| | | **1.1 Service Brake** | |
| | Information | Service brake is used in normal operation to control the vehicle speed and to decelerate the train quickly and efficiently from every speed and in any load condition. | |
| 2F_05.02.Zefiro-Europe.TRS.16 | Requirement | Service brakes shall be applied by:<br>– The driver via two control handles:<br>1. one master controller (drive/brake control handle) for blended brake<br>2. one DBV brake control handle for pneumatical brake<br>– the speed control & regulation function<br>– The ERTMS and other national ATP systems | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.150 | Requirement | The Service brake force shall be load compensated | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.103 | Requirement | ED braking and blending shall only be enabled if<br>– communication ok between the entities involved (BCU, DCUM) in blending control in the effected car<br>– DEP brake is available and without failure<br>In other cases ED brake shall be disabled in the effected car. | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.149 | Requirement | The brake force for the whole train shall be compensated by cross blending of ED between cars and DEP braking, thus assuring that the required deceleration is obtained in any case. Loss of braking performance in a car shall be compensated by a higher braking force in other cars as high as permitted by the maximum adhesion level. | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.117 | Requirement | The service brake function shall cross blend between ED and DEP over all cars in the train and make maximum use of available ED brake power on all motor axles before DEP is used for compensation. | SIL 0 |

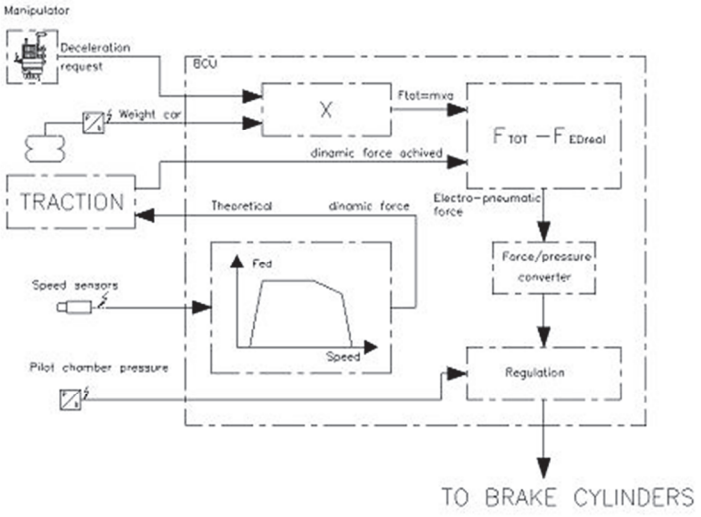
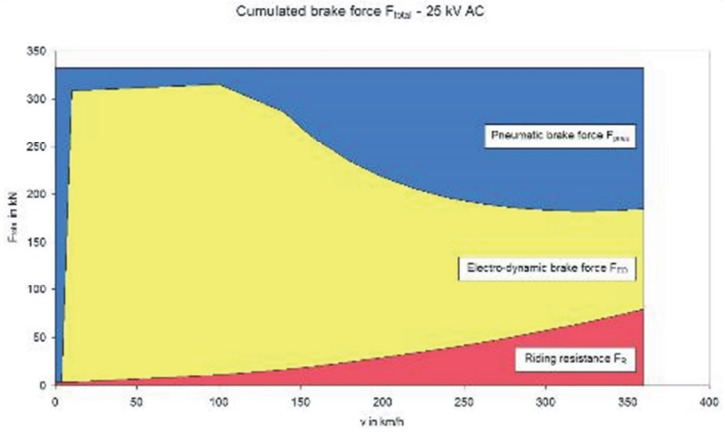
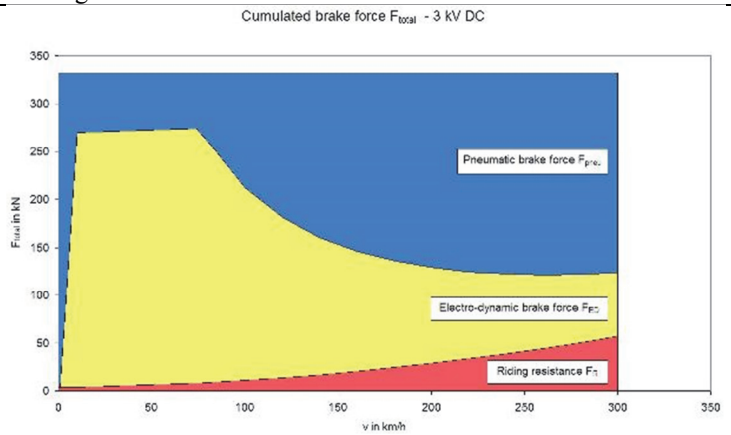

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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_05.02.Zefiro-Europe.TRS.152 | Requirement | During service braking the brake pipe shall be fixed at 5 bar by the DBV. The distributor valve shall remain released, ie. the IP brake shall not be used. | N/A |
| 2F_05.02.Zefiro-Europe.TRS.84 | Requirement | The service braking force demand is communicated to all BCU and DCUM along the train using communication buses and logical train lines. | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.157 | Requirement | The BCU's and DCUM's in a half train shall be connected via MVB for exchange of status and brake control signals. | N/A |
| 2F_05.02.Zefiro-Europe.TRS.180 | Requirement | If service brake is requested manually, the traction shall be blocked until traction is requested by the driver using the master controller. No automatical restore is allowed. | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.153 | Requirement | The total brake force per half train is calculated:<br>$F_{TOT} = a \cdot \sum_i w_i$ , i=1..4<br>a = requested deceleration at train level,<br>$w_i$ = weight on car i | N/A |
| 2F_05.02.Zefiro-Europe.TRS.154 | Requirement | The mechanical friction brake force during blending in a half train is calculated:<br>$F_F = F_{TOT} - (EDBA_1 + EDBA_2)$<br>$EDBA_1$ = EDBA achieved on first motor coach of half-train,<br>$EDBA_2$ = EDBA achieved on second motor coach of half-train. | N/A |
| 2F_05.02.Zefiro-Europe.TRS.155 | Requirement | If term $F_F \neq 0$, additional friction effort is needed to obtain the required deceleration at half-train level. In this case term $F_F$ is shared between motor and trailer axles: $F_F$ is applied on trailer axle first (where the adhesion is less stressed due to ED brake priority) till the adhesion limit and the remaining portion (if any) on all axles till the adhesion limit. | N/A |

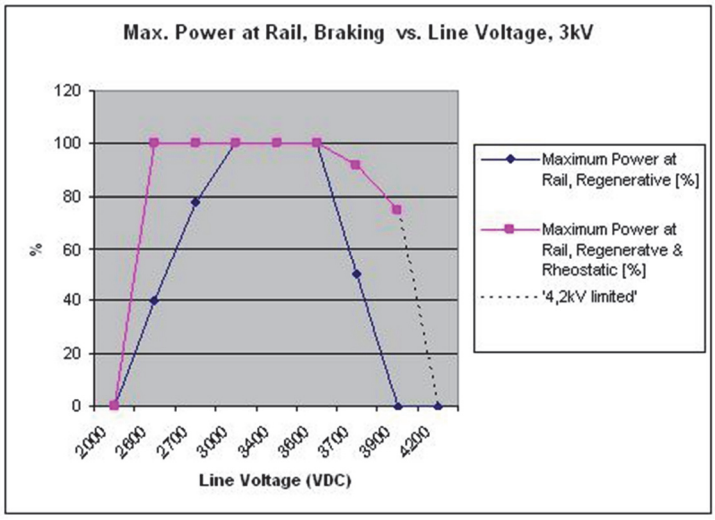

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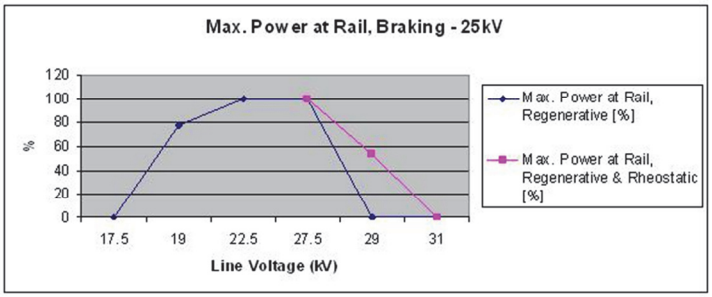
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| ID | Type | | SIL Needed |
|---|---|---|---|
| | Information | Diagram over service brake blending logic:<br>  | |
| 2F_05.02.Zefiro-Europe.TRS.158 | Requirement |  <br>braking curve at 25kV AC | N/A |
| 2F_05.02.Zefiro-Europe.TRS.159 | Requirement |  <br>braking curve at 3kV DC | N/A |


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<table>
<tr><th>ID</th><th>Type</th><th></th><th>SIL Needed</th></tr>
<tr><td></td><td>Information</td><td> 

max braking power 100% equals 5600kW at rail</td><td></td></tr>
<tr><td>2F_05.02.Zefiro-Europe.TRS.176</td><td>Requirement</td><td>
<table>
<tr><th>Line Voltage [V]</th><th>Maximum Power at Rail Regenerative [%]</th><th>Maximum Power at Rail Rheostatic [%]</th></tr>
<tr><td>2000</td><td>0</td><td>0</td></tr>
<tr><td>2600</td><td>40</td><td>100</td></tr>
<tr><td>2700</td><td>77,8</td><td>100</td></tr>
<tr><td>3000</td><td>100</td><td>100</td></tr>
<tr><td>3400</td><td>100</td><td>100</td></tr>
<tr><td>3600</td><td>100</td><td>100</td></tr>
<tr><td>3700</td><td>50</td><td>91,7</td></tr>
<tr><td>3900</td><td>0</td><td>75</td></tr>
<tr><td>4200*</td><td>0</td><td>0</td></tr>
<tr><td colspan="3">* Short term operatiion</td></tr>
</table>
</td><td>N/A</td></tr>
<tr><td></td><td>Information</td><td> 

Power limitation at 25kV AC</td><td></td></tr>
<tr><td>2F_05.02.Zefiro-Europe.TRS.178</td><td>Requirement</td><td>
<table>
<tr><th>Line Voltage [kV]</th><th>Maximum Power at Rail, Regenerative [%]</th><th>Maximum Power at Rail, Regenerative & Rheostatic [%]</th></tr>
<tr><td>17,5</td><td>0</td><td></td></tr>
<tr><td>19</td><td>77,8</td><td></td></tr>
<tr><td>22.5</td><td>100</td><td></td></tr>
<tr><td>25</td><td>100</td><td></td></tr>
<tr><td>27,5</td><td>100</td><td>100</td></tr>
<tr><td>29</td><td>0</td><td>53,33</td></tr>
<tr><td>31</td><td>0</td><td>0</td></tr>
</table>
</td><td>N/A</td></tr>
<tr><td></td><td></td><td><h1>1.2 Electro Dynamic Brake (ED)</h1></td><td></td></tr>
<tr><td></td><td>Information</td><td>Electro-dynamic braking uses the motors as generators, feeding energy either back into the catenary or into brake resistors.</td><td></td></tr>
</table>

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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_05.02.Zefiro-Europe.TRS.124 | Requirement | Regenerative braking shall be used whenever possible to the maximum extend. | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.95 | Requirement | Electro-dynamic braking shall work independently of the line voltage. The required braking power according to the nominal brake curve shall be compensated by rheostatic brake force by converting power into heat in the braking resistors if the line is not receptible or power feed back not allowed. | SIL 0 |
| | | **1.3 IP Brake** | |
| | Information | IP brake is a pure pneumatic brake mode managed by the DBV in the cabs controlling the distributor valve on each bogie brake control panel via the brake pipe. This IP brake shall be used by the driver if the master controller or the relevant components for the DEP are out of order.<br><br>The driver can only switch to IP brake at standstill. | |
| 2F_05.02.Zefiro-Europe.TRS.105 | Requirement | A driver's brake valve (DBV) controlling the IP brake shall be provided in the cabs. | N/A |
| 2F_05.02.Zefiro-Europe.TRS.121 | Requirement | A brake control pipe connecting all brake control panels distributor valves in the train according to UIC shall be provided. | N/A |
| 2F_05.02.Zefiro-Europe.TRS.127 | Requirement | The distributor valve on each brake panel shall apply the braking pressure in the brake cylinders following the brake pipe depression. | N/A |
| 2F_05.02.Zefiro-Europe.TRS.120 | Requirement | In IP brake mode the ED brake shall be disabled in the whole train. | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.167 | Requirement | If service brake is applied by brake pipe, traction shall be blocked in the whole train if the brake pipe pressure is less than 4,1 +/-0,2 bar (equivalent 50% brake force). Traction shall be enabled again if the brake pipe pressure exceeds 4,6 +/- 0,2 bar. | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.173 | Requirement | A leak indicator shall detect and report leaks on the brake control pipe. The leakage shall be reported to the driver via the diagnostics system. A luminous indication and an acoustic signal shall indicate the leakage if the cab is active and a driving direction is selected. | SIL 0 |
| | | **1.4 Holding Brake** | |
| | Information | Holding brake shall keep the train stationary at any stop in the maximum possible slope with the maximal possible load. | |
| | Information | The driver will be able to manually apply any brake by using the master controller. | |
| 2F_05.02.Zefiro-Europe.TRS.20 | Requirement | The holding brake shall apply a certain amount of brake cylinder pressure needed to keep the train stationary at 40 promille. It shall automatically be applied when the train comes to a stop. | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.132 | Requirement | The holding brake shall be automatically released when the applied traction is enough to start the train at the current slope in order to avoid rolling backward. The holding brake cylinder pressure shall be ramped down against the achieved traction force. | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.129 | Requirement | One push button shall be provided in the cab for holding brake release. The driver shall be able to manually release the holding brake by pushing and holding the button down (spring loaded button). | SIL 0 |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1.6 Emergency Brake** | |
| | Information | Emergency brake is a pneumatic brake controlled via the UIC brake pipe.<br>Emergency brake shall ensure the stopping distance of the train with the highest level of performance and integrity. | |
| 2F_05.02.Zefiro-Europe.TRS.151 | Requirement | For emergency brake, the brake pipe is vented (<=2,5bar). The ED brake shall be disabled (like IP brake) and not considered. | SIL 4 |
| 2F_05.02.Zefiro-Europe.TRS.80 | Requirement | An Emergency brake position of the master controller shall be provided. It shall be equivalent to the ultimate position of the braking position. In this position, the emergency or urgency brake shall be applied. | SIL 4 |
| 2F_05.02.Zefiro-Europe.TRS.184 | Requirement | The brake pipe shall be vented directly from the stop push buttons (mushroom) in the cab. | SIL 4 |
| 2F_05.02.Zefiro-Europe.TRS.29 | Requirement | The emergency brake shall provide braking with highest level of integrity. It shall be controlled and activated in a fail-safe way by the following safety related systems venting the brake pipe:<br>– ATP initiated brake<br>– Driver's Brake Valve ("DBV")<br>– Train disconnection | SIL 4 |
| 2F_05.02.Zefiro-Europe.TRS.315 | Requirement | The emergency brake shall provide braking with highest level of integrity. It shall be controlled and activated in a fail-safe way by the following safety related systems opening the electrical emergency brake loop or venting the brake pipe:<br>– Driver's safety (vigilance) device ("DSD")<br>– Power supply to the emergency brake control is cut off | SIL 2 |
| 2F_05.02.Zefiro-Europe.TRS.172 | Requirement | If the brake pipe is vented below 2,5 bar the emergency brake shall be applied. | SIL 4 |
| 2F_05.02.Zefiro-Europe.TRS.185 | Requirement | In order to reduce the thermal load on the brake disc, the pressure on the emergency braking shall be managed with a double stage valve, which reduces the pressure at high speed.<br>This management also allows the system to respect the adherence requested by the par. 4.2.4.1 of the HS Rolling Stock TSI. | N/A |
| 2F_05.02.Zefiro-Europe.TRS.160 | Requirement | For emergency brake, the emergency loop shall open to de-energise the application valves at least in the DM cars. | SIL 3 |
| 2F_05.02.Zefiro-Europe.TRS.199 | Requirement | At emergency braking also maximal DEP shall be applied via the BCU:s on the train. | SIL 2 |
| 2F_05.02.Zefiro-Europe.TRS.168 | Requirement | The emergency brake shall also inhibit the traction. | SIL 2 |
| 2F_05.02.Zefiro-Europe.TRS.200 | Requirement | The emergency brake application shall be recorded in the data recorder (Belgian standard). | SIL 0 |
| | Information | At activation of the stop button in cab<br>– the line circuit breaker shall be opened | |
| | Information | At emergency brake application from the stop button in cab<br>– the pantographs shall be lowered | |
| 2F_05.02.Zefiro-Europe.TRS.175 | Requirement | At emergency brake, the re-filling of the brake pipe shall be interrupted. | SIL 4 |
| | | **1.7 Parking Brake** | |
| 2F_05.02.Zefiro-Europe.TRS.37 | Requirement | The parking brake is a pure mechanical function which shall be installed on certain axles where a spring applies the brake pads if air pressure is low. This shall prevent the train from rolling when parked. | SIL 2 |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_05.02.Zefiro-Europe.TRS.34 | Requirement | Parking brake shall be applied:<br>– automatically when main reservoir pressure drops below threshold. | SIL 2 |
| 2F_05.02.Zefiro-Europe.TRS.196 | Requirement | Parking brake shall be applied:<br>– manually by a push-button on the driver's desk<br>– automatically when deactivating driver's cab. | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.134 | Requirement | Parking brake shall be possible to be released only if the main reservoir pressure is above threshold. | SIL 2 |
| 2F_05.02.Zefiro-Europe.TRS.197 | Requirement | Parking brake shall be released manually by a push-button on the drivers desk but only<br>– if the cab from where the release is commanded is activated AND<br>– if the main reservoir pressure is above threshold. | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.135 | Requirement | If the main reservoir air pressure is below threshold, and the parking brake is applied, a mechanical release shall be possible. | N/A |
| 2F_05.02.Zefiro-Europe.TRS.133 | Requirement | A push button function shall be provided in the cab for apply/release of the parking brake. | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.341 | Requirement | In case of undue parking brake applied at speed, driver shall be able to by-pass speed limitation. | SIL 2 |
| 2F_05.02.Zefiro-Europe.TRS.343 | Requirement | A protection versus and Undue Parking Brake Release Command by TCMS at vehicle level which could generated a total loss of Parking Brake Effort once Parking Brake has been applied shall be available. | SIL 2 |
| 2F_05.02.Zefiro-Europe.TRS.345 | Requirement | Diagnostics of inhibition of ASFA signalling system and of Vigilance function shall be granted. | SIL 0 |
| | | **1.8 Wheel Slide Protection** | |
| | Information | Wheel slide control protects the wheels from being locked by a brake force for which no adhesion is available. | |
| 2F_05.02.Zefiro-Europe.TRS.137 | Requirement | The WSP shall effect each axle separately | SIL 3 |
| 2F_05.02.Zefiro-Europe.TRS.138 | Requirement | On the motor axles, the slide control for ED brake shall be adjusted so that the slide control for ED is intervening first. Communication of slide status shall be exchanged between WSP and DCU/M. It shall be possible to reduce, keep or eliminate the ED depending on the WSP slide control logic in the brake computer. | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.44 | Requirement | Through anti-slide valves, controlled by WSP control unit (UIC 514-05 compliance), the system shall be able to make an optimum use of available adhesion during braking, optimize stopping distances and prevent locked (sliding) wheels and the resulting damages. | SIL 3 |
| | | **1.9 Detection of non-rotating axle (DNRA)** | |
| 2F_05.02.Zefiro-Europe.TRS.51 | Requirement | The DNRA control shall be separated from the Wheel Slide Protection System (WSP) | N/A |
| 2F_05.02.Zefiro-Europe.TRS.147 | Requirement | The DNRA shall be integrated in the brake control unit and shall be active in all conditions | N/A |
| 2F_05.02.Zefiro-Europe.TRS.48 | Requirement | DNRA system shall supervise each axle separately. When DNRA detects a wheel set non-rotating an alarm shall be presented to the driver. | SIL 2 |
| | Information | If DNRA is detected, service brake shall be applied immediately and after 10s emergency brake. | |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_05.02.Zefiro-Europe.TRS.342 | Requirement | In case of undue intervention of "DNRA - BMS" loop, driver shall be able to by-pass speed limitation. | SIL 2 |
| | | **1.10 Passenger Emergency Brake** | |
| | | **1.10.1 Passenger Emergency Request** | |
| | Information | Each passenger coach vestibule shall be equipped with a passenger emergency handle. | |
| 2F_05.02.Zefiro-Europe.TRS.53 | Requirement | It shall be possible for passengers or on board personnel to initiate an emergency braking. As requested by the HS Rolling Stock TSI para. 4.2.5.3, when an emergency handle is pulled a service brake shall be applied with a defined deceleration rate. | SIL 2 |
| 2F_05.02.Zefiro-Europe.TRS.139 | Requirement | The passenger emergency brake application shall be able to be inhibited manually by the driver depending on the position of the train. | SIL 2 |
| 2F_05.02.Zefiro-Europe.TRS.174 | Requirement | The driver shall be notified visually and audibly about an active Passenger Emergency brake. The audible part shall mute if the brake application is inhibited. The visual part will disappear when active brake handles are restored (rearmed). | SIL 2 |
| 2F_05.02.Zefiro-Europe.TRS.183 | Requirement | Activation of one or more additional passenger alarms shall have no additional effect until all alarm handles have been restored (rearmed). | N/A |
| | | **1.11 Mechanical Brake Cut Out** | |
| | Information | The cut out function may be used e.g. to release blocked wheels caused by a defect brake system or to release brakes mechanically for transport of the train without active brake system. | |
| 2F_05.02.Zefiro-Europe.TRS.57 | Requirement | There shall be a possibility to cut out the pneumatical brakes on a bogie basis. For each car two cut out switches for the pneumatic brake shall be provided. | N/A |
| 2F_05.02.Zefiro-Europe.TRS.144 | Requirement | The status of the cut out switches shall be monitored by the diagnostic system and the driver shall be alerted in case of any cut out switch in "cut out" position. | SIL 0 |
| | | **1.12 Sanding** | |
| | Information | Sanding devices shall be provided to improve braking and traction performance. | |
| 2F_05.02.Zefiro-Europe.TRS.188 | Requirement | Sanding devices shall only be activated in driving direction. | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.189 | Requirement | Sanding shall be disabled<br>– within switching zones (optional, if possible)<br>– during braking at speeds below 20 km/h. | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.327 | Requirement | The driver shall be able to switch off sanding in case automatic sanding function is provided. | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.325 | Requirement | Sanding shall be disabled at standstill | SIL 2 |
| 2F_05.02.Zefiro-Europe.TRS.190 | Requirement | Sanding shall be activated manually by the driver pressing a button on the driver's desk. | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.191 | Requirement | Sanding shall be activated automatically if any sliding is detected on the train. | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.193 | Requirement | The sanding devices shall be tested at standstill | SIL 0 |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **2 Brake Test** | |
| | Information | The brake test is performed regularly to check the status of the brake system with focus on safety. | |
| | Information | The duration of the entire coupling phase between 2 Trainsets must not be greater than 3 minutes, including the computerised brake test. | |
| 2F_05.02.Zefiro-Europe.TRS.11 | Requirement | The following test shall be provided:<br>– brake pipe venting<br>– emergency brake by electrical brake loop<br>– Emergency brake position of DBV handle and master controller.<br>– magnet track brake. | SIL 2 |
| 2F_05.02.Zefiro-Europe.TRS.206 | Requirement | The following test shall be provided:<br>– service brake by master controller<br>– parking brake | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.141 | Requirement | The Driver shall be reminded automatically by the control system to perform the applicable manual part of the brake test. | SIL 0 |
| 2F_05.02.Zefiro-Europe.TRS.142 | Requirement | Brake tests shall be performed at least every 24 hours or prior entering service. | N/A |
| | | **3 Towing** | |
| | Information | The train may be towed by another vehicle connected mechanically and pneumatically only. The braking demand from the towing vehicle is distributed to the train using the UIC brake pipe. The braking mode is similar the IP brake. | |
| 2F_05.02.Zefiro-Europe.TRS.88 | Requirement | In towing the brake command is given by the UIC brake pipe pressure manipulated from a towing vehicle. The braking function is like in IP braking. | N/A |
| | | **4 Events** | |
| 2F_05.02.Zefiro-Europe.TRS.317 | Requirement | IP Brake and brake pipe supervision: Diagnostics from the brake computer shall be stored in the train diagnostic system. | N/A |
| 2F_05.02.Zefiro-Europe.TRS.320 | Requirement | Parking Brake monitoring: Diagnostics from the brake computer shall be stored in the train diagnostic system. | N/A |

## 2.5 2F_06 Auxiliaries-VF

### 2.5.1 2F_06.01 Aux Electrical Supply-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 Electrical Auxiliary System (3~ 400V AC)** | |
| | | **1.1 General** | |
| | Information | There are 4 ACM per 8 car train connected to a 3~ 400V AC bus interconnecting 8 cars. There is no 3~ through the automatic coupler. | |
| 2F_06.01.Zefiro-Europe.TRS.119 | Requirement | The auxiliary loads shall be activated in sequence with time delays to minimize the peak load current. This applies to all operational scenarios. | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_06.01.Zefiro-Europe.TRS.133 | Requirement | A power management function shall ensure to take care of the ratio between available power and the load of consumers if power limitations apply. This is the case for external 3~ power supply and any degraded mode where more than one ACM are faulty. | N/A |
| | | **1.2 Activation and Synchronisation of ACM** | |
| 2F_06.01.Zefiro-Europe.TRS.105 | Requirement | Four ACM modules are feeding into one common 400V AC bus per 8 car train. The ACM modules need to synchronise to work in parallel. | N/A |
| 2F_06.01.Zefiro-Europe.TRS.106 | Requirement | For optimal control of loads during start-up and turn-on of high loads, a constant voltage-frequency ratio shall be used until the nominal voltage is reached. The amplitude of the auxiliary three-phase voltage shall be controlled by feedback of the actual phase voltage. | N/A |
| 2F_06.01.Zefiro-Europe.TRS.107 | Requirement | When high voltage (on DC-link) is detected, the ACM's shall be started. One ACM is to be defined to start first, the other ACM´s shall start with synchronising their 3~AC output voltage. | N/A |
| | | **1.3 Control of ACM** | |
| 2F_06.01.Zefiro-Europe.TRS.154 | Requirement | In case of fault on the 3 phase bus, the ACM shall automatically isolate. It shall be permanently isolated until a reset is received from the train control. | N/A |
| 2F_06.01.Zefiro-Europe.TRS.108 | Requirement | Each ACM shall be equipped with a load contactor. This contactor shall be able to disconnect the converter if a failure occurs. | N/A |
| | | **1.4 AC on Board Supply System** | |
| | | **1.4.1 Normal Mode** | |
| 2F_06.01.Zefiro-Europe.TRS.111 | Requirement | When the vehicle is connected to the catenary system and all ACMs are active and supplying all three-phase loads, the auxiliary electric system is in mode "Normal operation". | N/A |
| 2F_06.01.Zefiro-Europe.TRS.112 | Requirement | If one ACM fails, operation shall be continued with 3 of 4 ACMs supplying all loads. Redundancy for one ACM failure shall be provided, the remaining modules shall compensate the load if possible. | N/A |
| | | **1.4.2 Degraded Mode** | |
| 2F_06.01.Zefiro-Europe.TRS.115 | Requirement | If more than 1 ACM fails a suitable number of loads shall be automatically disconnected to compensate the loss of power supply. | N/A |
| | | **1.4.3 System Separation due to Short Circuits** | |
| 2F_06.01.Zefiro-Europe.TRS.234 | Requirement | The AC loads shall be supplied redundantly. The loads can either be supplied from the AC bus or from the local ACM feeding a local car AC bus. In case of short circuit the location of the short circuit can be isolated by individual separation contactors per car. | N/A |
| 2F_06.01.Zefiro-Europe.TRS.235 | Requirement | If short circuit is detected on the main AC bus the separation contactor in this car shall isolate the loads from the bus. The local loads shall be supplied from the local ACM on a two car basis. | SIL 0 |
| 2F_06.01.Zefiro-Europe.TRS.236 | Requirement | The local load contactor (placed in TT and T cars) shall close if the local AC loads have been separated from the AC bus. | SIL 0 |
| | | **1.4.4 Earth Fault Detection** | |
| 2F_06.01.Zefiro-Europe.TRS.120 | Requirement | The neutral point in auxiliary three-phase shall be directly connected to the carbody. | N/A |
| 2F_06.01.Zefiro-Europe.TRS.121 | Requirement | It shall be possible to disconnect the neutral conductor from the carbody. | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_06.01.Zefiro-Europe.TRS.122 | Requirement | A current relay shall monitor the current between the neutral conductor and the carbody, this function shall be an earth-fault protection | N/A |
| | | **1.4.5 Separation Contactors Between 4 Car Units** | |
| 2F_06.01.Zefiro-Europe.TRS.124 | Requirement | The 3~ AC bus of the two train halfs, the two 4 car units shall be able to be disconnected with separation contactors. This shall ensure that a short circuit in on 3~ AC system will not affect the whole train. | N/A |
| | | **1.4.6 External Power Supply** | |
| | Information | For power supply of a parked train (in the workshop or on the yard) a 3~ external connector can be inserted to the train's T cars to supply the 400V AC bus. | |
| | Information | Input plugs are provided at each car in the middle to enable connection to the stationary 3 AC 400 V network in the depots. | |
| 2F_06.01.Zefiro-Europe.TRS.132 | Requirement | Plugs of type CEE 3AC 400V 125 A according to EN 60309-1/2 with poles of type 3P+N+PE as well as pilot contacts shall be used and mounted in the T cars. | N/A |
| 2F_06.01.Zefiro-Europe.TRS.138 | Requirement | A separation contactor isolating the on board 3~ bus from the 3~ inlets in the T cars shall be provided. | N/A |
| 2F_06.01.Zefiro-Europe.TRS.137 | Requirement | The separation contactor shall close to supply the 3~ AC bus from the external inlet if no voltage is detected on the 3~ 400 V AC bus. | N/A |
| 2F_06.01.Zefiro-Europe.TRS.129 | Requirement | A 3~ AC load or separation contactor shall be provided in the cars where external inlets are located. | N/A |
| 2F_06.01.Zefiro-Europe.TRS.136 | Requirement | The separation contactor shall open to isolate the external inlet from the ACM if any ACM is active and any ACM load contactor closed. The inlet shall be disconnected from the 3~ bus as long as voltage is detected. It shall be ensured that 3 phase voltage is never active on a 3 phase inlet-plug while not connected to an external plug. | SIL 2 |
| 2F_06.01.Zefiro-Europe.TRS.131 | Requirement | The connection of an external power supply plug shall be recognised by the train and the traction shall be blocked as long as an external power supply is applied. | N/A |
| 2F_06.01.Zefiro-Europe.TRS.143 | Requirement | External supply inlets shall be interlocked so that only one can feed the 3AC train bus bar. | N/A |
| 2F_06.01.Zefiro-Europe.TRS.144 | Requirement | Traction shall be locked when at least one external supply cable is connected to the train (supply voltage is detected). | N/A |
| 2F_06.01.Zefiro-Europe.TRS.145 | Requirement | External supply shall start the battery charging. | N/A |
| 2F_06.01.Zefiro-Europe.TRS.146 | Requirement | A power management shall take care of the ratio between available power from external power supply and the load of the internal consumers. | N/A |
| 2F_06.01.Zefiro-Europe.TRS.147 | Requirement | In operation mode external power supply shall be mutually locked. | N/A |
| 2F_06.01.Zefiro-Europe.TRS.148 | Requirement | Phase rotation protection shall protect the auxiliary motors against operation in wrong direction | N/A |
| | | **1.4.7 Towed with generated power from traction motors** | |
| 2F_06.01.Zefiro-Europe.TRS.244 | Requirement | The Towing train shall accelerate to at least 50km/h or more to enable three phase loads on the bus in a Towed train. | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_06.01.Zefiro-Europe.TRS.245 | Requirement | The three phase loads in the Towed train shall be disconnected during towed when train speed is lower than 45 km/h. | N/A |
| 2F_06.01.Zefiro-Europe.TRS.246 | Requirement | In a Towed train the battery charging and essential braking, safety and comfort functions shall be maintained if possible. | N/A |
| | | **1.4.8 Earthing Switch** | |
| 2F_06.01.Zefiro-Europe.TRS.152 | Requirement | Manual earthing the 3AC-bus shall be provided for maintenance staff during maintenance procedures. It shall connect the active conductors to earth. | N/A |
| | | **1.5 230V, 50Hz Sockets** | |
| 2F_06.01.Zefiro-Europe.TRS.100 | Requirement | For each passenger seat a 1-phase 230 V 50 Hz supply socket of Italian type shall be installed. Additional 230 V 50 Hz sockets shall be foreseen for maintenance and cleaning purposes. | N/A |
| 2F_06.01.Zefiro-Europe.TRS.243 | Requirement | All 1-phase 230V 50 Hz supply sockets shall be disconnected in case of fire in the passenger compartment. | N/A |
| | | **1.6 Mini Circuit Breaker Supervision** | |
| 2F_06.01.Zefiro-Europe.TRS.128 | Requirement | All 3~ AC MCB shall be supervised. A tripped MCB shall be indicated by a diagnostic message indicating the location of the tripped MCB. | N/A |
| | | **1.7 Events** | |
| 2F_06.01.Zefiro-Europe.TRS.238 | Requirement | Diagnostics from the supplier shall be stored in the train diagnostic system. | N/A |

### 2.5.2 2F_06.03 Air Supply-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 Control of Main Compressor** | |
| 2F_06.03.Zefiro-Europe.TRS.11 | Requirement | All main compressors shall run during start up of the train when the main reservoirs are empty and when 3~ 400V AC supply is available. The shortest possible start up time shall be ensured. After achieving normal operational conditions the run time of the compressors shall be optimised. | N/A |
| 2F_06.03.Zefiro-Europe.TRS.15 | Requirement | When the pressure has decreased to limit 1 (850 kPa) the first compressor shall start. | SIL 0 |
| 2F_06.03.Zefiro-Europe.TRS.52 | Requirement | When the pressure has decreased to limit 2 (800 kPa) a second compressor shall start. | SIL 0 |
| 2F_06.03.Zefiro-Europe.TRS.53 | Requirement | If the pressure decreases below limit 3 (700 kPa), a warning message shall be presented in the cab. | N/A |
| 2F_06.03.Zefiro-Europe.TRS.54 | Requirement | If the pressure decreases below limit 4 (600 kPa), emergency brake shall be applied. | SIL 2 |
| 2F_06.03.Zefiro-Europe.TRS.55 | Requirement | All compressors shall stop when pressure has increased up to limit 1000 kPa. | SIL 0 |
| 2F_06.03.Zefiro-Europe.TRS.48 | Requirement | The compressor control shall equalize duty cycles of compressors in vehicle. | SIL 0 |
| 2F_06.03.Zefiro-Europe.TRS.49 | Requirement | Restart of compressors shall be prevented for at least 30 s (the time shall be adjustable). | N/A |
| 2F_06.03.Zefiro-Europe.TRS.69 | Requirement | It shall be possible to manually from the cab to force the running of the main compressors. | N/A |
| 2F_06.03.Zefiro-Europe.TRS.70 | Requirement | It shall be possible to manually block the start of each main air compressor from the IDU for maintenance mode. | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **2 Monitoring of Main Compressor** | |
| 2F_06.03.Zefiro-Europe.TRS.19 | Requirement | The TCMS shall monitor the main compressor to prevent damage due to high temperature and long running time. | SIL 0 |
| 2F_06.03.Zefiro-Europe.TRS.62 | Requirement | MR pipe shall be diagnostized when train is in operation. | SIL 0 |
| | | **3 Supervision** | |
| 2F_06.03.Zefiro-Europe.TRS.50 | Requirement | Leakages in the air supply system shall be detected if exceeding defined value. | SIL 0 |
| 2F_06.03.Zefiro-Europe.TRS.51 | Requirement | If Tilt system is in-active and the air consumption is high an event shall be generated. | SIL 0 |
| | | **4 Monitoring of Air Dryer** | |
| 2F_06.03.Zefiro-Europe.TRS.24 | Requirement | The TCMS system shall monitor that the air dryer switches between the two drying towers. | SIL 0 |
| 2F_06.03.Zefiro-Europe.TRS.77 | Requirement | In case of failure also to the secondary compressor, the reinclusion of the compressor with fault to the dryer in order to terminate the service shall be possible. | SIL 0 |
| | | **5 External Air Supply** | |
| | Information | The main reservoir pipe may be connected to an external pressurized air connection e.g. in the depot. | |
| 2F_06.03.Zefiro-Europe.TRS.29 | Requirement | When external air supply is connected:<br>– all compressors shall be disabled<br>– traction shall be blocked | SIL 0 |
| | | **6 Control of Auxiliary Compressor** | |
| | Information | The auxiliary compressor ensures that the pantograph can be raised during start up when there is neither main reservoir pressure nor 3~ AC supply available. | |
| 2F_06.03.Zefiro-Europe.TRS.35 | Requirement | The auxiliary compressor shall be started:<br>– depending on which pantograph is requested to raise AND<br>– main reservoir pressure too low for lifting the pantograph | N/A |
| 2F_06.03.Zefiro-Europe.TRS.42 | Requirement | The auxiliary compressor shall stop if the main compressors have taken over the air supply (i.e. 3~ AC and Main Reservoir Pressure available). | N/A |
| 2F_06.03.Zefiro-Europe.TRS.41 | Requirement | The aux compressor shall run if manually requested by push button in cab. | N/A |
| 2F_06.03.Zefiro-Europe.TRS.40 | Requirement | The aux compressor shall be disabled to run if the max run time is exceeded. | N/A |
| | | **7 Suspension Supervision** | |
| 2F_06.03.Zefiro-Europe.TRS.39 | Requirement | The air suspension of bogies shall be supervised and fault events shall be reported via the diagnostic system. | N/A |
| 2F_06.03.Zefiro-Europe.TRS.68 | Requirement | If the secondary suspension is not working, the train speed shall be limited. | SIL 0 |
| | | **8 Provision for Tilt system** | |
| | Information | There is a defined provision for a tilt system on the train. If this is ordered the parameters for the air supply system needs to be adapted according to the requirements in this section. | |
| | Information | Requirement on the Tilting control system:<br>The tilt function shall be activated if the train operates in Italy and the train speed ≥ 300km/h. | |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | Information | If the pressure decrease to limit 3 (800 kPa) the tilt function shall be deactivated.<br>When deactivated: the tilt function shall be reactivated when pressure has increased to limit 4 (900 kPa). | |
| | Information | If compressor run for 20 minutes continuously the tilt function shall be deactivated.<br>When deactivated: the tilt function shall be reactivated when at least one compressor has paused for at least 5 minutes. | |
| | Information | If MR pressure is below 800kPa the tilting function shall be disabled. If the MR pressure is increasing over 900kPa the tilting shall be enabled again. | |
| | Information | The tilting shall be disconnected and individual events be generated if either:<br>– any control valve for tilting is faulty or short circuit or disconnected<br>– any angular sensor is faulty<br>– any pressure sensor for tilting is faulty<br>– high leakage in the air supply system detected.<br>– the main compressors are running more than 20min<br>– secondary suspension is isolated in any car<br>– compressor does not start at entering a curve | |
| 2F_06.03.Zefiro-Europe.TRS.45 | Requirement | When the pressure has decreased to limit 1 (900 kPa) the first compressor shall start.<br>When the pressure has decreased to limit 2 (850 kPa) a second compressor shall start.<br><br>Compressors shall stop when pressure has increased up to limit 1000 kPa. | SIL 0 |
| | | **9 Events** | |
| | Information | Diagnostics from the supplier shall be stored in the train diagnostic system. | |

### 2.5.3 2F_06.04 Fire Protection and Extinction-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 System configuration** | |
| | Information | The high fire risk areas will be equipped with a fire detection system that can detect fire in an early stage and that initiates appropriate automatic actions. The fire detection system will exist in the following areas: | |
| 2F_06.04.Zefiro-Europe.TRS.96 | Requirement | Smoke/fire shall be detected in toilets. | SIL 2 |
| 2F_06.04.Zefiro-Europe.TRS.97 | Requirement | Smoke/fire shall be detected in the driver cabs. | SIL 2 |
| 2F_06.04.Zefiro-Europe.TRS.98 | Requirement | Smoke/fire shall be detected in passenger saloons, vestibules and gangways. | SIL 2 |
| 2F_06.04.Zefiro-Europe.TRS.101 | Requirement | Smoke/fire shall be detected in technical areas where MV or HV devices are installed. | SIL 2 |
| 2F_06.04.Zefiro-Europe.TRS.106 | Requirement | Smoke/fire shall be detected in electronic room. | SIL 2 |

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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **2 Supervision of the fire detection system** | |
| 2F_06.04.Zefiro-Europe.TRS.99 | Requirement | All smoke/fire detectors shall be supervised by the FPS | SIL 0 |
| 2F_06.04.Zefiro-Europe.TRS.45 | Requirement | Failure in the communication lines shall be detected. | SIL 2 |
| 2F_06.04.Zefiro-Europe.TRS.100 | Requirement | A failure on a single detector shall not affect the rest of the fire detection system. A single point of failure of the communication line of fire detectors shall not cause the function to fail. | N/A |
| 2F_06.04.Zefiro-Europe.TRS.104 | Requirement | The fire detection system shall operate in all train modes that have power supply available. | N/A |
| | | **3 Driver interface** | |
| | Information | If smoke is detected inside the train, the Fire Detection System gives an acoustic and an optical signal in the driver's cab. Information about where the smoke is detected shall be shown on the driver's display (IDU). | |
| | | **3.1 Smoke/heat alarm** | |
| 2F_06.04.Zefiro-Europe.TRS.85 | Requirement | When smoke/heat is detected in passanger area and/or in technical area this shall be indicated on the IDU as an event. | SIL 0 |
| 2F_06.04.Zefiro-Europe.TRS.86 | Requirement | When smoke/heat is detected this shall be indicated to the driver via an optical signal hard wire to the fire detection system. The indication shall be done irrespective of activated cab. | SIL 1 |
| 2F_06.04.Zefiro-Europe.TRS.87 | Requirement | When smoke/heat is detected an acoustic alarm shall be started in the activated cab. | SIL 0 |
| 2F_06.04.Zefiro-Europe.TRS.88 | Requirement | When smoke/heat is detected the HVAC shall be commanded to "fire mode" via a hard wired signal | SIL 1 |
| 2F_06.04.Zefiro-Europe.TRS.109 | Requirement | When smoke/heat is detected the gangway doors shall be commanded to "fire mode" via a hard wired signal | SIL 1 |
| 2F_06.04.Zefiro-Europe.TRS.114 | Requirement | When smoke/heat is detected in the passenger areas, power supply shall be interrupted to the equipment that can cause electrical shock. | SIL 0 |
| 2F_06.04.Zefiro-Europe.TRS.110 | Requirement | When smoke/heat is detected in the underfloor technical area power supply shall be interrupted to the technical area. | SIL 0 |
| | | **3.2 Mute function** | |
| | Information | The general buzzer in the cab will be used to play the sound alarm. | |
| 2F_06.04.Zefiro-Europe.TRS.73 | Requirement | The driver shall have the possibility to mute sound alarm in case of smoke/heat detection. | SIL 0 |
| | | **3.3 Reset function** | |
| 2F_06.04.Zefiro-Europe.TRS.41 | Requirement | The smoke/heat optical signal shall be switched off when the system is reset. | N/A |
| 2F_06.04.Zefiro-Europe.TRS.84 | Requirement | It shall not be possible to switch off smoke/heat alarm until the system is reset. | SIL 2 |
| | | **3.4 Display detectors status on IDU** | |
| 2F_06.04.Zefiro-Europe.TRS.64 | Requirement | The status of the fire detection system shall be indicated on the IDU. | SIL 0 |
| 2F_06.04.Zefiro-Europe.TRS.68 | Requirement | The status of the fire/smoke detectors shall be indicated on the IDU (detecting smoke/heat, detector faulty). | SIL 0 |
| | | **4 Fire Extinguisher** | |
| 2F_06.04.Zefiro-Europe.TRS.93 | Requirement | When smoke/heat is detected the fire extinguisher for that specific area shall be activated. | SIL 1 |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_06.04.Zefiro-Europe.TRS.94 | Requirement | The fire extinguisher shall not be activated for areas were no smoke/heat is detected. | N/A |
| 2F_06.04.Zefiro-Europe.TRS.95 | Requirement | Fire extinguishing system shall perform a continuous self monitoring in order to detect possible failures such as: loss of gas/water in cylinders, incorrect position of valves, etc.. | N/A |
| | | **5 Maintenance/Test Mode of fire detection system** | |
| 2F_06.04.Zefiro-Europe.TRS.81 | Requirement | When the fire detection system is running in Maintenance/Test mode the TCMS shall not show any extra faults or alarms from the fire detection system. | N/A |
| 2F_06.04.Zefiro-Europe.TRS.108 | Requirement | It shall be possible to test the fire detection system. | N/A |
| | | **6 Events** | |
| 2F_06.04.Zefiro-Europe.TRS.113 | Requirement | Diagnostics from the supplier shall be stored in the train diagnostic system. | N/A |

## 2.6 2F_08 Interiors and Passenger Comfort-VF

### 2.6.1 2F_08.01 Interior Lighting-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 Interior Lighting** | |
| | | **1.1 General Lighting** | |
| | Information | General lighting is the lighting in the ceiling of the passenger and vestibule area. It consists of direct lighting, which is the main contributing source of illumination, indirect lighting which contributes to providing a pleasant atmosphere and emergency lighting. | |
| 2F_08.01.Zefiro-Europe.TRS.11 | Requirement | General Lighting shall start automatically when a cab is activated and Battery Charging is active. | N/A |
| 2F_08.01.Zefiro-Europe.TRS.67 | Requirement | General Lighting shall remain in the current condition when cab is deactivated | N/A |
| 2F_08.01.Zefiro-Europe.TRS.66 | Requirement | It shall be possible to switch OFF/ON General Lighting from driver and from crew cabin and from all coaches. | N/A |
| 2F_08.01.Zefiro-Europe.TRS.109 | Requirement | It shall be possible to change the general lighting on train captain's command in all the defined operational modes (normal and dimmed - included "switched off"). | SIL 0 |
| | | **1.2 Tunnel Mode** | |
| 2F_08.01.Zefiro-Europe.TRS.15 | Requirement | General Lighting shall be activated in the train whenever the "tunnel mode" is active, even if General Lighting was switched OFF by driver or crew | N/A |
| | | **1.3 Emergency Lighting** | |
| | Information | The battery is sized for at least 3 hours of emergency light. The light can be turned off by driver or crew manually. | |
| 2F_08.01.Zefiro-Europe.TRS.26 | Requirement | Emergency lighting shall be powered directly from the battery | N/A |
| 2F_08.01.Zefiro-Europe.TRS.23 | Requirement | Emergency Lighting shall be active while "general lighting" is active (as part of lighting). | SIL 1 |

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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_08.01.Zefiro-Europe.TRS.68 | Requirement | If battery charging is disabled general lights shall be turned off with a time delay and emergency light shall remain on. | SIL 1 |
| | | **1.4 Safety lamps** | |
| | Information | Safety lamps are the same lamps as the emergency lamps but with different control unit. | |
| 2F_08.01.Zefiro-Europe.TRS.86 | Requirement | Safety lamps shall be activated automatically in case of the battery discharging if the system is in emergency lighting mode | SIL 0 |
| 2F_08.01.Zefiro-Europe.TRS.87 | Requirement | Safety lamps shall be possible to activate manually in case of battery power supply is cut off. The manual activation shall be possible locally for each lamp and globally at train level | N/A |
| 2F_08.01.Zefiro-Europe.TRS.89 | Requirement | Safety lamps shall in normal efficiency conditions be turned off. The internal battery shall then automatically recharge. | SIL 0 |
| | | **1.5 Reading Lamps** | |
| 2F_08.01.Zefiro-Europe.TRS.28 | Requirement | Reading lights shall be enabled when the train is active, except when the train is in loadshedding mode. | N/A |
| 2F_08.01.Zefiro-Europe.TRS.69 | Requirement | The individual reading lights ON/OFF shall be controlled by the passenger via a switch located at his seat. | N/A |
| 2F_08.01.Zefiro-Europe.TRS.77 | Requirement | It shall be possible to turn on all reading lights in the train with one central command, for testing. | N/A |
| | | **1.6 Toilet Lighting** | |
| 2F_08.01.Zefiro-Europe.TRS.40 | Requirement | Lavatory lighting shall be switched on when the toilet system is activated. | N/A |
| 2F_08.01.Zefiro-Europe.TRS.81 | Requirement | There shall be two spotlights connected to “emergency lighting” in the toilet lighting system. | SIL 1 |
| | | **1.7 Kitchen Lighting** | |
| 2F_08.01.Zefiro-Europe.TRS.44 | Requirement | It shall be possible to switch ON/OFF the kitchen lighting by a switch located in the kitchen. | N/A |
| 2F_08.01.Zefiro-Europe.TRS.82 | Requirement | The light for the working plate shall be possible to switch ON/OFF individually | N/A |
| 2F_08.01.Zefiro-Europe.TRS.83 | Requirement | There shall be at least two spotlights connected to "emergency lighting" in the kitchen lighting system. | SIL 1 |
| | | **1.8 Lighting in Crew Cabin** | |
| 2F_08.01.Zefiro-Europe.TRS.48 | Requirement | Lights in the crew cabin shall be connected at least to “emergency lighting”. | SIL 1 |
| 2F_08.01.Zefiro-Europe.TRS.70 | Requirement | Lights in the crew cabin shall be controlled ON/OFF by a switch. | N/A |
| | | **2 Events** | |
| 2F_08.01.Zefiro-Europe.TRS.105 | Requirement | Diagnostics from the supplier shall be stored in the train diagnostic system. | N/A |

### 2.6.2 2F_08.02 Toilet-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 Toilet** | |
| | Information | All the V300ZEFIRO toilets (Standard, Universal and Bistro staff toilet) will use a closed circuit system, working on the basis of air under pressure. | |

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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1.1 Status of Toilets** | |
| 2F_08.02.Zefiro-Europe.TRS.70 | Requirement | Suitable devices, located outside the toilet, shall give the status: occupied, out of order or free for use. | N/A |
| 2F_08.02.Zefiro-Europe.TRS.71 | Requirement | Suitable devices, located in the passenger compartment, shall show the current toilet status: occupied, out of order or free for use. | N/A |
| | | **1.2 Standard Toilet** | |
| | Information | The door of the standard toilet is a manual sliding door. The door will be locked manually if the toilet has a technical fault, empty water tank or full retention tank. | |
| | | **1.3 Universal Toilet for Disabled** | |
| | Information | The universal toilet is an automatic sliding door with obstacle detection system. The control of the universal door is done by the local door control. | |
| | Information | Opening and closing operation of the automatic door will take place by means of dedicated command buttons located inside and outside the toilet. Inside the toilet there are a close and an open button, and outside there is an open button. In addition, an automatic timed re-closing function is foreseen. | |
| 2F_08.02.Zefiro-Europe.TRS.76 | Requirement | The activation of the alarm in the UNI toilet shall be signalled in the crew cabin and in the drivers cab. | N/A |
| | | **2 Tank/pipe control and supervision** | |
| | | **2.1 Fresh Water supply supervision** | |
| 2F_08.02.Zefiro-Europe.TRS.26 | Requirement | Information on the level in the fresh water tanks shall be monitored. Information about low level shall be shown on the IDU in the crew cabin. | N/A |
| | | **2.2 Retention tank supervision** | |
| 2F_08.02.Zefiro-Europe.TRS.30 | Requirement | Information on the level in the retention tank shall be monitored to determine if the toilet door needs to be locked to deny access to passengers.<br>Information about high level shall be shown on the IDU in the crew cabin with an instruction to lock the toilet door. | N/A |
| | | **2.3 Heating of Water Pipes** | |
| 2F_08.02.Zefiro-Europe.TRS.74 | Requirement | Heating of waste water tanks shall be available to ensure that the waste water does not freeze. | N/A |
| | | **2.4 Complete Freeze Drainage** | |
| 2F_08.02.Zefiro-Europe.TRS.43 | Requirement | In a situation where the temperature outside the train is low and it is a risk of freezing in the water tanks an automatic or manual drainage of the water tanks shall be activated | N/A |
| | | **2.5 Partial Freeze Drainage** | |
| | Information | Compressed air is needed for some parts of the fresh water tubing system. The water that is held in that piping must be drained with compressed air. | |
| 2F_08.02.Zefiro-Europe.TRS.72 | Requirement | If the air pressure is too low the water shall be drained<br>If the air pressure is too low the water that is held in the fresh water tubing piping must be drained | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **3 Supervision** | |
| | | **3.1 TCMS Communication** | |
| 2F_08.02.Zefiro-Europe.TRS.78 | Requirement | The Toilet system shall continue to work in case there is a communication failure with TCMS | N/A |
| | | **4 Events** | |
| 2F_08.02.Zefiro-Europe.TRS.82 | Requirement | Diagnostics from the supplier shall be stored in the train diagnostic system. | N/A |

### 2.6.3 2F_08.03 Catering-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 Tank/pipe control and supervision** | |
| | | **1.1 Fresh Water supply supervision** | |
| 2F_08.03.Zefiro-Europe.TRS.11 | Requirement | Fresh Water supply in the bistro area shall be supervised.<br>Information on the level in the fresh water tank shall be shown on the IDU (only provisions):<br>• FWT level 0% (only provisions)<br>• FWT level 25% (only provisions)<br>• FWT level 50% (only provisions)<br>• FWT level 75% (only provisions)<br>• FWT level 100% (only provisions) | N/A |
| | | **1.2 Retention tank supervision** | |
| 2F_08.03.Zefiro-Europe.TRS.13 | Requirement | Retention tank in the bistro area shall be supervised.<br>Information on the level in the retention tank shall be shown on the IDU (only provisions):<br>– WWT level 50% (only provisions)<br>– WWT level 75% (only provisions)<br>– WWT level 100% (only provisions) | N/A |
| | | **1.3 Heating of Water Pipes** | |
| | | **1.4 Heating of waste water tanks** | |
| 2F_08.03.Zefiro-Europe.TRS.20 | Requirement | Heating of waste water tanks in the bistro area shall be done to prevent pipe damage. | N/A |
| | | **1.5 Freeze Drainage** | |
| 2F_08.03.Zefiro-Europe.TRS.17 | Requirement | Freeze drainage of water pipes in the bistro area shall be done to prevent pipe damage. | N/A |
| | | **2 Events** | |
| 2F_08.03.Zefiro-Europe.TRS.22 | Requirement | Diagnostics from the supplier shall be stored in the train diagnostic system. | N/A |


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### 2.6.4 2F_08.04 Interior Doors-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 Internal saloon doors** | |
| | | **1.1 Open internal saloon doors** | |
| | Information | In normal functioning the open command is given by special presence sensors (infrared type). In addition, there are opening buttons placed on both sides of the door. The door automatically closes after a pre-set time (software adjustable).<br><br>If electrical function fails it will be possible to open the doors manually. | |
| | Information | The internal saloon door opens automatically when someone is approaching (by presence sensors). | |
| | Information | The internal saloon door opens when the opening pushbutton on either side of the door is pressed. | |
| 2F_08.04.Zefiro-Europe.TRS.84 | Requirement | The 2 galley doors shall be synchronized and operated in parallel by the activation of one of the presence sensors or by one of the pushbuttons. | SIL 0 |
| | Information | If electrical function fails it is possible to open the internal saloon doors manually. | |
| | | **1.2 Close internal saloon door** | |
| | Information | The internal saloon door will automatically close when not used in a defined period | |
| | | **1.2.1 Obstacle detection during closing** | |
| | Information | Obstacle detection is active during closing of internal saloon door. | |
| | | **1.3 Lock internal saloon Door** | |
| | Information | The internal saloon doors are controlled with a RIC type one key (square key) to following positions/status:<br>– Locked in closed position<br>– Locked in open position<br>– Status Automatic (not locked) | |
| | | **2 Gangway Doors** | |
| | | **2.1 Open gangway door** | |
| | Information | Evacuation in other coaches, throughout the train is done through the gangway doors. | |
| | Information | The gangway doors are normally open. | |
| | | **2.2 Lock Gangway Door** | |
| | Information | It is possible to control the gangway doors with a RIC type one key (square key) to following positions/status:<br>– Locked in closed position<br>– Locked in open position<br>– Status Automatic (open) | |
| | | **2.3 Gangway Doors - Fire mode** | |
| | Information | Gangway doors are connected to the fire detection system, so in case of fire detection in the passenger area of a car, the gangway doors on both sides of the car will close automatically. | |
| | Information | The closing force of gangway door is limited in order not to cause injury. | |

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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_08.04.Zefiro-Europe.TRS.48 | Requirement | Both gangway doors of both gangways of the car where a fire has been detected shall automatically close. | SIL 1 |
| | | **3 Supervision** | |
| | | **3.1 Diagnostic data** | |
| 2F_08.04.Zefiro-Europe.TRS.63 | Requirement | Diagnostic data exchange between TCMS and internal doors shall be done by means of hardwired lines. | SIL 0 |

### 2.6.5 2F_08.05 Wheelchair Lifts-VF

Included in Exterior Door and Step System, see section 2.14

## 2.7 2F_09 Train Control-VF

### 2.7.1 2F_09.01 Automatic Train Control-VF

N/A

### 2.7.2 2F_09.02 Train Protection-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 Train Protection** | |
| | Information | Train Protection system includes the following functionalities:<br>• Automatic train protection<br>• Automatic driver vigilance device<br>• Measurement and visualization to the driver of train speed and time<br>• Juridical Recorder Unit | |
| | | **1.1 Driver´s Safety Device (DSD)** | |
| | | **1.1.1 Driver´s Vigilance Device (DVD)** | |
| | Information | The Driver's Vigilance Device is part of the signalling system | |
| 2F_09.02.Zefiro-Europe.TRS.105 | Requirement | The driving personnel shall confirm presence and vigilance by activating the safety pedal or pushbuttons. | SIL 2 |
| 2F_09.02.Zefiro-Europe.TRS.145 | Requirement | Emergency brake shall be applied if the driver does not confirm the presence after first an optical warning and after that a acoustic warning. | SIL 2 |
| 2F_09.02.Zefiro-Europe.TRS.147 | Requirement | The DVD shall be interfaced with the Cab radio through MVB for the generation of the GSM-R alarm in case of DVD intervention | N/A |
| 2F_09.02.Zefiro-Europe.TRS.148 | Requirement | The DVD shall get digital input to detect the driver desk status | SIL 2 |
| 2F_09.02.Zefiro-Europe.TRS.254 | Requirement | The DVD test shall be started manually by the driver in the active cab. | SIL 0 |
| | | **1.1.2 Driver Isolation Device** | |
| 2F_09.02.Zefiro-Europe.TRS.322 | Requirement | It shall be possible to Bypass the EB applied by the DVD in case of a faulty DVD system. | SIL 2 |

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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1.2 Automatic Train Protection System (ATP)** | |
| | | **1.2.1 Visualization** | |
| 2F_09.02.Zefiro-Europe.TRS.323 | Requirement | The train speed shall be displayed on the diagnostic display in case of absence of the ATP systems (both ETCS and ASFA). | SIL 2 |
| 2F_09.02.Zefiro-Europe.TRS.324 | Requirement | The JRU system shall provide the time to the train for synchronization to all equipment onboard. | N/A |
| | | **1.2.2 ATP Direction control** | |
| 2F_09.02.Zefiro-Europe.TRS.325 | Requirement | The requirements of signals necessary for ATP to control the driving direction of the train during ATP system supervision shall be defined.<br>The defined driving directions could be:<br>– Forward Direction<br>– Backward Direction | SIL 2 |
| | | **1.2.3 ATP Isolation** | |
| 2F_09.02.Zefiro-Europe.TRS.120 | Requirement | ATP isolation shall be possible using a sealed isolation switch. | N/A |
| 2F_09.02.Zefiro-Europe.TRS.326 | Requirement | It shall be possible to isolate ASFA. | SIL 0 |
| 2F_09.02.Zefiro-Europe.TRS.327 | Requirement | It shall be possible to isolate LZB via ATP. | SIL 0 |
| | | **1.2.4 ATP Start up Mission during Cab activation** | |
| 2F_09.02.Zefiro-Europe.TRS.328 | Requirement | ATP shall be able to manage cab. Active Signal Information which allows the ATP to start its Mission. | SIL 0 |
| | | **1.2.5 ATP Emergency Brake control** | |
| 2F_09.02.Zefiro-Europe.TRS.329 | Requirement | Every ATP equipment shall be integrated in the train's electrical and pneumatical circuits in such a manner that it can order Emergency Brake. | SIL 4 |
| 2F_09.02.Zefiro-Europe.TRS.330 | Requirement | ASFA shall be able to command an Emergency Braking. | SIL 4 |
| 2F_09.02.Zefiro-Europe.TRS.331 | Requirement | LZB shall be able via ATP to command an Emergency Braking. | SIL 4 |
| 2F_09.02.Zefiro-Europe.TRS.182 | Requirement | In parallel with the pneumatic Emergency brake order to the Braking system an electrical signal shall be sent to the Braking system to apply Emergency brake. | SIL 3 |
| | | **1.2.6 ATP Service Brake control** | |
| 2F_09.02.Zefiro-Europe.TRS.132 | Requirement | ATP shall be able to order Service Brake. | SIL 0 |
| | | **1.2.7 Brake pressure** | |
| 2F_09.02.Zefiro-Europe.TRS.333 | Requirement | ATP shall be able to manage Brake Pressure Information. | SIL 0 |
| | | **1.2.8 Traction Cut-off** | |
| 2F_09.02.Zefiro-Europe.TRS.334 | Requirement | ATP shall be able to command a traction cut-off in the whole train. | SIL 2 |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1.2.9 Main circuit breaker command** | |
| 2F_09.02.Zefiro-Europe.TRS.335 | Requirement | The ATP shall give additional information to manage the opening or closing of the main circuit breaker. | SIL 0 |
| | | **1.2.10 Phase transition** | |
| 2F_09.02.Zefiro-Europe.TRS.336 | Requirement | ATP (ETCS) shall command the Entry/Exit to the location of change phase. | SIL 0 |
| | | **1.2.11 Voltage transition** | |
| 2F_09.02.Zefiro-Europe.TRS.337 | Requirement | ATP (ETCS) shall command the Entry/Exit to the location of change of traction system. | SIL 0 |
| | | **1.2.12 Tunnel Entry/Exit** | |
| 2F_09.02.Zefiro-Europe.TRS.285 | Requirement | ATP shall command the Tunnel Entry/Exit section. | SIL 0 |
| | | **1.2.13 LZB button** | |
| 2F_09.02.Zefiro-Europe.TRS.339 | Requirement | LZB shall acquire the status of its buttons via TCMS (through ATP). | SIL 0 |
| | | **1.2.14 LZB TRANS LAMP** | |
| 2F_09.02.Zefiro-Europe.TRS.341 | Requirement | ETCS shall manage LZB-TRANS lamp on driver desk. | SIL 4 |
| | | **1.2.15 Sleeping** | |
| 2F_09.02.Zefiro-Europe.TRS.343 | Requirement | ATP shall be able to manage Sleeping Information (predisposition). | SIL 0 |
| | | **1.2.16 Train Integrity** | |
| 2F_09.02.Zefiro-Europe.TRS.345 | Requirement | ATP shall be able to manage Train Integrity Information (predisposition). | SIL 0 |
| | | **1.3 Juridical Recorder Unit (JRU)** | |
| 2F_09.02.Zefiro-Europe.TRS.346 | Requirement | The JRU shall be able to storage all the Information needed at least by TSIs and Spanish NNTRs. | N/A |
| 2F_09.02.Zefiro-Europe.TRS.175 | Requirement | A faulty JRU shall be reported to the driver. | N/A |
| | | **1.4 Cab Radio** | |
| 2F_09.02.Zefiro-Europe.TRS.347 | Requirement | The Cab Radio shall be compliant to TSIs (CCS) and Spanish NNTRs. | N/A |

### 2.7.3 2F_09.03 Conventional Train Control-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 Train Control** | |
| | | **1.1 Cab Activation** | |
| 2F_09.03.Zefiro-Europe.TRS.347 | Requirement | Two Position: 0 = Cab off (stable position) 1 = Cab activated (stable position). It is possible to extract the key only in OFF position | N/A |
| 2F_09.03.Zefiro-Europe.TRS.174 | Requirement | Cab activation is made by means of the driver's key. | N/A |
| 2F_09.03.Zefiro-Europe.TRS.175 | Requirement | Activation of a second cab in a train shall be prevented. | SIL 2 |

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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_09.03.Zefiro-Europe.TRS.222 | Requirement | Sufficient brake shall be applied e.g. by opening the emergency brake loop in case the cab is deactivated. This shall avoid roll back in case of not occupied cab. | SIL 2 |
| | | **1.2 Demand of Driving Direction** | |
| 2F_09.03.Zefiro-Europe.TRS.61 | Requirement | When a cab will be activated this cab will be defined as head of the train. From this results the normal driving direction (forward direction). For shunting trips it is be possible to select reverse direction. | N/A |
| 2F_09.03.Zefiro-Europe.TRS.62 | Requirement | The active cab shall always define the forward direction.<br>The driving direction shall be actively chosen by the driver using a "forward" or a "Reverse" button in the driver's desk. | N/A |
| 2F_09.03.Zefiro-Europe.TRS.351 | Requirement | Direction selector pushbuttons<br>Forward:Lit when selected,<br>Neutral: Lit when selected (neutral selected by default when at the desk activation)<br>Backward: Lit when selected | N/A |
| 2F_09.03.Zefiro-Europe.TRS.230 | Requirement | Driving in reverse direction by system failure shall be avoided. | SIL 1 |
| 2F_09.03.Zefiro-Europe.TRS.148 | Requirement | Change of direction shall only be allowed when:<br>– at standstill (v=0)<br>– master controller in position "0" | SIL 0 |
| 2F_09.03.Zefiro-Europe.TRS.149 | Requirement | Maximum reverse speed shall be limited to 40km/h. | SIL 0 |
| | | **1.3 Traction/Brake reference (Master Controller)** | |
| 2F_09.03.Zefiro-Europe.TRS.188 | Requirement | A master controller for traction and braking effort demand shall be provided. The driver sets the demand of traction or braking power in different levels of controller positions. | SIL 1 |
| 2F_09.03.Zefiro-Europe.TRS.209 | Requirement | The ultimate position of the traction/brake shall cause emergency braking. | SIL 4 |
| 2F_09.03.Zefiro-Europe.TRS.187 | Requirement | A speed controller for speed set (cruise control) shall be provided.<br>Positions<br>4: increase speed set point<br>3: neutral<br>2: decrease set point<br>1: Off (speed set disabled)<br>The speed set point (target speed) shall be integrated over time. The time base depends on the time the lever is in the + or - position.<br>I.e in the + position >1s=fast increase, <1s = slow increase<br>in the - position >1s=slow decrease, >1s = fast decrease | N/A |
| 2F_09.03.Zefiro-Europe.TRS.361 | Requirement | The master controller functionality shall be redundant. | N/A |
| | Information | Either of the controller can be used if the other one is set in inactive position | |
| 2F_09.03.Zefiro-Europe.TRS.197 | Requirement | Two controllers, the ordinary master controller and the speed controller work independently. | N/A |
| | Information | The speed controller is inactive in "0" position | |
| | Information | The traction brake controller is inactive in Coasting position. | |
| 2F_09.03.Zefiro-Europe.TRS.359 | Requirement | The master controllers brake position has priority even if the speed controller is not in "0" position. This because braking has always priority. | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_09.03.Zefiro-Europe.TRS.334 | Requirement | The traction demand shall be blocked to all motors in the train if any of following conditions is true:<br>– master controller is in brake position<br>– ATP service brake is active<br>– too high speed in washing- or coupling mode OR<br>– fault in speed calculation | SIL 2 |
| 2F_09.03.Zefiro-Europe.TRS.371 | Requirement | In case of Master Controller fault, Master controller back-up mode (sealed latched switch) can be selected. In this mode the driver can drive the train by using the Speed controller. | N/A |
| 2F_09.03.Zefiro-Europe.TRS.370 | Requirement | It shall be possible to perform a test of the Master controller from the IDU. In the test each position of the Master controller shall be tested by manually moving the Master controller based on instructions provided on the IDU. | N/A |
| | | **1.4 Traction Safe** | |
| 2F_09.03.Zefiro-Europe.TRS.340 | Requirement | Traction shall only be possible in the whole train when all conditions for traction release are fulfilled. For that case there shall be a hard-wired train line "traction safe". | SIL 2 |
| 2F_09.03.Zefiro-Europe.TRS.389 | Requirement | Use of the emergency brake, for whatever reason, shall automatically cut all traction power, without the facility of reapplying traction power while the emergency brake is being applied. | SIL 2 |
| 2F_09.03.Zefiro-Europe.TRS.373 | Requirement | Traction shall be disabled if the pressure in the brake pipe is less than 3,5 bars. | SIL 2 |
| | | **1.5 Traction Safe Override** | |
| 2F_09.03.Zefiro-Europe.TRS.343 | Requirement | It shall be possible for the driver to manually override Traction safe in case of malfunction to clear the track and start traction. | SIL 2 |
| | | **1.6 Traction demand** | |
| 2F_09.03.Zefiro-Europe.TRS.336 | Requirement | The traction demand level shall be distributed from the active cab to all DCUM along the train. | SIL 1 |
| 2F_09.03.Zefiro-Europe.TRS.337 | Requirement | The distribution of traction demand level shall be ensured to at least 50% of all DCUM along the train for minimum 15 minutes in case of fire. | N/A |
| 2F_09.03.Zefiro-Europe.TRS.338 | Requirement | No traction force shall be applied if not demanded. | N/A |
| | | **1.7 Train Speed Calculation** | |
| 2F_09.03.Zefiro-Europe.TRS.158 | Requirement | The speed sensors from brake- and propulsion system in a 8 car unit will be used to calculate the train speed. The strategy of the calculation is to use the highest speed value when braking and coasting and the second lowest value when accelerating. | N/A |
| 2F_09.03.Zefiro-Europe.TRS.199 | Requirement | The speed signals from all available DCUM and WSP on one MVB shall be used for train speed calculation. | SIL 2 |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_09.03.Zefiro-Europe.TRS.200 | Requirement | The speed calculation for<br>– visualisation in the cab<br>– doors interlocking<br>– DSD function<br>shall be provided by ATP or other SIL2 computer, e.g. the Brake Controller BCU or WSP.<br>In case of ATP cut out :<br>– TCMS shall calculate the speed by means of the available BCU.<br>– The speed provided by DIS system shall be used for visualization in the cab | SIL 2 |
| 2F_09.03.Zefiro-Europe.TRS.161 | Requirement | The speed sensors shall be continually supervised and faulty sensors shall be excluded from the speed calculation | SIL 0 |
| | | **1.8 Speed Limitation** | |
| 2F_09.03.Zefiro-Europe.TRS.155 | Requirement | Depending on the status of the train speed limits are applied. The most restrictive speed limit shall be used.<br>The following speed limits are defined:<br>– driving without ATP (ATP cut out)<br>– coupling speed<br>– washing speed<br>– IMS warning (manual speed reduction by driver)<br>– IMS alarm (manual speed reduction by driver)<br>– one or more WSP cut out or not working<br>– one or more BCU cut out or not working (EP brake)<br>– overheat in propulsion system<br>– overheat in line voltage system<br>– limited power from catenary<br>– pantograph error (DC current limitation) | N/A |
| 2F_09.03.Zefiro-Europe.TRS.223 | Requirement | If ATPs are cut out, the speed shall be limited to 80 km/h. | SIL 2 |
| | | **1.9 Speed Calibration** | |
| 2F_09.03.Zefiro-Europe.TRS.212 | Requirement | Speed calibration shall ensure the consideration of wheel diameters in the train speed signal calculation. | N/A |
| 2F_09.03.Zefiro-Europe.TRS.154 | Requirement | The axle speed signals from the BCU/WSP and DCUM shall represent a rotational speed only. | SIL 2 |
| 2F_09.03.Zefiro-Europe.TRS.201 | Requirement | The wheel diameters shall be stored in the TCMS for calibration of the speed signals received from BCU/WSP and DCUM (rotational frequencies). The calibration shall be done using the speed from the ATP system. | SIL 0 |
| 2F_09.03.Zefiro-Europe.TRS.203 | Requirement | The calibration procedure shall only be activated manually from a driver or maintainer after a workshop visit affecting the wheel diameters. | SIL 0 |
| | | **1.10 Distance Counter** | |
| 2F_09.03.Zefiro-Europe.TRS.163 | Requirement | An odometer function logging the travelled distance shall be provided for maintenance use. The data shall be stored in non volatile memory and be available for the Telediagnostica application. | N/A |
| | | **1.11 Master Reset** | |
| 2F_09.03.Zefiro-Europe.TRS.165 | Requirement | Different systems use interlocks for blocking the use of faulty equipment. That interlock has a reset function that is available for the driver/maintainer. The reset is called “Master Reset”. | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_09.03.Zefiro-Europe.TRS.213 | Requirement | A "Reset" function shall be provided for the maintenance crew to reset applied interlocks of faulty equipment. The reset accessible to maintainers shall be unlimited but be protected against unauthorised access. | SIL 0 |
| | | **1.12 Drivers Desk** | |
| 2F_09.03.Zefiro-Europe.TRS.168 | Requirement | Control from driver desk buttons and switches shall only be active in the active cab (excluding the emergency stop pushbutton and the driver key which are active in all cabs). | SIL 2 |
| 2F_09.03.Zefiro-Europe.TRS.169 | Requirement | Indicators, lamps and buzzers controlled by hardware directly, shall be disabled in the non-active cab a part from the parking mode lamp and door release indication. | N/A |
| 2F_09.03.Zefiro-Europe.TRS.171 | Requirement | Test of hardware controlled lamps, indicators shall be done upon manual command to ensure the correct functionality. (SIL 2 by means of the performance of the manual test by the driver -> driver´s manual) | SIL 2 |
| 2F_09.03.Zefiro-Europe.TRS.214 | Requirement | Test of TCMS controlled lamps, indicators shall be done upon command to ensure the correct functionality. | SIL 0 |
| | | **1.13 Indication of Serious Fault** | |
| 2F_09.03.Zefiro-Europe.TRS.226 | Requirement | A "serious fault" shall be indicated to the driver via a dedicated lamp on the driver's desk. | SIL 2 |
| 2F_09.03.Zefiro-Europe.TRS.376 | Requirement | TCMS shall illuminate the "Serious fault" lamp on the driver's desk if a 'serious fault' has occurred. | SIL 2 |
| | | **1.14 TCMS configuration waiting indication** | |
| 2F_09.03.Zefiro-Europe.TRS.382 | Requirement | One lamp indicating that the train configuration is changing shall be installed on Driver's Desk | N/A |
| | | **1.15 Cab doors** | |
| | Information | Cab doors are located in the electronic room and are under manual operation. | |
| 2F_09.03.Zefiro-Europe.TRS.236 | Requirement | If a cab door is open the green loop shall be interrupted | SIL 2 |
| | | **1.16 Driving Mode** | |
| 2F_09.03.Zefiro-Europe.TRS.388 | Requirement | Entering "Driving Mode" shall be possible only if the following condition are fulfilled:<br>• battery voltage is activated, AND<br>• pantograph is up, AND<br>• main circuit breaker is closed, AND<br>• line voltage is available, AND<br>• 1 cab is active | N/A |
| 2F_09.03.Zefiro-Europe.TRS.390 | Requirement | Normal driving mode shall be ended when:<br>• battery voltage is switched off, OR<br>• line voltage is not available, OR<br>• 1 cab is deactivated | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1.17 Train Off Mode** | |
| 2F_09.03.Zefiro-Europe.TRS.393 | Requirement | Train off mode shall be reached from driving mode by the following actions:<br>– by pressing the 'Main circuit breaker off' button<br>– by pressing the 'Pantograph down' button<br>– by deactivating a cab with the key switch<br>– by pressing the "Battery OFF" push button<br><br>From train off mode the switch-over to driving mode or workshop mode is possible. | N/A |
| | | **1.18 Towed mode** | |
| | Information | Towing may be performed together with another V300 Zefiro train or any other vehicle with appropriate coupler adapter. During towing operation no electrical connection between the two vehicles is foreseen. The only connection is the mechanical coupler and the two air brake pipes. | |
| | Information | The towing V300 Zefiro is operating in driving mode. Possible restrictions in speed and the usage of IP brake during towing shall be managed manually by the driver. | |
| 2F_09.03.Zefiro-Europe.TRS.215 | Requirement | The towed V300 Zefiro shall be put into "towed" mode: the cab shall be activated and a dedicated switch for towing shall be activated. | N/A |
| 2F_09.03.Zefiro-Europe.TRS.217 | Requirement | The switching into "towed" mode shall be interlocked until mechanical coupling is confirmed from the coupler sensor. | N/A |
| 2F_09.03.Zefiro-Europe.TRS.216 | Requirement | The following restrictions apply in towed mode:<br>– brake control via the UIC brake pipe (IP braking), i.e. mechanical brake only.<br>– No traction, line circuit breaker open<br>– no dynamic braking<br>– three phase supply only available from backup braking of the propulsion system (if possible). | N/A |
| | | **1.19 Washing mode** | |
| | Information | Washing mode is a low speed regulation mode for passing through a washing plant. | |
| 2F_09.03.Zefiro-Europe.TRS.218 | Requirement | The Washing mode shall be activated from the IDU. | N/A |
| 2F_09.03.Zefiro-Europe.TRS.82 | Requirement | Washing speed control shall only be possible to selected when:<br>– the Master Controller is in position coast AND<br>– train speed is 0 AND<br>– traction is possible | N/A |
| 2F_09.03.Zefiro-Europe.TRS.397 | Requirement | It shall be possible for the speed at which the train passes through the train washing plant to be adapted to suit each washing plant, i.e. between 2 and 5 km/h. | N/A |
| | | **1.20 Parking Mode** | |
| | Information | Parking mode is the state of the train when the cab is deactivated and all power supply systems remain active. This may be short time for cab change in a station or the stabling of the train for up to two days. | |
| 2F_09.03.Zefiro-Europe.TRS.132 | Requirement | In parking mode, heating, cooling, saloon lighting and passenger door control shall remain in operation. | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_09.03.Zefiro-Europe.TRS.131 | Requirement | In parking mode the line voltage shall remain activated. | N/A |
| 2F_09.03.Zefiro-Europe.TRS.133 | Requirement | The duration while the train is in parking mode shall be monitored. When the maximum allowed time is reached, the automatic exit shall be triggered (switch to 'train off mode'). | N/A |
| 2F_09.03.Zefiro-Europe.TRS.134 | Requirement | The maximum allowed time shall be a parameter within the TCMS. | N/A |
| 2F_09.03.Zefiro-Europe.TRS.144 | Requirement | The trainset must have, at the front and back ends, a signal visible from the outside constituted (eg a luminous strip of red LEDs), positioned in the lower centre zone of the front window. | N/A |
| 2F_09.03.Zefiro-Europe.TRS.145 | Requirement | The outside front and back ends visible signal must remain on with a steady light in the Parking mode. | N/A |
| 2F_09.03.Zefiro-Europe.TRS.310 | Requirement | If fire or smoke alarm is active a message shall be sent to the Telediagnostica application to trigger an alarm message to the workshop. | N/A |
| | | **1.20.1 Enter Parking mode** | |
| 2F_09.03.Zefiro-Europe.TRS.124 | Requirement | Parking mode shall be entered by:<br>– pressing the 'Parking' push button<br>– keeping the Parking push button pressed and reset panto switches<br>– keeping the Parking push button pressed and rotate and extract the key from the driver's desk<br>– releasing the push button; the rolling stock is in Parking mode and the energy conversion systems must remain active depending on the configuration set on the driver's desk prior to entering into Parking mode. | SIL 0 |
| 2F_09.03.Zefiro-Europe.TRS.140 | Requirement | Conditions for entering the Parking mode:<br>– train in standstill *<br>– Master controller in "0" position on enabled driver desk<br>– Speed controller in "0" position on enabled driver desk<br>– DBV in normal position on enabled driver desk<br>– Train direction set on "neutral" on enabled driver desk<br>– Parking Brake applied * (note: at least 2/3 of parking applied per unit)<br>– main reservoir pressure not below 5.5 bar *<br>– no fire alarm detected *<br>– fire detection system is efficient *<br>– no main circuit breaker trip *<br>– presence of line voltage *<br>– no external power supply inserted *<br><br>* Conditions above shall be verified on both units in multiple configuration | N/A |
| 2F_09.03.Zefiro-Europe.TRS.143 | Requirement | Parking push button shall begin to flash when pressed and shall remain steadily on when the key is removed from the driver's desk. | N/A |
| | | **1.20.1.1 Enter Parking mode interrupted** | |
| 2F_09.03.Zefiro-Europe.TRS.125 | Requirement | The release of the push button before the light on it is lit steady, must interrupt the process for entering into Parking and must configure the trainset based on the new position of the driver's desk levers. | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_09.03.Zefiro-Europe.TRS.126 | Requirement | If the configuration is the initial one, the trainset must behave as if the procedure for entering the Parking mode has not been started. | N/A |
| | | **1.20.2 Manual Exit** | |
| 2F_09.03.Zefiro-Europe.TRS.135 | Requirement | Parking mode shall be finished by:<br>– pressing the 'Parking' push button<br>– keeping the Parking push button pressed and insert and rotate the key to the driver's desk<br>– checking that the light is off and by confirming the exit of the rolling stock from the Parking mode and releasing the push button | N/A |
| 2F_09.03.Zefiro-Europe.TRS.147 | Requirement | When the operator exits from parking mode by pressing the Parking push button, it shall flash. | N/A |
| | | **1.20.3 Automatic Exit** | |
| 2F_09.03.Zefiro-Europe.TRS.137 | Requirement | In the absence of at least one of the conditions for entering into Parking mode, the Parking mode must be automatically switched off (on each unit in case of multiple configuration). | N/A |
| 2F_09.03.Zefiro-Europe.TRS.138 | Requirement | Upon automatic exit from Parking mode, the following must occur:<br>– the start of the 10 minute timed period;<br>– the activation of the internal and external flashing signals; | N/A |
| 2F_09.03.Zefiro-Europe.TRS.146 | Requirement | The external Parking signal must flash upon automatic exit from Parking mode (for example when the protections are triggered) and during the timing prior to the cut-off of the train batteries. | N/A |
| 2F_09.03.Zefiro-Europe.TRS.139 | Requirement | During the timed period, the trainset must remain with the panto up and, after 10 minutes, the following must occur, in succession:<br>a) controlled closure (therefore with sensor edge switched on) of all of the access doors, 60 seconds prior to the expiration of the timed period;<br>b) cut-off of the train batteries<br>c) train enters train OFF mode | N/A |
| 2F_09.03.Zefiro-Europe.TRS.309 | Requirement | Conditions for leaving the Parking mode:<br>– main reservoir pressure below 5.5 bar *<br>– train speed greater that 0 km/h *<br>– parking brake released * (note: not less than 2/3 of parking brake applied per unit)<br>– fire extinguish system is activated *<br>– fire detection system is not efficient *<br>– main circuit breaker trip *<br>– no line voltage *<br>– external power supply inserted *<br><br>* Related to each unit in case of multiple configuration | N/A |
| 2F_09.03.Zefiro-Europe.TRS.384 | Requirement | If the driver commands the opening of the LCB from one of the cabs, the immediate exit from the parking mode shall be commanded.<br>The train will go into the "disabled" state with all LCB open and all pantographs down | SIL 0 |
| | | **1.21 Coupling** | |
| | Information | Coupling mode is a low speed regulation mode to couple two V300 Zefiro trains. | |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | Information | The coupling operation shall be a controlled process with a predetermined speed when the driver selects the coupling function. | |
| | Information | It will only be possible to connect up to two trains for multiple operation. | |
| 2F_09.03.Zefiro-Europe.TRS.385 | Requirement | It must be possible to couple the Trainsets in multiple compositions, without any loss of command and control system functionality or other functionalities as compared to those of the single Trainset and without any restriction or limitation on circulation. | N/A |
| 2F_09.03.Zefiro-Europe.TRS.219 | Requirement | There shall be a selector switch on the Driver's desk to select coupling speed control. | N/A |
| 2F_09.03.Zefiro-Europe.TRS.86 | Requirement | Coupling speed control shall only be possible to be select when:<br>– the Master Controller is in position coast AND<br>– train speed is 0 AND<br>– traction is possible | N/A |
| 2F_09.03.Zefiro-Europe.TRS.323 | Requirement | Minimum speed shall be 0.5 km/h and maximum speed shall be 3km/h. | SIL 0 |
| 2F_09.03.Zefiro-Europe.TRS.396 | Requirement | A new train configuration shall be always acknowledged by the driver | N/A |
| | | **1.22 Emergency Operation** | |
| 2F_09.03.Zefiro-Europe.TRS.192 | Requirement | The emergency operation mode is used as backup for traction and brake control in case of a major incident like fire. The redundancy of the normal communication paths via MVB/WTB is not enough because no fire safe communication bus cables are available on the market. | N/A |
| 2F_09.03.Zefiro-Europe.TRS.194 | Requirement | The master controller shall be connected directly to this backup communication path though being independent. In emergency operation only the traction/brake controller and the indirect brake lever (DBV) can be used.<br>The emergency operation mode shall allow the train to run even in case of fault on the Emergency Brake Loop. | N/A |
| 2F_09.03.Zefiro-Europe.TRS.231 | Requirement | In case of an incident, the activation of braking operation shall not cause braking down to stand still. Any evacuation drive must be ensured with minimum time delay. | N/A |
| 2F_09.03.Zefiro-Europe.TRS.196 | Requirement | The driver shall confirm the emergency operation by a dedicated button or switch. Emergency operation shall be active only if this switch is activated by the driver. | N/A |
| | | **1.23 HMI Power Reset** | |
| 2F_09.03.Zefiro-Europe.TRS.395 | Requirement | HMI is in fail safe state shall be automatic recovered by TCMS via Power cycles | N/A |

### 2.7.4 2F_09.04 TCMS - Operational-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 Architecture** | |
| 2F_09.04.Zefiro-Europe.TRS.226 | Requirement | The TCMS System shall provide a distributed communication network for train and vehicle level control and supervision. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.227 | Requirement | The TCMS system shall be built up by distributed intelligent computing units, touch screen displays, communication units and digital and analog I/O units. | N/A |

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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1.1 Components** | |
| 2F_09.04.Zefiro-Europe.TRS.255 | Requirement | The CCU-O shall use a VCU-Compact type with MVB and serial channels. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.263 | Requirement | The CCU-O shall contain the vehicle control application. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.376 | Requirement | The CCU-S shall contain the safe vehicle control application. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.256 | Requirement | The CCU-D shall use a VCU-Compact type with MVB and serial channels. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.264 | Requirement | The CCU-D shall contain the train diagnostic database. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.257 | Requirement | A Gateway shall be used as a communication bridge between the local MVB and the train-wide WTB. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.258 | Requirement | I/O devices shall be used to interface a subset of connected systems and the vehicles hardware circuits. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.259 | Requirement | An IDU shall be used for the interface to the driver. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.260 | Requirement | IP-Ring Switches shall be used to build up the on board Ethernet Consist Network. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.270 | Requirement | IP-Ring Switches shall be used to connect end devices which are using an IPT interface. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.261 | Requirement | IP-Train Switches shall be used to connect the Ethernet Consist Network to the Ethernet Train Backbone (ETB). | N/A |
| 2F_09.04.Zefiro-Europe.TRS.262 | Requirement | IP-Train Repeaters shall be used to ensure that the maximum allowed cable lengths of 90 meters between IP-Train switches/IP-Train Repeaters are followed for the Ethernet Train Backbone. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.268 | Requirement | There shall be a MVB used for train operation control in each 4-car units. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.269 | Requirement | There shall be a WTB used to interconnect the car MVB (for multiple operation of two 8-car trains). | N/A |
| | | **1.2 Interfaces** | |
| 2F_09.04.Zefiro-Europe.TRS.271 | Requirement | The following interfaces shall be used to interact with other systems on the train:<br>– MVB<br>– IP<br>– IO | N/A |
| 2F_09.04.Zefiro-Europe.TRS.272 | Requirement | The Propulsion (including DCU/L, DCU/M, DCU/F and DCU/A) shall use an MVB interface for operational control and supervision, IP for service access and diagnostics. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.273 | Requirement | Brakes shall use MVB and IO | N/A |
| 2F_09.04.Zefiro-Europe.TRS.274 | Requirement | Auxiliary Electric System shall use IO | N/A |
| 2F_09.04.Zefiro-Europe.TRS.275 | Requirement | Train Operation Control shall use IO | N/A |
| 2F_09.04.Zefiro-Europe.TRS.276 | Requirement | Passenger Doors shall use IP | N/A |
| 2F_09.04.Zefiro-Europe.TRS.277 | Requirement | Passenger Information System and CCTV shall use IP | N/A |
| 2F_09.04.Zefiro-Europe.TRS.278 | Requirement | ATP shall use MVB and IO | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_09.04.Zefiro-Europe.TRS.279 | Requirement | Bogie Monitoring shall use IP and IO | N/A |
| 2F_09.04.Zefiro-Europe.TRS.280 | Requirement | Air Supply shall use IO | N/A |
| 2F_09.04.Zefiro-Europe.TRS.281 | Requirement | Carbody Fittings shall use IO | N/A |
| 2F_09.04.Zefiro-Europe.TRS.282 | Requirement | Interior shall use IO | N/A |
| 2F_09.04.Zefiro-Europe.TRS.283 | Requirement | Toilet shall use IP | N/A |
| 2F_09.04.Zefiro-Europe.TRS.284 | Requirement | Coupler shall use IO | N/A |
| 2F_09.04.Zefiro-Europe.TRS.285 | Requirement | HVAC shall use IP | N/A |
| 2F_09.04.Zefiro-Europe.TRS.286 | Requirement | Line Voltage shall use MVB and IO | N/A |
| 2F_09.04.Zefiro-Europe.TRS.297 | Requirement | The pantograph control shall use IP | N/A |
| 2F_09.04.Zefiro-Europe.TRS.287 | Requirement | Battery shall use IP and IO | N/A |
| 2F_09.04.Zefiro-Europe.TRS.288 | Requirement | Fire Detection shall use IP | N/A |
| 2F_09.04.Zefiro-Europe.TRS.289 | Requirement | LIM shall use MVB | N/A |
| 2F_09.04.Zefiro-Europe.TRS.290 | Requirement | Train Radio shall use MVB | N/A |
| | | **2 Communication** | |
| | | **2.1 On Board system** | |
| 2F_09.04.Zefiro-Europe.TRS.207 | Requirement | TCMS shall be used for train control, supervision and diagnostics. | SIL 0 |
| 2F_09.04.Zefiro-Europe.TRS.292 | Requirement | TCMS communication between all devices in the TCMS System shall be supervised; An event shall be generated if communication with a specific device is not present. | SIL 0 |
| 2F_09.04.Zefiro-Europe.TRS.293 | Requirement | Bus coupler shall be supervised and errors reported. | SIL 0 |
| | | **2.2 Redundancy** | |
| 2F_09.04.Zefiro-Europe.TRS.209 | Requirement | A single failure on the TCMS system shall not stop the train from continuing in operation.<br>If one important unit (e.g. CCU-O, I/O, GW) fails a redundant unit shall handle the functionality of the failing unit. | SIL 0 |
| | Information | CCU-O: | |
| 2F_09.04.Zefiro-Europe.TRS.229 | Requirement | The two CCU-Os shall implement the Leader Follower concept. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.236 | Requirement | The two CCU-Os shall be connected to the same line redundant MVB. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.237 | Requirement | The two CCU-Os shall be connected to the Ethernet Consist Network. | N/A |
| | Information | CCU-S: | |
| 2F_09.04.Zefiro-Europe.TRS.378 | Requirement | The two CCU-S in each consist shall implement the Leader Follower concept. | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | Information | MVB Bus Administrator | |
| 2F_09.04.Zefiro-Europe.TRS.379 | Requirement | The MVB bus administrator shall be redundant. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.380 | Requirement | The usage of MVB bandwidth up to 95% shall be used for process data. | N/A |
| | Information | Ethernet Consist Networks: | |
| 2F_09.04.Zefiro-Europe.TRS.231 | Requirement | The Ethernet Consist Networks shall be implemented as a ring to provide redundant traffic pathways. | N/A |
| | Information | Ethernet Train Bus: | |
| 2F_09.04.Zefiro-Europe.TRS.233 | Requirement | The Ethernet Train Bus (ETB) shall be implemented using redundant communication cables. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.303 | Requirement | The focal point control (direction control of the network traffic in the ring) shall be located in the train switches | N/A |
| | Information | GW: | |
| 2F_09.04.Zefiro-Europe.TRS.235 | Requirement | The two Gateways (GW) shall implement the Leader Follower concept. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.238 | Requirement | The two Gateways (GW) shall be connected to the same line redundant MVB. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.239 | Requirement | The two Gateways (GW) shall be connected to the two WTB cables. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.240 | Requirement | The two Gateways (GW) shall be connected to the Ethernet Consist Network. | N/A |
| | Information | WTB: | |
| 2F_09.04.Zefiro-Europe.TRS.242 | Requirement | The Wire Train Bus (WTB) shall be implemented using redundant communication cables. | N/A |
| | Information | MVB: | |
| 2F_09.04.Zefiro-Europe.TRS.244 | Requirement | The MVB shall be implemented using line redundant communication cables. | N/A |
| | Information | Discrete IO: | |
| 2F_09.04.Zefiro-Europe.TRS.246 | Requirement | Critical functions to the train operation shall be implemented using redundant I/O. | N/A |
| | | **2.3 Self diagnosis / Self reset** | |
| 2F_09.04.Zefiro-Europe.TRS.248 | Requirement | During boot-up, the device shall perform self diagnosis of its main components of the on board system (boards, RAM, etc.), Power On Self Test - POST. | SIL 0 |
| 2F_09.04.Zefiro-Europe.TRS.249 | Requirement | Moreover, all active system processes shall be cyclically checked on the basis of configuration, after minor failures and, if necessary, restarted. | SIL 0 |
| 2F_09.04.Zefiro-Europe.TRS.251 | Requirement | In case of serious non-recoverable damage the complete system shall be reset and a controlled shutdown shall be performed. | SIL 0 |
| 2F_09.04.Zefiro-Europe.TRS.250 | Requirement | The device shall not require manual activities to restore non-blocking hardware or software failures of its component. | SIL 0 |
| 2F_09.04.Zefiro-Europe.TRS.383 | Requirement | Hardwired IDU restart push-buttons or MCB shall be accessible for restart of each IDU separately. | N/A |
| | | **2.4 Time Control** | |
| 2F_09.04.Zefiro-Europe.TRS.211 | Requirement | TCMS system time shall be synchronized from the GPS. | SIL 0 |
| 2F_09.04.Zefiro-Europe.TRS.296 | Requirement | The TCMS units and other IP connected devices shall synchronise their system time against the TCMS on board time master. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.298 | Requirement | The time used for event logging in the fire detection system shall be synchronised with the TCMS system time | N/A |

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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **3 Train Integrity** | |
| | | **3.1 Monitoring of Train Configuration** | |
| 2F_09.04.Zefiro-Europe.TRS.222 | Requirement | The TCMS configuration shall be dynamic and support operation of 8 and 16 car trains. | N/A |
| | Information | The duration of the entire coupling phase between 2 Trainsets must not be greater than 3 minutes, including the computerised brake test. | |
| 2F_09.04.Zefiro-Europe.TRS.294 | Requirement | The train configuration shall be available for other systems. | N/A |
| 2F_09.04.Zefiro-Europe.TRS.295 | Requirement | A unique car id shall be configured for each car (UIC id), which corresponds to the id painted on the outside of the car. | N/A |

### 2.7.5 2F_09.05 Additonal On Board Control-VF

N/A

## 2.8 2F_10 TCMS - Monitoring-VF

### 2.8.1 2F_10.01 Diagnostics-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 Diagnostics** | |
| | | **1.1 General requirements** | |
| | Information | TCMS includes a Train Diagnostic System, TDS, which stores predefined faults and events that occur in the train.<br>The events which are relevant for the driver is presented on the IDU in the activated driver's cabin. It is possible for the driver, crew, and maintenance personnel to see all active events on every IDU in the train. | |

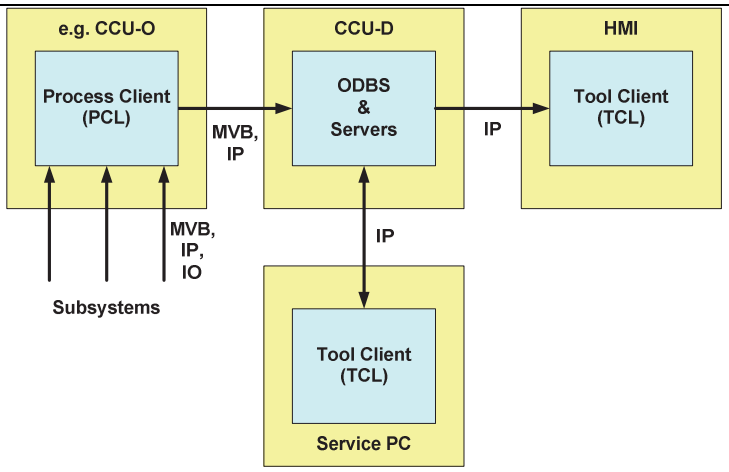

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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_10.01.Zefiro-Europe.TRS.66 | Requirement | Events shall cover all systems listed in below including TCMS itself:<br>– Air Supply System<br>– ATP System<br>– Auxiliary Electric System<br>– Battery System<br>– Brake System<br>– CCTV System<br>– Driver's Safety Device<br>– Exterior Doors<br>– Fire Detection System<br>– Heating, Ventilation and Air Conditioning System<br>– High Voltage System<br>– Infotainment<br>– Interior Doors<br>– Interior Lighting<br>– Passenger Information System<br>– Propulsion System<br>– Toilets<br>– Bogie Monitoring<br>and other systems. | N/A |
| 2F_10.01.Zefiro-Europe.TRS.92 | Requirement | The events shall be classified based on severity and character. | N/A |
| 2F_10.01.Zefiro-Europe.TRS.94 | Requirement | The diagnostic system shall also store other data (called condition data) about the various systems such as operation time and number of operations. | N/A |
| 2F_10.01.Zefiro-Europe.TRS.129 | Requirement | The diagnostic system shall be redundant. | N/A |
| 2F_10.01.Zefiro-Europe.TRS.95 | Requirement | It shall be possible to perform remote upload of the stored diagnostic data to an off-board system. | N/A |
| 2F_10.01.Zefiro-Europe.TRS.96 | Requirement | Gathered event data and condition data shall be stored in a database located in battery backed up memory or with equivalent security so it can be used for operational, maintenance or vehicle performance measurements purposes. | N/A |
| 2F_10.01.Zefiro-Europe.TRS.97 | Requirement | The location and origin of collected data shall be visible. It shall be possible to determine car and unit. | N/A |
| 2F_10.01.Zefiro-Europe.TRS.134 | Requirement | TDS shall support multiple language (namely French, Spanish, Italian, English) event texts. | N/A |
| 2F_10.01.Zefiro-Europe.TRS.133 | Requirement | Spanish shall be the default language. It shall be possible to switch between languages in run-time. | N/A |
| 2F_10.01.Zefiro-Europe.TRS.100 | Requirement | It shall be possible to clear the TDS Event database without uploading the database first. | N/A |
| 2F_10.01.Zefiro-Europe.TRS.101 | Requirement | The TDS module shall provide the global environmental signals that are generic for all events. | N/A |

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| ID | Type | | SIL Needed |
|---|---|---|---|
| | Information |  Figure: Train Diagnostic System (TDS) | |
| | | **1.2 Event handling** | |
| | Information | Event data are used to store events of different reasons, e.g. disturbances in the control system or information that a specified status value has been reached. Anything that can be represented as active or passive can be stored as event data. | |
| 2F_10.01.Zefiro-Europe.TRS.103 | Requirement | Each event shall have the following attributes:<br>– Code/identity<br>– Category<br>– Start time<br>– End time<br>– Number of occurrences<br>– Texts | N/A |
| 2F_10.01.Zefiro-Europe.TRS.68 | Requirement | Environmental data with pre- and post-history shall be attached to each group of events. | N/A |
| | Information | The ODBS filling level approaches maximum, the data upload shall be triggered from the train. | |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1.3 Event categorization** | |
| 2F_10.01.Zefiro-Europe.TRS.70 | Requirement | Events shall use the following categories listed in priority order:<br>0. Serious fault events<br>Information to the driver about the serious fault that has occurred. Automatically displayed on IDU when it has occurred.<br>1. Advisory<br>Information to the driver that he has done an incorrect maneuver. No acknowledge for this event. Automatically displayed on IDU when it has occurred.<br>2. Alarm<br>Major event, i.e. stopping events that require the relevant user's full attention. Automatically displayed on IDU when it has occurred.<br>3. Warning<br>Warnings or service effecting event, i.e. train performance reduction event. Automatically displayed on IDU when it has occurred.<br>4. Maintenance<br>Minor event intended for maintenance personnel. No service effecting event. Available in event lists on IDU for maintainers.<br>5. Other<br>Informative Event, Non effecting event. Available in event history list on IDU. | N/A |
| | Information | Serious fault events events have the highest priority. This means that Serious fault events always take precedence over Advisory-, Alarm- and/or Warning events even if it already has been presented.<br><br>Advisory events takes precedence over Alarm- and/or Warning events even if it already has been presented.<br><br>Alarm events takes precedence over Warning events even if it already has been presented | |
| 2F_10.01.Zefiro-Europe.TRS.105 | Requirement | Events with higher priority shall take precedence over events with lower priority, even if a lower priority event has already been presented on the IDU. | N/A |
| | | **1.4 Event presentation** | |
| | Information | The events which are relevant for the driver are presented on the IDU in the activated driver's cabin. It is possible for the driver, crew, and maintenance personnel to see all active events on every IDU in the train. | |
| 2F_10.01.Zefiro-Europe.TRS.73 | Requirement | It shall be possible for the driver, crew and maintenance personnel to see all active events on every IDU in the train (relevant for user category). | N/A |
| | Information | Occurred events which cause an operational impact to the vehicle is presented to the driver combined with a driver advice. | |
| 2F_10.01.Zefiro-Europe.TRS.131 | Requirement | Serious fault events shall be presented on the IDU and they shall be acknowledged by pressing a button on the IDU. When the event has been acknowledged it shall be removed from the screen. | N/A |
| 2F_10.01.Zefiro-Europe.TRS.106 | Requirement | Advisory event shall be displayed on the IDU. As soon as the driver corrects the fault, the Advisory event shall disappear automatically from the screen. | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_10.01.Zefiro-Europe.TRS.130 | Requirement | Alarm events shall be presented on the IDU and they shall be acknowledged by pressing a button on the IDU before it is removed from the screen. | N/A |
| 2F_10.01.Zefiro-Europe.TRS.110 | Requirement | Warning events shall be presented on the IDU and they shall be acknowledged by pressing a button on the IDU before it is removed from the screen. | SIL 0 |
| 2F_10.01.Zefiro-Europe.TRS.107 | Requirement | Maintenance personal shall be able to acknowledge all events both individually and as a group except for serious fault events that shall be acknowledged individually. | N/A |
| | | **1.5 Buzzer control** | |
| | Information | The cab is fitted with a buzzer to make the driver aware of different events that can occur on the train. | |
| 2F_10.01.Zefiro-Europe.TRS.76 | Requirement | New Alarm event shall trigger the buzzer in the active cab. | SIL 0 |
| | Information | It's possible to mute the buzzer by a pushbutton until next Alarm | |
| | | **1.6 Condition data handling** | |
| | Information | Condition data handle two subtypes of data; parameters and counters.<br>Parameters are used as inputs to the process control application, e.g. the setting of values whereas counters are used to count the state of a subset of process values such as mileage, number of door openings etc. Each parameter has a default value, which is stored in the ODBS and used as the start value.<br><br>Counters are used to count the number of cycles as e.g. the number of door openings. Each counter has a counter limit, which is stored in the ODBS and used to compare the counter value to a predefined counter limit. | |
| 2F_10.01.Zefiro-Europe.TRS.78 | Requirement | Condition data of type parameters and counters shall be supported. | N/A |
| | | **2 Software version handling** | |
| 2F_10.01.Zefiro-Europe.TRS.80 | Requirement | The TCMS system shall include a version reporting system used for application software version (for presentation on the IDU). | N/A |
| 2F_10.01.Zefiro-Europe.TRS.117 | Requirement | The stored software versions shall be possible to upload to a connected service LapTop and Remote (via MCG). | N/A |
| | | **3 Maintenance** | |
| 2F_10.01.Zefiro-Europe.TRS.116 | Requirement | From Each cab IP service port there is possible to upload diagnostic data, download software, check communication, configuration, software version numbers, monitor and temporary changing parameters and initiate function tests. | N/A |
| | | **3.1 Software Download** | |
| 2F_10.01.Zefiro-Europe.TRS.61 | Requirement | It shall be possible to download application software and other information to the TCMS units (including the DCU/A, /F, /L, /M and LIM) from a connected service LapTop. | SIL 0 |
| 2F_10.01.Zefiro-Europe.TRS.122 | Requirement | The MTVD tool shall be used for downloading and version check of downloaded applications. | N/A |
| | | **3.2 Upload and analysis** | |
| 2F_10.01.Zefiro-Europe.TRS.46 | Requirement | The event information stored in the train diagnostic database shall be possible to upload to an off-board maintenance and service system for long term storage and additional off line analysis. | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_10.01.Zefiro-Europe.TRS.112 | Requirement | The diagnostic system shall interface the Train Telediagnostica System for remote upload of stored event data. | N/A |
| 2F_10.01.Zefiro-Europe.TRS.121 | Requirement | The MAVIS tool shall be used for analysis of uploaded events on a service LapTop. | N/A |
| | | **3.3 Subscription of signals** | |
| | Information | It is possible to subscribe on signals using the TSSP. | |
| 2F_10.01.Zefiro-Europe.TRS.115 | Requirement | It shall be possible to subscribe to sub-system signals on MVB and IPT, e.g. from the IDU. | N/A |
| 2F_10.01.Zefiro-Europe.TRS.114 | Requirement | A TSSP server shall be included to provide the functionality to subscribe to onboard signals from a ground based system (via the Train Telediagnostica System). | N/A |
| | | **3.4 Service Laptop** | |
| | Information | The DCUTerm tool on a service laptop cab be used to view signals, parameters. It can also be used to temporarily change parameters and do analyze in a graphical format. | |
| 2F_10.01.Zefiro-Europe.TRS.119 | Requirement | It shall be possible to view signals and parameters from a connected service LapTop. | N/A |
| 2F_10.01.Zefiro-Europe.TRS.120 | Requirement | It shall be possible to temporarily change parameters in the TCMS software from a connected service LapTop. | N/A |
| 2F_10.01.Zefiro-Europe.TRS.123 | Requirement | It shall be possible to analyze signals status over time in graphical format. | N/A |

### 2.8.2 2F_10.02 Monitoring-VF

N/A

### 2.8.3 2F_10.03 Visualisation-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 General** | |
| | Information | There are two IDU units located in each cab and one IDU unit in the crew compartment. The primary IDU is located in front of the driver in the cab and back-up IDU located on the left hand side of the driver in the cab. The primary IDU contains all functions described in the TRS. The back-up IDU is connected to power but used as spare. It is not intended to include control of functions. In case the primary IDU fails the back-up IDU will contain full functionality. | |
| | | **1.1 Graphical Strategy for IDU** | |
| | Information | General requirements for display unit (IDU/HMI) | |
| 2F_10.03.Zefiro-Europe.TRS.11 | Requirement | The screen shall be divided into 3 areas:<br>• Upper part shall present main data (e.g. time, date, train speed) and the train configuration<br>• Middle part shall present the menu data currently selected<br>• • Lower part shall present event text and acknowledge feature for Events as well as buttons for navigating on the IDU | N/A |
| 2F_10.03.Zefiro-Europe.TRS.266 | Requirement | The train speed shall be displayed on the IDU. | SIL 2 |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1.2 Audible message** | |
| 2F_10.03.Zefiro-Europe.TRS.181 | Requirement | If an event occurs, requiring train staff attention, this shall be indicated in all cars. An audible message shall be used to direct the train crew to the IDU in the crew compartment. | SIL 0 |
| | | **1.3 Reset of IDU** | |
| | Information | The handshaking mechanism between CCU-S and the IDU's Safety Processor is described within SFRS TCMS Operational, see 3EST000225-7917 chapter 3.4 Fail Safe State Control. | |
| | Information | The handshaking mechanism between CCU-S and the IDU's Safety Processor is described within SFRS TCMS Operational, see chapter 3.4 Fail Safe State Control. | |
| | | **1.4 Back-up IDU** | |
| | Information | The back-up IDU is located on the backwall in the cab. It is mounted behind a transparent cover so the touch screen can not be operated. It is connected to power and Ethernet network. The back-up is in normal operation Off and not used, it is spare.<br><br>The following scenario is foreseen:<br>If the primary IDU fails the driver replaces the primary with the back-up IDU. This will only be done at standstill. If the primary IDU fails at speed the back-up IDU will get all functionality. This makes it possible for the driver to view an alarm if one is coming at speed, i.e. before it is possible to replace the primary IDU with the back-up IDU. | |
| 2F_10.03.Zefiro-Europe.TRS.229 | Requirement | In case the primary IDU in the activated cab is failing the back-up IDU shall contain all functionality | N/A |
| | | **2 Menu structure and navigation** | |
| 2F_10.03.Zefiro-Europe.TRS.234 | Requirement | The IDU functionality shall be designed as a structured menu tree where specific system menus and diagnostics menus can be reached depending on user and location of IDU. | N/A |
| | | **3 Menus** | |
| | | **3.1 IDU - Start + Main Menus** | |
| | | **3.1.1 Login menu** | |
| 2F_10.03.Zefiro-Europe.TRS.21 | Requirement | Access to the IDU shall be protected via a login procedure. | N/A |
| 2F_10.03.Zefiro-Europe.TRS.277 | Requirement | The IDU shall support multiple language (namely French, Spanish, Italian, English) selectable during the login. | N/A |
| 2F_10.03.Zefiro-Europe.TRS.278 | Requirement | Spanish shall be the default language. | N/A |
| | | **3.1.2 Activation menu** | |
| 2F_10.03.Zefiro-Europe.TRS.59 | Requirement | The activation menu shall present the activation status of the train in a graphical format. | SIL 0 |
| 2F_10.03.Zefiro-Europe.TRS.184 | Requirement | The activation menu shall present to the driver the status of the main components related to activation of the train. | SIL 0 |
| | | **3.1.3 Main menu** | |
| 2F_10.03.Zefiro-Europe.TRS.63 | Requirement | The main menu shall act as the menu used as the main access point for navigating the IDU structure. | SIL 0 |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **3.1.4 Station menu** | |
| 2F_10.03.Zefiro-Europe.TRS.67 | Requirement | The station menu shall present the door status to the driver when stopping at a station | SIL 0 |
| | | **3.1.5 Driving menu** | |
| 2F_10.03.Zefiro-Europe.TRS.71 | Requirement | The driving menu shall only present and allow access to necessary information and functions when driving the train. | SIL 0 |
| | | **3.2 Menus for Diagnostics** | |
| | Information | The faults and events which are relevant for the driver are presented on the IDU in the activated driver's cabin. It will be possible for the driver, crew, and service personnel to see all active faults and events on every IDU in the train. | |
| | | **3.2.1 Active event overview menu** | |
| 2F_10.03.Zefiro-Europe.TRS.47 | Requirement | The active event overview menu shall provide an clear overview of which systems on the train have active events and also provide navigation possibilities for the systems with active alarms/faults. | SIL 0 |
| | | **3.2.2 Active event menu** | |
| 2F_10.03.Zefiro-Europe.TRS.75 | Requirement | The active event menu shall present the active events on a train for the driver, crew or maintainer. | SIL 0 |
| 2F_10.03.Zefiro-Europe.TRS.193 | Requirement | The active event menu shall in a clearly arranged format (e.g. list format) present the active events for the selected area. | SIL 0 |
| | | **3.2.3 Event history overview menu** | |
| 2F_10.03.Zefiro-Europe.TRS.205 | Requirement | The event history overview menu shall provide selection and navigation possibility to view event history for a selected system. | SIL 0 |
| | | **3.2.4 Event history menu** | |
| 2F_10.03.Zefiro-Europe.TRS.84 | Requirement | The event history menu shall present the event history on a train for the driver, crew or maintainer. | SIL 0 |
| 2F_10.03.Zefiro-Europe.TRS.191 | Requirement | The event history menu shall in a clearly arranged format (e.g. in a list format) present the history of the events for the selected area. | N/A |
| 2F_10.03.Zefiro-Europe.TRS.279 | Requirement | The event history menu shall present history events from category 1 - 4 for driver and crew. | SIL 0 |
| 2F_10.03.Zefiro-Europe.TRS.280 | Requirement | The event history menu shall present history events from category 1 - 5 for the maintainer. | SIL 0 |
| | | **3.2.5 Expanded event menu** | |
| 2F_10.03.Zefiro-Europe.TRS.255 | Requirement | The expanded event menu shall present detailed information of the selected event. | N/A |
| | | **3.2.6 Events** | |
| 2F_10.03.Zefiro-Europe.TRS.93 | Requirement | The event menus shall present vital event related information to the driver | SIL 0 |
| 2F_10.03.Zefiro-Europe.TRS.197 | Requirement | Events shall be displayed differently depending of the priority of the event. Higher priority events always take precedence over an event of lower priority. | SIL 0 |

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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **3.2.6.1 Serious fault event** | |
| | Information | The serious fault function is described in Conventional Train control, chapter Serious fault indication.<br><br>In parallel to a serious fault an ODBS event is also stored. The ODBS event has got an own category, serious fault events, to distinguish them from the other events and to make it possible to announce them on the IDU with highest priority. | |
| 2F_10.03.Zefiro-Europe.TRS.272 | Requirement | The Serious fault event shall be presented in the active cab when an Serious fault event has occurred. | SIL 0 |
| 2F_10.03.Zefiro-Europe.TRS.273 | Requirement | The Serious fault event shall be removed when the driver has acknowledged the event using the Acknowdge pushbutton on the display. | SIL 0 |
| | | **3.2.6.2 Advisory** | |
| 2F_10.03.Zefiro-Europe.TRS.208 | Requirement | The Advisory event shall be presented in the active cab when the driver has done an incorrect manuevre. | SIL 0 |
| 2F_10.03.Zefiro-Europe.TRS.213 | Requirement | The Advisory event shall be removed when the driver has corrected the faulty manuevre. | SIL 0 |
| | | **3.2.6.3 Alarm** | |
| 2F_10.03.Zefiro-Europe.TRS.210 | Requirement | The Alarm event shall be presented in the active cab and/or in the crew compartment when an Alarm event has occurred. | SIL 0 |
| 2F_10.03.Zefiro-Europe.TRS.214 | Requirement | The Alarm event shall be removed when the user has acknowledged the event using the Acknowdge pushbutton on the display. | SIL 0 |
| | | **3.2.6.4 Warning** | |
| 2F_10.03.Zefiro-Europe.TRS.212 | Requirement | Warning event information shall be presented on the lower part of the screen area of the display in the active cab and/or in the crew compartment when a Warning event has occurred. | SIL 0 |
| 2F_10.03.Zefiro-Europe.TRS.215 | Requirement | The Warning event information shall be removed when the user has acknowledged the event using the Acknowdge pushbutton on the display. | SIL 0 |
| | | **3.2.7 Serious fault acknowledge** | |
| 2F_10.03.Zefiro-Europe.TRS.267 | Requirement | The serious fault indication on driver's desk shall be acknowledged using the acknowledge button on the IDU frame. | SIL 0 |
| | | **3.2.8 System messages** | |
| 2F_10.03.Zefiro-Europe.TRS.97 | Requirement | System message menus shall present information to the user e.g. when the user attempts to activate the cab while another cab is already active or when shutting down the IDU. | SIL 0 |
| | | **3.2.8.1 Train configuration** | |
| 2F_10.03.Zefiro-Europe.TRS.260 | Requirement | At train inauguration the train configuration menu shall be displayed. When the user has acknowledged the train configuration the menu disappears. | SIL 2 |
| | | **3.2.8.2 Lamp test** | |
| 2F_10.03.Zefiro-Europe.TRS.262 | Requirement | At lamp test the IDU shall display a white menu covering the entire screen so that the user can detect if there are any pixel faults. | SIL 0 |
| | | **3.2.8.3 Power On** | |
| 2F_10.03.Zefiro-Europe.TRS.220 | Requirement | When the IDU unit is powered on a startup menu shall be displayed indicating that the application is starting | SIL 0 |

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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **3.2.8.4 Communication failure** | |
| 2F_10.03.Zefiro-Europe.TRS.222 | Requirement | A communication error message menu shall be displayed if the communication to the CCU in the leading cab is lost. | SIL 0 |
| | | **3.2.8.5 Another cab already active** | |
| 2F_10.03.Zefiro-Europe.TRS.224 | Requirement | Another cab already active message menu shall be displayed if an attempt to activate the cab is made when another cab is already active. | SIL 0 |
| | | **3.2.8.6 Shut down** | |
| 2F_10.03.Zefiro-Europe.TRS.226 | Requirement | A shut down message menu shall be displayed when the train is ordered to shut down. | SIL 0 |
| | | **3.2.8.7 Waiting for train configuration** | |
| 2F_10.03.Zefiro-Europe.TRS.253 | Requirement | A waiting for train configuration message menu shall be displayed when the train configuration can not be retrieved or is not valid. | SIL 0 |
| | | **3.3 Menus for Instruments** | |
| | | **3.3.1 Traction-brake menu** | |
| 2F_10.03.Zefiro-Europe.TRS.51 | Requirement | The traction-brake menu shall present the current traction/brake status in a graphical format and allow cut-out and cut-in of dynamic brake | SIL 0 |
| | | **3.3.2 Interlocks menu** | |
| 2F_10.03.Zefiro-Europe.TRS.101 | Requirement | The interlocks menu shall present on IDU if there are any interlocks resulting in traction block, emergency brake or speed limitation | SIL 0 |
| | | **3.3.3 System overview menu** | |
| 2F_10.03.Zefiro-Europe.TRS.109 | Requirement | The system overview menu shall provide navigation buttons to all the system menus of the IDU. | SIL 0 |
| | | **3.3.4 High voltage system menu** | |
| 2F_10.03.Zefiro-Europe.TRS.113 | Requirement | The high voltage system menu shall present status and allow control of the high voltage system via IDU. | SIL 0 |
| | | **3.3.5 Traction system menu** | |
| 2F_10.03.Zefiro-Europe.TRS.117 | Requirement | The traction system menu shall present status and allow manual control of the traction system via IDU. | SIL 0 |
| 2F_10.03.Zefiro-Europe.TRS.189 | Requirement | It shall be possible to cut-out and cut-in main components on the traction system menu (e.g. line converters). | SIL 0 |
| | | **3.3.6 HVAC system menu** | |
| 2F_10.03.Zefiro-Europe.TRS.125 | Requirement | The HVAC system menu shall present status and allow manual control of the HVAC system. | SIL 0 |
| | | **3.3.7 Battery system menu** | |
| 2F_10.03.Zefiro-Europe.TRS.129 | Requirement | The battery system menu shall present the battery system status on the IDU. | SIL 0 |
| | | **3.3.8 Brake system menu** | |
| 2F_10.03.Zefiro-Europe.TRS.133 | Requirement | The brake system menu shall present status and allow to control of the brake system. | SIL 0 |
| | | **3.3.9 Tank supervision menu** | |
| 2F_10.03.Zefiro-Europe.TRS.137 | Requirement | The Tank supervision menu shall present status and allow control of the Tanks on the train (toilet, water supply, waste water system) | SIL 0 |

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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **3.3.10 Fire detection system menu** | |
| 2F_10.03.Zefiro-Europe.TRS.281 | Requirement | The fire detection system menu shall present the status of the fire detection system on the IDU. | SIL 0 |
| | | **3.3.11 Bogie monitoring system menu** | |
| 2F_10.03.Zefiro-Europe.TRS.146 | Requirement | The bogie monitoring system menu shall present status and allow control of the bogie monitoring system. | SIL 0 |
| | | **3.3.12 Air supply system menu** | |
| 2F_10.03.Zefiro-Europe.TRS.150 | Requirement | The air supply system menu shall present status and allow control of the air supply system. | SIL 0 |
| | | **3.3.13 Interior lights system menu** | |
| 2F_10.03.Zefiro-Europe.TRS.276 | Requirement | The interior lights system menu shall present status and allow manual control of the interior lights system. | N/A |
| | | **3.3.14 Front system menu** | |
| 2F_10.03.Zefiro-Europe.TRS.154 | Requirement | The front system menu shall present status of the air front couplers and front hatches. | SIL 0 |
| | | **3.4 Menus for Settings, Tests and Maintenance** | |
| | | **3.4.1 Settings menu** | |
| 2F_10.03.Zefiro-Europe.TRS.55 | Requirement | The Settings menu shall provide possibility to navigate to the different setting functions and also adjust the intensity of the IDU screen. | SIL 0 |
| 2F_10.03.Zefiro-Europe.TRS.236 | Requirement | It shall be possible to switch between languages in run-time. | SIL 0 |
| | | **3.4.2 Master reset menu** | |
| 2F_10.03.Zefiro-Europe.TRS.158 | Requirement | The Master reset menu shall allow activation of master reset on the train via IDU. | SIL 0 |
| | Information | Master reset is not allowed when train is at speed or when master reset have been performed more times then allowed. | |
| | | **3.4.3 Country menu** | |
| | Information | The country menu is used to set the country where the train goes in service. This information is used for systems like ATP that are different in different countries. | |
| 2F_10.03.Zefiro-Europe.TRS.164 | Requirement | The country menu shall present status and allow to set country. | SIL 0 |
| | | **3.4.4 Program version menu** | |
| 2F_10.03.Zefiro-Europe.TRS.166 | Requirement | The program version menu shall present the program versions of TCMS and sub-system devices. | SIL 0 |
| | | **3.4.5 Brake test menu** | |
| 2F_10.03.Zefiro-Europe.TRS.170 | Requirement | The brake test menu shall allow activation of a brake test, give advice how to perform the test and present the test result via IDU. | SIL 2 |
| | | **3.4.6 Master controller test menu** | |
| 2F_10.03.Zefiro-Europe.TRS.174 | Requirement | The master controller test menu shall allow activation of a master controller test, give advice how to perform the test and present the test result | SIL 0 |

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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **3.4.7 Test overview menu** | |
| 2F_10.03.Zefiro-Europe.TRS.178 | Requirement | The test overview menu shall allow activation of different applicable train tests. | SIL 0 |

## 2.9 2F_11 Passenger Information and Entertainment-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 General PIS Control** | |
| | | **1.1 External Interfaces** | |
| | Information | The PIS control unit shall include an USB interface and Bluetooth or WLAN card for the download of PIS database for the automatic management of announcement and displays. | |
| | Information | Bluetooth or WLAN is used for the interface with the standard handheld terminal of the train staff. | |
| | Information | The system control unit shall be connected to the Train to Wayside communication solution. | |
| 2F_11.Zefiro-Europe.TRS.209 | Requirement | The TCMS shall send a signal to PIS control unit before power down. | N/A |
| 2F_11.Zefiro-Europe.TRS.178 | Requirement | The system shall be connected to power supply line controlled via TCMS. | N/A |
| 2F_11.Zefiro-Europe.TRS.233 | Requirement | At train power off phase, all the servers shall be capable to perform a graceful shutdown. | N/A |
| | | **1.2 Internal Interfaces** | |
| | Information | The PIS control unit will also be interfaced with the TCMS, using a dedicated Ethernet hardware interface to be connected on TCMS Ethernet Network. This interface is used to exchange data like between the systems e.g. train signals as speed, door release, diagnostic data, etc.. | |
| | Information | The PIS shall have a separate Ethernet Network. | |
| | Information | The passenger information and video surveillance system shall be implemented using a distributed system. | |
| 2F_11.Zefiro-Europe.TRS.175 | Requirement | The PIS system shall support trains operating in multiple. | N/A |
| | Information | The PIS network shall guarantee communication for single faults of communication equipment. | |
| | | **1.3 PIS MMI in staff area** | |
| | Information | The train staff area located in the special car is equipped with a MMI controlled by the PIS master control unit and allows the train staff to manage the PIS and video surveillance system. | |
| | Information | A MMI controlled by the PIS master control unit shall be provided in the train staff area. | |
| | Information | It shall be possible to send predefined messages to loudspeakers and displays from the PIS MMI in staff area. | |
| | Information | It shall be possible to send real time messages to displays from the PIS MMI in staff area. | |
| | Information | It shall be possible to schedule and manage video and music from the PIS MMI in staff area. | |
| | Information | It shall be possible to modify parameter configuration for the PIS and entertainment system from the PIS MMI in staff area. | |

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| ID | Type | | SIL Needed |
|---|---|---|---|
| | Information | The PIS MMI in staff area shall include functions for system diagnostic | |
| | | **2 Public Announcements** | |
| | Information | Each area of the vehicle accessible to passengers shall be equipped with loudspeakers connected to a centralised information system for sending of messages in case of fire. | |
| | Information | The audible information must be clearly perceived (intelligibility of the language) in any operating condition and at any point in the rolling stock (including restroom and inter-vehicle gangways). | |
| | Information | This will be achieved with 6 loudspeakers for the passenger saloon and 1 loudspeaker for each vestibule and each toilet for every car. One loudspeaker is foreseen for the staff compartment. The vestibule loudspeakers are audible also in the gangway area. | |
| | Information | The loudspeaker system of each coach is subdivided into two lines, one driven by the amplifier of the local control unit, the other by the one of the control unit of the adjacent car. The chosen strategies allows to have the availability of the functionalities on at least half of the loudspeaker in case of failure of one amplifier or of one local control unit. | |
| | Information | The communication system shall be designed in such a manner that it continues to operate at least half (distributed throughout the train) of its loudspeakers in the event of a failure in one of its transmission elements, or another mean shall be available to inform the passengers. | |
| 2F_11.Zefiro-Europe.TRS.92 | Requirement | The intercom function shall be realized through the Bus UIC 568 to which the cab radio, the PIS handsets of each coach and the passenger emergency device are connected. | N/A |
| | Information | The noise level of each coach shall be measured through dedicated noise sensors connected to the local control unit. | |
| | Information | The measured noise level shall be distributed to adjacent local control unit to allow the volume adjustment of both loudspeaker lines. | |
| | Information | The PA volume shall be set, in the passenger saloon, between 5dBA and 10 dBA above the background noise level. | |
| | | **2.1 Announcement from Driver to Passengers** | |
| | Information | There shall be one PIS handset in each driver's cab. This is separate from the GSM-R voice handset. | |
| | Information | It shall be possible to make a public announcement in the train from the PIS handset in the active cab. | |
| | Information | All announcements from the PIS handset shall be heard in all of the loudspeakers in the passenger compartments and in the other cab. | |
| | | **2.2 Announcement from Crew to Passengers** | |
| | Information | There shall be one handset in the crew cabin and one handset in each couple of vestibules. | |
| | Information | There shall be one handset in the bar-bistro area. | |
| | Information | It shall be possible to make a public announcement in the train from the handsets in the crew cabins, vestibules and bar-bistro area. | |
| | Information | All announcements from handset in crew cabin, vestibule and bar-bistro area shall be heard in all of the loudspeakers in the passenger compartments and in all cabs. | |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | Information | Each local control unit shall be connected to the Bus UIC 568 in order to realize the back up for the public address functionalities. | |
| | Information | In case of Ethernet unavailability the public address functionality is assured by the bus UIC in all cases (public address generated through the cab radio or through the dedicated PIS handsets). | |
| | | **2.3 Announcement from Ground to Passengers** | |
| | Information | It shall be possible to make a public announcement in the train from ground control (through the cab radio). | |
| | | **2.4 Automatic Station Information via DVA** | |
| | Information | The ability to send pre-recorded digitized and stored PA is provided. The train position is given from the GPS system or calculated from distance pulses and door released/opening signals from the TCMS.<br>After entering route settings the driver/staff can select a predefined text, and a predefined DVA message can be connected to a predefined text, and will in that case be announced at the same time as the predefined text is scrolled on the internal displays.<br>Pre recorded announcements are automatically triggered by the train position. | |
| | Information | The system shall be used to announce the destination and next stop of the train, or on departure from, each stop. | |
| | Information | The system shall be used to announce the next stop of the train at least two minutes before the arrival of the train at that stop. | |
| | Information | If the next station is less than two minutes planned journey time away, the next station shall be announced immediately following departure from the previous station. | |
| | Information | All digital voice announcements shall be in Spanish, but the system shall be capable of giving announcements in more than one language. (The choice and number of languages shall be the responsibility of the Railway Undertaking having regard to the clientele of an individual train service). | |
| | Information | The system may operate on a manual, an automated or pre-programmed basis. If the system is automated, it shall be possible to suppress, or correct, incorrect or misleading information. | |
| | | **3 Intercommunication between train staff and between train staff and passengers** | |
| | | **3.1 Intercommunication between train personnel** | |
| | Information | Via the handsets it shall be possible to place an intercom call to any other handset present in the train.<br>It shall also be possible to place a global IC to all the handsets. | |
| | Information | The intercommunication shall be managed via GSM-R. | |
| | | **3.2 Intercommunication between train personnel and ground** | |
| | Information | It shall be possible for the train crew and ground control to communicate between one another. | |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **3.3 Intercommunication from passenger to train personnel** | |
| | | **3.3.1 Passenger Emergency Communication system** | |
| 2F_11.Zefiro-Europe.TRS.43 | Requirement | There shall be an interface between the emergency brake handle and the passenger emergency intercom to allow communication between the passenger and the driver. | N/A |
| | Information | The emergency call device shall have a visual and audible indication that the device has been operated. | |
| | Information | There shall be an acknowledgement which is recognisable by the person who triggered the signal. | |
| | Information | It shall not be possible for the passengers to reset the alarm. | |
| | Information | There shall be one Emergency Speech Unit with a brake handle in each vestibule which makes it possible for the passenger to speak to the crew in full duplex. | |
| 2F_11.Zefiro-Europe.TRS.130 | Requirement | Pulling down the emergency brake handle shall activate the Emergency Speech Unit. | SIL 0 |
| | | **3.3.2 Emergency call from Toilet** | |
| | Information | Two emergency call buttons with illuminated indicator shall be arranged in the toilet for PRM. | |
| 2F_11.Zefiro-Europe.TRS.203 | Requirement | When an Emergency Call Button is pressed the system shall show an alarm on the PIS MMI in the crew cabin. The alarm shall include the position of the emergency call in the train | SIL 0 |
| | Information | The crew has to act in a defined way when emergency or attendance events happen, to ensure that the passenger in toilet gets help.<br>After the problem or services is addressed, the crew can press the emergency call button again to reset it. | |
| | | **3.4 Request from wheelchair user** | |
| | Information | The PIS will include equipment by the PRM doors (two doors, one on each side of the bistro car) via which PRM can call the train crew for assistance. | |
| 2F_11.Zefiro-Europe.TRS.81 | Requirement | The PIS control unit will be interfaced with the door dedicated to the access of people with reduced mobility (two PRM doors on the special car) in order to forward the request to the train staff on the dedicated MMI on the special car. | N/A |
| 2F_11.Zefiro-Europe.TRS.227 | Requirement | When a PRM help request button in the wheelchair space is pressed the system shall show an alarm on the PIS MMI in the crew cabin. The alarm shall include the position of the emergency call in the train | N/A |
| 2F_11.Zefiro-Europe.TRS.228 | Requirement | When a PRM help request button in the wheelchair space is pressed the system shall send an alarm to the train staff handheld terminal. The alarm shall include the position of the emergency call in the train | N/A |
| | | **4 Infotainment** | |
| | | **4.1 Entertainment and Scheduling Functions** | |
| | Information | Advertisements shall not be combined with the routeing and information systems. | |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | Information | Both entertainment content and commercial advertising content can be selected and scheduled. Dynamic pages with real time data are also available and can be scheduled. Dynamic pages are also available and dynamic pages with real time info can be scheduled. | |
| | Information | The visualization on the displays can be programmed by defining the schedule of the content to be visualized for each window. | |
| | Information | The new content will be sent to the entertainment server of car 3 to be stored on the local disk. | |
| | Information | Downloading of scheduling and media content shall be supported via USB key, through Ethernet network on train and from groundbased system for storage on the entertainment server. | |
| | Information | The displays shall be able to visualize up to four windows. | |
| | Information | Information on all LCD displays shall be synchronized. | |
| | Information | The audio shall be distributed over Ethernet. | |
| | Information | A GPS signal shall be received and evaluated to determine the position of the train. | |
| | Information | A scheduling can be requested to visualize the current position of the train. A georeferenced map will be displayed and the train will be presented over the map. The layers and the zoom of the map will be defined. An example of possible visualization way is done in the following figure. | |
| | | **4.2 WiFi internet/intranet connections on board** | |
| | Information | A local wifi connection on board trains has to be foreseen, both to server for access to the content on board, both to the Internet. | |
| | Information | Internet access is provided to the passenger in case of connectivity from train to Internet. | |
| | Information | Three separate GSM/UMTS/HSDPA/HSUPA modems with three different broadband suppliers shall be used for trains access to Internet. | |
| | Information | Each car shall be equipped with one dedicated Intranet/Internet server and one WiFi access point. | |
| | Information | The train access to Internet shall include a hybrid system satellite/UMTS. UMTS applies in Italy, and option for hybrid system satellite applies abroad. | |
| | Information | The system shall automatically chooses the satellite or ground connection according to the signal availability and to configured priorities. | |
| | Information | Change of active telephone or internet connection shall be transparent to the passengers connection. | |
| | | **5 Displays** | |
| | Information | Each station name (which may be abbreviated), or words of messages, shall be displayed for a minimum of 2 seconds. | |
| | Information | If a scrolling display is used (either horizontal or vertical), each complete word shall be displayed for a minimum of 2 seconds and the horizontal scrolling speed shall not exceed 6 characters per second. | |
| | | **5.1 Exterior Display** | |
| 2F_11.Zefiro-Europe.TRS.31 | Requirement | Automatic management of internal and external displays by the OBoE system | N/A |

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| ID | Type | | SIL Needed |
|---|---|---|---|
| | Information | All text presented on the displays shall be in Spanish, but the system shall be capable of presenting text in more than one language. (The choice and number of languages shall be the responsibility of the Railway Undertaking having regard to the clientele of an individual train service). | |
| | Information | The final destination or route shall be displayed outside of the train on the platform side adjacent to at least one of the passenger access doors on a minimum of alternate vehicles of the train. | |
| | Information | Where trains operate in a system, in which dynamic visual information is given on the station platforms within a distance of 50 meters, and destination or route information is also provided on the front of the train, it is not mandatory to provide information on the sides of each vehicle. | |
| 2F_11.Zefiro-Europe.TRS.232 | Requirement | The coach number for each coach shall be derived, also in the case of multiple unit, and displayed on the external displays. | N/A |
| | | **5.2 Interior Display** | |
| | Information | The next station shall be displayed at least two minutes before arrival at the station concerned. | |
| | Information | If the next station is less than two minutes planned journey time away, the next station shall be displayed immediately following departure from the previous station. | |
| | Information | The display shall revert to show the final destination as soon as the train has stopped. | |
| | Information | It shall be possible for the crew to select a predefined text which will be presented on the display. | |
| | Information | All text presented on the displays shall be in Spanish, but the system shall be capable of presenting text in more than one language. (The choice and number of languages shall be the responsibility of the Railway Undertaking having regard to the clientele of an individual train service). | |
| | | **6 Events** | |
| 2F_11.Zefiro-Europe.TRS.224 | Requirement | Diagnostics from the supplier shall be stored in the train diagnostic system. | N/A |

### 2.10 2F_12 Train to Wayside Communication-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 General information** | |
| | Information | Telediagnostica is a system for remote diagnostics incorporating an on-board part on the train and an off-board part in the customer workshop.<br>The on-board part is made up of two different devices: MCG (Mobile Communication Gateway) and TU (Telediagnostica Unit).<br>Telediagnostica system scope is to collect and provide data to support CBM analysis. | |
| | Information | The on board Telediagnostica application software is located in the MCG and TU. They shall be independent on the TCMS software. | |
| | Information | The MCG is used for the remote upload of diagnostic data and for the wayside train monitoring (e.g. sending the current train status and position to a wayside system). | |

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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **2 Train To Wayside Communication** | |
| | | **2.1 Self diagnosis** | |
| 2F_12.Zefiro-Europe.TRS.164 | Requirement | During boot-up, the device will perform diagnosis activities of its main components of the on board system (boards, RAM, etc.). | N/A |
| 2F_12.Zefiro-Europe.TRS.165 | Requirement | Moreover, all active system processes will be cyclically checked on the basis of configuration, after minor failures and, if necessary, restarted. In case of serious non-recoverable damage the complete system will be reset or a controlled shutdown will be performed. | N/A |
| 2F_12.Zefiro-Europe.TRS.166 | Requirement | The device does not require manual activities to restore non-blocking hardware or software failures of its component. | N/A |
| | | **2.2 Event Data and Condition Data upload** | |
| 2F_12.Zefiro-Europe.TRS.85 | Requirement | The on-board system shall read the content of diagnostic database (ODBS) to upload it to the wayside system. | N/A |
| 2F_12.Zefiro-Europe.TRS.102 | Requirement | The on-board system shall archive TCMS diagnostic data in order to be uploaded manually using an USB stick. | N/A |
| 2F_12.Zefiro-Europe.TRS.168 | Requirement | The on-board unit shall upload diagnostic data (and eventually states and alarms) using LTE communication interfaces.. | N/A |
| 2F_12.Zefiro-Europe.TRS.110 | Requirement | The on-board unit shall remove from its permanent memory diagnostic data files already uploaded to the ground server. | N/A |
| | | **2.3 Remote monitoring in real time** | |
| 2F_12.Zefiro-Europe.TRS.51 | Requirement | The maintainers shall get remote access to a representation of the active train driver desk status in real time. | N/A |
| 2F_12.Zefiro-Europe.TRS.58 | Requirement | The maintainers shall get remote read access to a representation of the active cab HMI in real time. | N/A |
| 2F_12.Zefiro-Europe.TRS.60 | Requirement | The refresh time of signals transferred to wayside for remote driver desk/HMI session shall be scaled on the available bandwidth and priority management. | N/A |
| 2F_12.Zefiro-Europe.TRS.69 | Requirement | The maintainers shall get read access to current diagnostic status of the train. | N/A |
| | | **2.4 Process data logging** | |
| 2F_12.Zefiro-Europe.TRS.112 | Requirement | The on-board unit shall be able to log sets of signals acquired from the vehicle busses. | N/A |
| 2F_12.Zefiro-Europe.TRS.123 | Requirement | Data logging results shall be uploaded on wayside servers. | N/A |
| 2F_12.Zefiro-Europe.TRS.125 | Requirement | It shall be possible to configure data logging sample time | N/A |
| 2F_12.Zefiro-Europe.TRS.126 | Requirement | It shall be possible to configure data logging duration | N/A |
| 2F_12.Zefiro-Europe.TRS.127 | Requirement | It shall be possible to configure data logging periodicity | N/A |
| 2F_12.Zefiro-Europe.TRS.128 | Requirement | It shall be possible to configure data logging application (fleets of vehicles or individual vehicles) | N/A |
| 2F_12.Zefiro-Europe.TRS.129 | Requirement | It shall be possible to configure data logging triggering start conditions | N/A |
| | | **2.5 Operating counters** | |
| 2F_12.Zefiro-Europe.TRS.135 | Requirement | The on-board system shall allow to remotely configure new condition data. | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **2.6 Upgrading of SW configuratio** | |
| 2F_12.Zefiro-Europe.TRS.138 | Requirement | It shall be possible to change the on-board system configuration. | N/A |
| | | **2.7 Connection to maintenance Laptop** | |
| 2F_12.Zefiro-Europe.TRS.145 | Requirement | It shall be possible to update the configuration of the on-board system using a maintenance PC equipped with proper management software. | N/A |
| 2F_12.Zefiro-Europe.TRS.146 | Requirement | It shall be possible to access to data logging files using a maintenance laptop equipped with proper management software. | N/A |
| 2F_12.Zefiro-Europe.TRS.147 | Requirement | It shall be possible to access signals monitoring in real time using a maintenance laptop equipped with proper management software. | N/A |
| 2F_12.Zefiro-Europe.TRS.59 | Requirement | The maintainers shall get read access to all available process data on the TCMS and propulsion system using DCUTerm tool (Only by wired connection). | N/A |
| 2F_12.Zefiro-Europe.TRS.67 | Requirement | The maintainers shall get read access to parameters in the TCMS and propulsion system using the DCUTerm tool (Only by wired connection). | N/A |
| | | **2.8 Execution of diagnostic algorithms** | |
| 2F_12.Zefiro-Europe.TRS.158 | Requirement | The on-board unit shall be able to execute specific diagnostic algorithms in order to generate diagnostic events triggered by programmable conditions. | N/A |
| | | **2.9 Downloading data using USB stick** | |
| 2F_12.Zefiro-Europe.TRS.150 | Requirement | The vehicle diagnostic shall also be uploaded using a removable memory with USB interface, just inserting it in the on-board unit. | N/A |
| 2F_12.Zefiro-Europe.TRS.151 | Requirement | The signals data logging shall also be uploaded using a removable memory with USB interface, just inserting it in the on-board unit. | N/A |

### 2.11 2F_13 CCTV-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 General CCTV Control** | |
| | | **1.1 Architecture** | |
| | Information | The delivery shall include a video surveillance system. | |
| | Information | The recording of video streams shall guarantee 100 hours of recordings. | |
| | Information | Download of information from a centralized location shall be supported through the PIS network. | |
| | Information | Video-surveillance devices which provide for a terminal in the driver's cab or other devices equipped, in any case, with video terminal in driver's cab, may be enabled, in the same driver's cab at speeds = 0 km/h. | |
| | Information | There are four cameras for the intermediate cars, and three cameras for the front cars. | |
| | Information | Front and intermediate cars shall be equipped with one camera for each vestibule and two cameras for passenger saloon. | |
| | Information | Special car shall be equipped with one camera for the passenger saloon, one camera for each vestibule and one camera for the bar-bistro area. | |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_13.Zefiro-Europe.TRS.40 | Requirement | Each image shall be recorded together with the information such as:<br>* Train number<br>* Date and time<br>* Vehicle number | N/A |
| | | **1.2 PIS MMI in staff area** | |
| | Information | The train staff area located in the special car is equipped with a MMI controlled by the PIS master control unit and allows the train staff to manage the PIS and video surveillance system. | |
| | Information | The PIS MMI in staff area shall be used to manage the video surveillance system. | |
| | Information | It shall be possible the stream video from every camera in the train to the PIS MMI in staff area. | |
| | Information | It shall be possible to use up to four camera steams in parallel to the PIS MMI in staff area. | |
| | Information | It shall be possible to cyclically visualize all camera streams on the PIS MMI in staff area. | |
| | Information | It shall be possible to modify parameter configuration for the CCTV system from the PIS MMI in staff area. | |
| 2F_13.Zefiro-Europe.TRS.37 | Requirement | In case of fire alarm detected onboard the Staff MMI shall visualize the image of the nearest camera to the location of where the fire has been detected. | N/A |
| | | **2 Events** | |
| 2F_13.Zefiro-Europe.TRS.39 | Requirement | Diagnostics from the supplier shall be stored in the train diagnostic system. | N/A |

### 2.12 2F_14 Exterior Door and Step Systems-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 Exterior Doors: Overall Door Control** | |
| | Information | There are two types of exterior doors:<br>– Standard door<br>– PRM Doors for wheelchair users; two in the bistro car.<br><br>Both door types comply with the TSI PRM standard, however only the door type for wheelchair users is called 'PRM door'. | |
| | Information | Each local door system (both standard and PRM door) is controlled by its own door control unit which is connected to TCMS. | |
| | | **1.1 Door Release** | |
| | | **1.1.1 Release exterior door by driver** | |
| 2F_14.Zefiro-Europe.TRS.64 | Requirement | Permission to open exterior doors shall be activated by the driver by pushing the door release push buttons. | SIL 2 |
| 2F_14.Zefiro-Europe.TRS.222 | Requirement | It shall only be possible to release the doors when the train speed is 0 km/h. | SIL 2 |
| 2F_14.Zefiro-Europe.TRS.175 | Requirement | The doors shall be kept closed and locked until the driver releases them. | SIL 2 |
| 2F_14.Zefiro-Europe.TRS.180 | Requirement | The doors shall only be possible to open when the speed is equal to 0 km/h. | SIL 2 |
| 2F_14.Zefiro-Europe.TRS.177 | Requirement | Door release shall be cancelled automatically when train speed > 0 km/h | SIL 0 |

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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_14.Zefiro-Europe.TRS.223 | Requirement | Door release shall be cancelled when close request is given by crew using one local panel. | SIL 0 |
| 2F_14.Zefiro-Europe.TRS.224 | Requirement | Door release shall be cancelled when close request is given by driver. | N/A |
| 2F_14.Zefiro-Europe.TRS.174 | Requirement | In the event of loss of power to the door controls, the doors shall be kept locked by the locking mechanism. | N/A |
| 2F_14.Zefiro-Europe.TRS.213 | Requirement | All doors shall automatically close when speed > 0 km/h is detected. | SIL 0 |
| 2F_14.Zefiro-Europe.TRS.302 | Requirement | In washing mode door release shall not be possible. | N/A |
| 2F_14.Zefiro-Europe.TRS.311 | Requirement | Undue Door Release Command (several doors are unlocked (with train crew not correctly informed of this door status) or released or opened in inappropriate areas or situations) shall have a risk controlled to an acceptable level. | SIL 3 |
| | | **1.1.2 Side Selection for Door Release** | |
| 2F_14.Zefiro-Europe.TRS.66 | Requirement | Permission shall be given individually for each side (side selective) of the train<br>→ 2 push-buttons (1x for left side, 1x for right side). | SIL 2 |
| 2F_14.Zefiro-Europe.TRS.225 | Requirement | The push button in the cab shall be lighted when the doors are released | SIL 0 |
| | | **1.1.3 Acoustic Signal for Door Release** | |
| 2F_14.Zefiro-Europe.TRS.166 | Requirement | When the doors are released a signal, different from the audible door closing alarm, shall be given that is clearly audible to persons inside the train. This alert signal shall sound for a minimum of five seconds unless the door is operated, in which case it may cease after 3 seconds. | SIL 0 |
| | | **1.2 Close External Doors** | |
| | | **1.2.1 Close external doors by driver command** | |
| 2F_14.Zefiro-Europe.TRS.187 | Requirement | A general close command shall be issued if the door close push button in the cab is pressed. | SIL 0 |
| 2F_14.Zefiro-Europe.TRS.219 | Requirement | There shall be one close pushbutton closing all doors of the train. | N/A |
| 2F_14.Zefiro-Europe.TRS.188 | Requirement | When pressing the door close push button in the cab the release signal for both sides shall be withdrawn. | SIL 0 |
| 2F_14.Zefiro-Europe.TRS.217 | Requirement | The visual indication that the door is released shall cease when the door release has been withdrawn.<br>3 seconds after the door release has been withdrawn the door shall start to close. | N/A |
| 2F_14.Zefiro-Europe.TRS.189 | Requirement | A general closing command shall override any other command. | SIL 2 |
| | Information | The doors will not be possible to open again once a general closing command has been issued.<br>This is ensured since the release has been withdrawn. | |
| | | **1.2.2 Audible warning** | |
| 2F_14.Zefiro-Europe.TRS.49 | Requirement | An audible warning sound at each door shall be activated 3s before the doors start to close and continue until the door is closed. The sound shall be different in tone to that used when the door is released. | SIL 0 |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1.2.3 Green Loop** | |
| 2F_14.Zefiro-Europe.TRS.136 | Requirement | All doors must be closed and locked in order for the train to depart. | SIL 2 |
| 2F_14.Zefiro-Europe.TRS.298 | Requirement | It shall only be possible to drive the train if the on-board lift is stowed. | N/A |
| 2F_14.Zefiro-Europe.TRS.191 | Requirement | The closed status of all doors and steps shall be indicated to the driver by a lamp in driver's desk. | SIL 2 |
| | | **1.3 Step operation** | |
| | Information | The opening of the door leaf is interlocked to the complete extraction of the step.<br><br>During the closing phase, the step is immediately closed as soon as the translation movement of the door leaf is ended.<br><br>The sequence between door and step movement is obtained through the use of interlock contacts.<br>The door system provides a signal to the train that the doors are closed and locked (or isolated) and the relevant steps are closed.<br>In the case of the moving step extending beyond that permitted by the gauging rules, the will be immobilised whilst the step is extended.<br>The step will be fitted with a device capable of stopping the movement of that step, if its front edge comes into contact with any object or person whilst the step is in movement. | |
| 2F_14.Zefiro-Europe.TRS.139 | Requirement | The steps shall only extend if the train is at standstill. | SIL 1 |
| | | **1.3.1 Door Status during Cab activation/deactivation** | |
| 2F_14.Zefiro-Europe.TRS.53 | Requirement | When a cab is deactivated, each door (including step) status shall remain until a cab is activated again and driver gives a new release/close order. | N/A |
| | | **1.4 Supervision and fault handling** | |
| 2F_14.Zefiro-Europe.TRS.52 | Requirement | Status of each door shall be shown on the IDU. | SIL 0 |
| 2F_14.Zefiro-Europe.TRS.305 | Requirement | Visualization of door status (vehicle green loop) is needed on IDU. | SIL 0 |
| | Information | A fault log is stored in the TDS (Train Diagnostic System).<br>Each door system provides its status and diagnostic signals (failure codes) via the TCMS. The TDS stores an individual fault log and malfunction events for each door.<br>A detailed fault log is stored on the DCU and can be downloaded via local service interface. | |
| 2F_14.Zefiro-Europe.TRS.57 | Requirement | There shall be one set of events stored locally in the DCUs and one set that shall be stored in the TDS. | SIL 0 |
| 2F_14.Zefiro-Europe.TRS.236 | Requirement | The control system shall be designed in such a manner that at least two independent faults must occur at the same time before the respective door opens un-commanded if the door is not released. | N/A |
| 2F_14.Zefiro-Europe.TRS.242 | Requirement | After shut down, power failure, isolation or emergency operation, powering up of the door system shall be in a controlled manner to prevent injury. | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1.5 Special functions for PRM door** | |
| | Information | The PRM door is equipped with a system featuring a wheelchair lift | |
| | | **1.5.1 PRM door opening** | |
| | Information | The PRM door opening command is activated by crew members through the RIC key operated switch at each door way (inside and outside). | |
| 2F_14.Zefiro-Europe.TRS.229 | Requirement | The counter clockwise rotation shall provoke the opening of the door. | N/A |
| | Information | The door opening time is adjustable by software. The average opening time is 6 ± 1 s for a 1050mm free opening. | |
| | | **1.5.2 PRM door closing** | |
| 2F_14.Zefiro-Europe.TRS.194 | Requirement | PRM door shall only be possible to close when the PRM lift is in "stowed position". | SIL 0 |
| 2F_14.Zefiro-Europe.TRS.230 | Requirement | The PRM door shall be closed when the general closing command is issued. | N/A |
| 2F_14.Zefiro-Europe.TRS.231 | Requirement | The clockwise rotation of the PRM door RIC key shall close the PRM door. | SIL 0 |
| | Information | The average closing time is 6 ± 1 s for a completely open door. The door cannot be closed both locally or centrally if relevant lift consent is not available. | |
| | | **1.5.3 Request to open door for wheelchair users** | |
| | Information | There will be a possibility for wheelchair users to inform train staff on the dedicated MMI on the special car and on the staff mobile handset to request the opening of the door for wheelchair users. | |
| | | **2 Local Door Control** | |
| | | **2.1 Cab doors** | |
| | Information | Cab doors are located in the electronic room and are under manual operation. | |
| 2F_14.Zefiro-Europe.TRS.245 | Requirement | If a cab door is open the green loop shall be interrupted | SIL 2 |
| | | **2.2 Open external door** | |
| | Information | The push button is located in door area but not on the door leaf. | |
| 2F_14.Zefiro-Europe.TRS.70 | Requirement | Passengers shall have the possibility to open a released standard door by using the push button from inside and outside | SIL 0 |
| 2F_14.Zefiro-Europe.TRS.232 | Requirement | An active push button shall give acknowledgement of its activation. | SIL 0 |
| 2F_14.Zefiro-Europe.TRS.216 | Requirement | The push button shall be illuminated when the door is released (possible to open). | SIL 0 |
| | | **2.3 Close external door** | |
| 2F_14.Zefiro-Europe.TRS.104 | Requirement | Closing command of the local standard door shall be activated by passengers through push buttons locate inside the coach. | SIL 0 |
| 2F_14.Zefiro-Europe.TRS.45 | Requirement | If the door release signal is present, the door shall start to close after an adjustable time from the opening. In this case this closing can be interrupted by a local opening command. | SIL 0 |
| 2F_14.Zefiro-Europe.TRS.218 | Requirement | If door released is withdrawn, the door shall automatically close. | SIL 0 |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **2.4 Local crew operation** | |
| | Information | Every door is equipped with a three positions crew switch to activate the general closing by the use of a RIC-key. | |
| 2F_14.Zefiro-Europe.TRS.74 | Requirement | The clockwise rotation of the crew switch shall provoke the closure of all doors excluding the door under control. | SIL 0 |
| 2F_14.Zefiro-Europe.TRS.195 | Requirement | The counter clockwise rotation of the crew switch shall provoke the cancellation of the general closing command previously given from the same door. | SIL 2 |
| | Information | Near the crew switch there is a white lamp indicating that all the doors, excluding the door under control, are closed. The lamp is switched off when the door under control closes. | |
| 2F_14.Zefiro-Europe.TRS.198 | Requirement | When the doors are closed by the crew switch the white lamp by the door shall be illuminated when all doors excluding the local door is closed. The indication lamp shall be turned off when also the local door is closed. | SIL 0 |
| | Information | The door under control has to be closed by the local closing push-button within 5-15 s (configurable parameter) from the switching on of the white lamp. | |
| 2F_14.Zefiro-Europe.TRS.196 | Requirement | The door under local command shall close if the close push button is pushed within 5-15 s (configurable parameter) from switching on the white lamp. | SIL 0 |
| | | **2.5 Status indication of each door** | |
| 2F_14.Zefiro-Europe.TRS.146 | Requirement | Near each door, inside the vestibules, the following lamps shall be present<br>– Yellow, to indicate the status of door system active AND door closed AND not released<br>– Green, to indicate the status of door system active AND released<br>– White: to indicate the status of door isolated | SIL 0 |
| | | **2.6 Obstacle Detection** | |
| 2F_14.Zefiro-Europe.TRS.79 | Requirement | The obstacle detection shall be active during closing and opening. | SIL 1 |
| | Information | The obstacle detection shall check if the door movement is obstructed in any way. In case of that the door shall stop. | |
| | Information | At standstill, after the door has been open by the obstacle detection, it closes again after 3 seconds (cyclically till the removal of the obstacle). | |
| 2F_14.Zefiro-Europe.TRS.203 | Requirement | During obstacle detection sequences the warning buzzer shall always be activated | SIL 0 |
| 2F_14.Zefiro-Europe.TRS.201 | Requirement | The door shall try to close a number of attempts before the door control stops any further door movements and sends an error message. Then the door shall be free to be pushed aside by hand. | SIL 0 |
| 2F_14.Zefiro-Europe.TRS.204 | Requirement | Obstacle detection parameters shall be adjustable acting on door central unit. | N/A |
| | | **2.7 Isolation of Doors** | |
| | | **2.7.1 Manual isolation of single Doors** | |
| | Information | It is possible to manually close and lock each door through a door isolation device. This device is activated by use of a RIC Key. | |
| 2F_14.Zefiro-Europe.TRS.3 | Requirement | It shall be possible to manually isolate one door from inside and outside, which electrically bypasses the part of the green loop for this door, independent from the automatic lock. | SIL 2 |

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| ID | Type | | SIL Needed |
|---|---|---|---|
| | Information | A dedicated lamp is switched on when the isolation procedure is correctly completed. It is possible to see the status of this door on the IDU. | |
| | Information | On the shelf plate a service toggle switch is mounted. Operation of this switch makes it possible to switch off the voltage supply of the door system for maintenance purposes.<br>In this case no failure codes can be generated by the door control unit, no information of this door can be sent to the train control unit via Ethernet-Bus and no out of service indicators can be illuminated. Door "green loop" will not be bypassed.<br>The door can only be opened with the emergency egress/access device. | |
| | | **2.8 Emergency opening exterior doors** | |
| | Information | Every door shall have inside and outside a manual emergency opening device (emergency handle). | |
| 2F_14.Zefiro-Europe.TRS.207 | Requirement | It shall only be possible to operate the internal egress device successful at speed < 10 km/h. | N/A |
| 2F_14.Zefiro-Europe.TRS.209 | Requirement | For train speeds v≥10km/h the internal emergency opening shall be prevented. | SIL 2 |
| | Information | In case of fire the vehicle will normally be evacuated through the passengers access doors. The passengers access doors (clearly identified as “exits”) which can be used for evacuation in case of an emergency, will be opened by hand (with or without current) from both sides. The manual opening from outside is possible through the square service key. | |
| | Information | The external access device is intended to be also for crew or authorized personnel use. This device unlocks the individual door from outside the vehicle. The device is designed to be service key operated; the service key allows to act on an handle device that unlocks the door leaf, in compliance with EN 14752 requirements. | |
| 2F_14.Zefiro-Europe.TRS.93 | Requirement | The operation of the emergency handle causes also the step extraction, if the power supply is available. | N/A |
| 2F_14.Zefiro-Europe.TRS.208 | Requirement | Through the RIC key it shall also be possible to open a door without removing the seal of the emergency egress device. | SIL 2 |
| 2F_14.Zefiro-Europe.TRS.210 | Requirement | The external emergency handle to open exterior door from outside shall work independently from train speed. | SIL 2 |
| 2F_14.Zefiro-Europe.TRS.211 | Requirement | An activation of the emergency opening device shall result in:<br>1. Unlocking of the door system<br>2. The green loop shall be interrupted,<br>3. The door shall send an “emergency opened” status to train control<br>4. Energizing of the door motor shall be prevented<br>5. Permanent warning sound shall be activated (when power supply is available) | SIL 2 |
| 2F_14.Zefiro-Europe.TRS.212 | Requirement | Emergency opening shall not be possible when door system is isolated, i.e. locked out of service. | N/A |
| | Information | If the voltage supply of the door was switched off before the emergency device was reset, the door control unit can’t monitor the current door position. Therefore the door will be closed with an initialization routine until the door has reached the closed position (reference position for door position sensor). | |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_14.Zefiro-Europe.TRS.238 | Requirement | An emergency egress device that has been operated shall be indicated as follows:<br>– to the train system (on board crew or control centre);<br>– locally at the related door (visually and/or audible).<br>A clear proof of operation of the device shall be provided.(e.g. break glass, lead seal, data recording.) The exact method shall be specified. | N/A |
| 2F_14.Zefiro-Europe.TRS.240 | Requirement | Under power supply failure condition the access device shall remain functional and allow access, provided that the door has not been isolated in service. | N/A |
| 2F_14.Zefiro-Europe.TRS.241 | Requirement | An emergency access device that has been operated shall be reported on the train system | SIL 0 |
| 2F_14.Zefiro-Europe.TRS.301 | Requirement | Status of released door s shall be maintained after the trains have been coupled/decoupled | SIL 2 |
| | | **3 Events** | |
| 2F_14.Zefiro-Europe.TRS.291 | Requirement | Diagnostics from the supplier shall be stored in the train diagnostic system. | SIL 0 |

## 2.13 2F_15 Heating and Ventilation and Air Conditioning-VF

### 2.13.1 2F_15.01 Cab HVAC-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 Priority of cab operational modes** | |
| 2F_15.01.Zefiro-Europe.TRS.10 | Requirement | The Safety relevant mode "Fire inside" shall have highest priority. | SIL 1 |
| 2F_15.01.Zefiro-Europe.TRS.78 | Requirement | The priority order below the Fire Inside mode shall be:<br>• HVAC OFF mode | N/A |
| | | **2 HVAC cab - Fire inside mode** | |
| | Information | In case fire is detected the cab HVAC in the non activated cab is automatically switched off via hardwired train line signalled via the fire detection system. The cab HVAC in the activated cab is switched off by the driver using a switch on the driver's desk via train line. | |
| 2F_15.01.Zefiro-Europe.TRS.18 | Requirement | The cab HVAC shall be switched off by hardwired train line when fire has been detected. | SIL 1 |
| | | **3 HVAC cab - Off mode** | |
| 2F_15.01.Zefiro-Europe.TRS.15 | Requirement | The driver shall be able to shut down HVAC cab manually via the IDU. | N/A |
| 2F_15.01.Zefiro-Europe.TRS.146 | Requirement | The driver shall be able to switch off the cab HVAC. | SIL 1 |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_15.01.Zefiro-Europe.TRS.16 | Requirement | The TCMS shall switch off the cab HVAC in all cabs in case of the following states:<br><br>Normal switch off (for whole train):<br>The TCMS shall switch off the HVAC system in case of the following states (input via IDU or connected with train status):<br>– Wash mode → from train operational mode<br>– Shut down → from train operational mode(opening of battery contactor)<br>– HVAC off order per train<br>– HVAC off order from TCMS, if 400 Volt are not available in preconditioning mode/ frost protection mode. | SIL 0 |
| | | **4 HVAC cab - Automatic mode** | |
| | Information | In normal operation the cab HVAC system automatically controls the interior climate in the cab according to ambient conditions. The set point temperature is controlled by the driver. | |
| 2F_15.01.Zefiro-Europe.TRS.24 | Requirement | In automatic mode the cab HVAC system controller shall automatically switch between the modes Cooling, Ventilation and Heating. The interior temperature shall be set depending on the ambient temperature. | N/A |
| 2F_15.01.Zefiro-Europe.TRS.25 | Requirement | When no other mode is requested the cab HVAC system shall automatically operate in normal mode. | N/A |
| | | **5 HVAC cab - Communication missing mode** | |
| 2F_15.01.Zefiro-Europe.TRS.40 | Requirement | When the TCMS fails the cab HVAC system shall operate automatically with the latest valid operation mode. | N/A |
| | | **6 HVAC cab - Preconditioning mode** | |
| | Information | Preheating / pre-cooling is set before the train enters into service. The modes are automatically deactivated. | |
| 2F_15.01.Zefiro-Europe.TRS.44 | Requirement | The preconditioning mode of HVAC cab unit shall be activated and deactivated in parallel with preconditioning mode of HVAC saloon unit. | N/A |
| | | **7 HVAC cab - Tunnel mode** | |
| 2F_15.01.Zefiro-Europe.TRS.76 | Requirement | When signal from differential pressure switch is activated and if no other mode with higher priority is set, all HVAC systems shall operate in tunnel mode.<br>The driver shall also be able to activate and de-activate tunnel mode manually. | N/A |
| 2F_15.01.Zefiro-Europe.TRS.74 | Requirement | The air inlet and outlet flaps shall be possible to close when running in tunnel mode. | N/A |
| | | **8 HVAC Cab - Recirculation mode** | |
| 2F_15.01.Zefiro-Europe.TRS.151 | Requirement | When fire has been detected in one car the HVAC units in the others cars shall be set in recirculation mode. | SIL 0 |
| | | **9 HVAC Cab - Compressor start** | |
| 2F_15.01.Zefiro-Europe.TRS.149 | Requirement | The HVAC compressors shall be started sequentially to ensure that the start-up load is not too high. | N/A |
| | | **10 HVAC cab - Interface with HVAC saloon** | |
| 2F_15.01.Zefiro-Europe.TRS.45 | Requirement | The HVAC saloon shall be interfaced with the HVAC cab. | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| 2F_15.01.Zefiro-Europe.TRS.42 | Requirement | The nominal level of comfort shall then be provided in case of fault to the cab HVAC. | N/A |
| | | **11 HVAC Cab - Maintenance mode** | |
| 2F_15.01.Zefiro-Europe.TRS.154 | Requirement | It shall be possible to set the HVAC in test mode. | N/A |
| | | **12 HVAC cab - Control of temperature** | |
| 2F_15.01.Zefiro-Europe.TRS.50 | Requirement | The cab room temperature shall be adjusted from the comfort curve. | N/A |
| 2F_15.01.Zefiro-Europe.TRS.48 | Requirement | The driver shall be able to adjust the cab room temperature offset in the activated cab. | N/A |
| | | **13 HVAC cab - Control of airflow** | |
| 2F_15.01.Zefiro-Europe.TRS.52 | Requirement | The driver shall be able to change the air flow level in the cab. | N/A |
| 2F_15.01.Zefiro-Europe.TRS.53 | Requirement | The air flow level positions shall be Stop, Auto, 1, 2, 3 and 4. | N/A |
| 2F_15.01.Zefiro-Europe.TRS.54 | Requirement | The air flow level and temperature set point in a non active cab shall be set to Auto regardless of the switch position. | N/A |
| | | **14 HVAC cab - Condition data** | |
| 2F_15.01.Zefiro-Europe.TRS.64 | Requirement | The status of the HVAC system shall be indicated on the IDU. | SIL 0 |
| | Information | It shall be possible to view the software version of the HVAC application for each HVAC computer on the IDU. | |
| | | **15 HVAC cab - Fault supervision** | |
| 2F_15.01.Zefiro-Europe.TRS.60 | Requirement | The HVAC shall supervise itself and a fault indication shall be set in case of component faults. | N/A |
| 2F_15.01.Zefiro-Europe.TRS.155 | Requirement | Complete diagnostic of the cab HVAC system, including predictive diagnostic functions and CBM, shall be fully implemented in each cab HVAC unit controller. The data shall be transmitted to the TCMS via the saloon HVAC bus interface. | N/A |
| | | **16 Events** | |
| 2F_15.01.Zefiro-Europe.TRS.143 | Requirement | Diagnostics from the supplier shall be stored in the train diagnostic system. | N/A |

### 2.13.2 2F_15.02 Saloon HVAC-VF

| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **1 Priority of operational modes** | |
| 2F_15.02.Zefiro-Europe.TRS.11 | Requirement | The Safety relevant mode "Fire Inside" shall have highest priority | SIL 1 |
| 2F_15.02.Zefiro-Europe.TRS.99 | Requirement | The priority order below the Fire Inside mode shall be:<br>HVAC OFF mode<br>Battery mode | N/A |
| | | **2 HVAC saloon - Fire inside mode** | |
| 2F_15.02.Zefiro-Europe.TRS.15 | Requirement | The saloon HVAC shall be switched off by hardwired train line when<br>fire has been detected | SIL 1 |
| | Information | Intumescent grids are used and they are not controlled by train computer or electrically. | |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **3 HVAC saloon - Off mode** | |
| 2F_15.02.Zefiro-Europe.TRS.22 | Requirement | The driver shall be able to shut down HVAC Saloon manually via the IDU. | N/A |
| 2F_15.02.Zefiro-Europe.TRS.100 | Requirement | The driver shall be able to switch off the saloon HVAC. | SIL 1 |
| 2F_15.02.Zefiro-Europe.TRS.19 | Requirement | Normal switch off (for whole train):<br>The TCMS shall switch off the HVAC system in case of the following states (input via IDU or connected with train status):<br>– Wash mode → from train operational mode<br>– Shut down → from train operational mode(opening of battery contactor)<br>– HVAC off order per train<br>– HVAC off order from TCMS, if 400 Volt are not available in preconditioning mode/ frost protection mode.<br><br>Local switch off (for car):<br>– Locally switch located on the HVAC control panel (per car)<br>– The switch could be used also to switch off the HVAC unit by crew in case of failure on communication link. | SIL 0 |
| | | **4 HVAC saloon - Automatic mode** | |
| | Information | In normal operation the HVAC system automatically controls the interior climate according to ambient conditions. | |
| 2F_15.02.Zefiro-Europe.TRS.24 | Requirement | In automatic mode shall the controller automatically switch between the modes Cooling, Ventilation or Heating. The interior temperature shall be set depending on the ambient temperature. | N/A |
| 2F_15.02.Zefiro-Europe.TRS.25 | Requirement | When no other mode is requested the saloon HVAC system shall automatically operate in Automatic mode. | N/A |
| | | **4.1 Heating mode** | |
| | Information | The HVAC system provides a supply air flow over floor and window ducts in the heating mode . | |
| | | **4.2 Ventilation mode** | |
| | Information | In ventilation mode shall the HVAC system increase the amount of fresh air to reach the specified Tim values as long as possible without switching to cooling mode. | |
| | | **4.3 Cooling mode** | |
| | Information | The cooling mode shall be automatically activated when the specified Tim cannot be reached in the ventilation mode. | |
| | | **5 HVAC saloon - Platform mode** | |
| | Information | When the train is stopping at a platform and as soon as the doors are released, the HVAC system will operate in platform mode. | |
| 2F_15.02.Zefiro-Europe.TRS.28 | Requirement | As soon as any doors are released, and if no other mode with higher priority is set, all saloon HVAC systems shall operate in platform mode.<br>As soon as there is a door close command from the driver, HVAC shall leave platform mode. | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **6 HVAC saloon - Tunnel mode** | |
| | Information | In tunnel mode the HVAC will use only return air for operation, dampers for fresh air and exhaust air are closed.<br><br>Tunnel mode is activated automatically by differential pressure transducers. | |
| 2F_15.02.Zefiro-Europe.TRS.33 | Requirement | When signal from differential pressure transducer is activated and if no other mode with higher priority is set, all HVAC systems shall operate in tunnel mode.<br>The driver shall also be able to activate and de-activate tunnel mode manually. | N/A |
| 2F_15.02.Zefiro-Europe.TRS.34 | Requirement | The air inlet and outlet flaps shall be possible to close when running in tunnel mode. | N/A |
| | | **7 HVAC saloon - Battery mode** | |
| | Information | When 3AC auxiliary power is not available the HVAC system will ensure minimum ventilation so that the CO2 level in the air does not rise to extreme level. | |
| 2F_15.02.Zefiro-Europe.TRS.38 | Requirement | Battery mode shall be requested when<br>– no 3AC auxiliary power available AND<br>– no other mode with higher priority is set | SIL 0 |
| 2F_15.02.Zefiro-Europe.TRS.39 | Requirement | Battery mode shall work for 30 min when battery capacity is sufficient and no other mode will be requested | SIL 0 |
| | | **8 HVAC saloon - Maintenance mode** | |
| | Information | The user can set the saloon HVAC in maintenance mode. | |
| 2F_15.02.Zefiro-Europe.TRS.43 | Requirement | It shall be possible to set the HVAC in test mode. | N/A |
| | | **9 HVAC saloon - Frost protection mode** | |
| 2F_15.02.Zefiro-Europe.TRS.190 | Requirement | It shall be possible for TCMS to set HVAC saloon in Frost protection mode | N/A |
| | | **10 HVAC saloon - Communication missing mode** | |
| | Information | When there is a total communication failure (both saloon HVAC units loose contact to IP in car) the HVAC saloon works in normal ventilation mode. If at the same time the 3AC auxiliary power is not available the HVAC saloon unit works in battery mode since battery mode has higher priority. If the emergency HVAC Off is activate from the fire detection system, the emergency HVAC off is activated. | |
| 2F_15.02.Zefiro-Europe.TRS.52 | Requirement | In case of communication failure of the IP link of both saloon HVAC units, the HVAC shall work in normal ventilation mode until a mode with higher priority is activated by a hardware input. | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **11 HVAC saloon - Preconditioning mode** | |
| | Information | Before starting passenger operation the saloon HVAC system operates in pre-conditioning mode to reach sufficient temperature inside the train.<br>When operating in pre-heating or pre-cooling mode the functionality of the HVAC system is as follows:<br>– Outdoor air dampers are in fully closed position (= return air damper fully open position)<br>– Exhaust air dampers are in fully closed position | |
| 2F_15.02.Zefiro-Europe.TRS.61 | Requirement | The HVAC saloon system shall switch from Preconditioning mode to normal operation when<br>– requested by the driver via the IDU OR<br>– if there is any indication, that there are passengers on bord (e.g required platform mode, train moving etc.) OR<br>– Preconditioning mode shall be automatically deactived if the internal targets in terms of comfort are achieved. | N/A |
| | | **12 HVAC saloon - Reduced power supply mode** | |
| 2F_15.02.Zefiro-Europe.TRS.63 | Requirement | If reduced auxiliary power supply is available the saloon HVAC system shall work at reduced cooling and heating performances. | N/A |
| | | **13 HVAC Saloon - Recirculation mode** | |
| | Information | When fire is detected in one car the other cars will run in recirculation mode. | |
| 2F_15.02.Zefiro-Europe.TRS.187 | Requirement | When fire has been detected in one car the HVAC units in the others cars shall be set in recirculation mode. | SIL 0 |
| | | **14 HVAC Saloon - Compressor start** | |
| 2F_15.02.Zefiro-Europe.TRS.70 | Requirement | The HVAC compressors shall be started sequentially to ensure that the start-up load is not too high. | N/A |
| | | **15 HVAC saloon - Interface with HVAC cab** | |
| 2F_15.02.Zefiro-Europe.TRS.79 | Requirement | The HVAC Saloon shall be interfaced with HVAC cab. | N/A |
| 2F_15.02.Zefiro-Europe.TRS.72 | Requirement | The nominal level of comfort in the cab shall be provided by HVAC Saloon in case of fault in the HVAC Cab | N/A |
| | | **16 HVAC saloon - Two HVAC units in each car** | |
| | Information | In any car there is two saloon HVAC units. If one unit fails the other saloon HVAC will regulate the comfort in the car. If one units controller fails the unit is shut off. This avoids the situation of two units working in different modes. | |
| 2F_15.02.Zefiro-Europe.TRS.84 | Requirement | The two saloon HVAC units in each car shall communicate with each other. In case one of the two saloon HVAC units fails in a car the other saloon HVAC unit shall regulate the comfort in the saloon. | N/A |
| 2F_15.02.Zefiro-Europe.TRS.57 | Requirement | When there is a fault in one of HVAC controller units the HVAC unit shall be controlled by the other HVAC units controller. The comfort level according to Trenitalia CTP for failure mode shall then apply. | N/A |


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| ID | Type | | SIL Needed |
|---|---|---|---|
| | | **17 HVAC saloon - Comfort control Data** | |
| | Information | Each car has a local control panel for the saloon HVAC. It is possible to set the offset from the nominal set-point in the passenger area and to diagnose the main functions and failures of the HVAC equipment. The offset can also be set at train level via the IDU. | |
| 2F_15.02.Zefiro-Europe.TRS.66 | Requirement | Train-staff shall be able to set offset from the nominal set-point value in the passenger area via the local control panel. There shall be 5 offsets (-2 K, -1 K, 0, +1 K, +2 K). The offset shall be possible to set independently for each car. | N/A |
| 2F_15.02.Zefiro-Europe.TRS.71 | Requirement | The driver or train-staff shall be able to set the temperature offset via the IDU. The temperature offset shall be set at train level. There shall be 5 offsets (-2 K, -1 K, 0, +1 K, +2 K). | N/A |
| 2F_15.02.Zefiro-Europe.TRS.78 | Requirement | The saloon HVAC controllers shall send the outdoor air temperatures to TCMS. The TCMS shall calculate the average outdoor air temperature based on all valid temperatures received from the HVAC controllers. The calculated average outdoor air temperature shall be sent back to the HVAC controllers. | N/A |
| | | **18 HVAC Saloon – Condition data** | |
| 2F_15.02.Zefiro-Europe.TRS.191 | Requirement | The status of the HVAC system shall be indicated on the IDU. | SIL 0 |
| | | **19 HVAC saloon - Fault Supervision** | |
| 2F_15.02.Zefiro-Europe.TRS.82 | Requirement | The HVAC shall supervise itself and a fault indication shall be set in case of component faults. | N/A |
| 2F_15.02.Zefiro-Europe.TRS.87 | Requirement | Complete diagnostic of the saloon HVAC system, including predictive diagnostic functions and CBM, shall be fully implemented in each saloon HVAC unit controller. The data shall be transmitted to the TCMS via the saloon HVAC bus interface. | N/A |
| | | **20 Events** | |
| 2F_15.02.Zefiro-Europe.TRS.184 | Requirement | Diagnostics from the supplier shall be stored in the train diagnostic system. | N/A |


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## 3 Revision

| Rev. | Description | Date | Name |
|---|---|---|---|
| — | FVRS exported from DOORS according to table below: | 2021-02-03 | Michele Ciabatti |

| RBS location | Module Name in DOORS | Baseline version |
|---|---|---|
| 2F_02.02 | Wash Wipe-VF.TRS | 2.0 |
| 2F_02.03 | Windscreen-VF.TRS | 1.1 |
| 2F_02.04 | Coupler-VF.TRS | 2.1 |
| 2F_02.06 | Exterior Lighting-VF.TRS | 2.0 |
| 2F_02.07 | Horn-VF.TRS | 2.0 |
| 2F_03 | Bogies-VF.TRS | 2.0 |
| 2F_04.01 | Line Voltage-VF.TRS | 2.1 |
| 2F_04.02 | Battery System-VF.TRS | 2.0 |
| 2F_05.01 | Drive Control-VF.TRS | 2.0 |
| 2F_05.02 | Brake Control-VF.TRS | 3.0 |
| 2F_06.01 | Aux Electrical Supply-VF.TRS | 2.0 |
| 2F_06.03 | Air Supply-VF.TRS | 2.1 |
| 2F_06.04 | Fire Protection and Extinction-VF.TRS | 2.0 |
| 2F_08.01 | Interior Lighting-VF.TRS | 2.0 |
| 2F_08.02 | Toilet-VF.TRS | 2.0 |
| 2F_08.03 | Catering-VF.TRS | 1.3 |
| 2F_08.04 | Interior Doors-VF.TRS | 2.0 |
| 2F_09.02 | Train Protection-VF.TRS.-.ILSA | 1.1 |
| 2F_09.03 | Conventional Train Control-VF.TRS | 2.0 |
| 2F_09.04 | TCMS - Operational-VF.TRS | 2.0 |
| 2F_10.01 | Diagnostics-VF.TRS | 2.1 |
| 2F_10.03 | Visualisation-VF.TRS | 3.1 |
| 2F_11 | Passenger Information and Entertainment-VF.TRS | 2.0 |
| 2F_12 | Train to Wayside Communication-VF.TRS | 2.0 |
| 2F_13 | CCTV-VF.TRS | 1.1 |
| 2F_14 | Exterior Door and Step Systems-VF.TRS | 2.0 |
| 2F_15.01 | Cab HVAC-VF.TRS | 2.0 |
| 2F_15.02 | Saloon HVAC-VF.TRS | 2.0 |


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| Rev. | Description | Date | Name |
|---|---|---|---|
| _A | FVRS exported from DOORS according to table below: | 2021-06-04 | Aniello Memoli |

| RBS location | Module Name in DOORS | Baseline version |
|---|---|---|
| 2F_02.02 | Wash Wipe-VF.TRS | 2.0 |
| 2F_02.03 | Windscreen-VF.TRS | 1.1 |
| 2F_02.04 | Coupler-VF.TRS | 2.1 |
| 2F_02.06 | Exterior Lighting-VF.TRS | 2.0 |
| 2F_02.07 | Horn-VF.TRS | 2.0 |
| **2F_03** | **Bogies-VF.TRS** | **3.0** |
| **2F_04.01** | **Line Voltage-VF.TRS** | **3.1** |
| 2F_04.02 | Battery System-VF.TRS | 2.0 |
| 2F_05.01 | Drive Control-VF.TRS | 2.0 |
| **2F_05.02** | **Brake Control-VF.TRS** | **4.2** |
| **2F_06.01** | **Aux Electrical Supply-VF.TRS** | **3.0** |
| **2F_06.03** | **Air Supply-VF.TRS** | **3.0** |
| **2F_06.04** | **Fire Protection and Extinction-VF.TRS** | **3.0** |
| **2F_08.01** | **Interior Lighting-VF.TRS** | **3.0** |
| **2F_08.02** | **Toilet-VF.TRS** | **3.0** |
| **2F_08.03** | **Catering-VF.TRS** | **2.0** |
| **2F_08.04** | **Interior Doors-VF.TRS** | **3.0** |
| **2F_09.02** | **Train Protection-VF.TRS** | **3.0** |
| **2F_09.03** | **Conventional Train Control-VF.TRS** | **3.0** |
| **2F_09.04** | **TCMS – Operational-VF.TRS** | **3.0** |
| **2F_10.01** | **Diagnostics-VF.TRS** | **3.0** |
| **2F_10.03** | **Visualisation-VF.TRS** | **4.0** |
| **2F_11** | **Passenger Information and Entertainment-VF.TRS** | **3.1** |
| **2F_12** | **Train to Wayside Communication-VF.TRS** | **3.0** |
| **2F_13** | **CCTV-VF.TRS** | **2.0** |
| **2F_14** | **Exterior Door and Step Systems-VF.TRS** | **3.1** |
| **2F_15.01** | **Cab HVAC-VF.TRS** | **3.0** |
| **2F_15.02** | **Saloon HVAC-VF.TRS** | **3.0** |


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| Rev. | Description | Date | Name |
|---|---|---|---|
| _B | FVRS exported from DOORS according to table below: | 2022-02-14 | Aniello Memoli |

| RBS location | Module Name in DOORS | Baseline version |
|---|---|---|
| 2F_02.02 | Wash Wipe-VF.TRS | 2.0 |
| 2F_02.03 | Windscreen-VF.TRS | 1.1 |
| 2F_02.04 | Coupler-VF.TRS | 2.1 |
| 2F_02.06 | Exterior Lighting-VF.TRS | 2.0 |
| 2F_02.07 | Horn-VF.TRS | 2.0 |
| 2F_03 | Bogies-VF.TRS | 3.0 |
| 2F_04.01 | Line Voltage-VF.TRS | 3.1 |
| 2F_04.02 | Battery System-VF.TRS | 2.0 |
| 2F_05.01 | Drive Control-VF.TRS | 2.0 |
| 2F_05.02 | Brake Control-VF.TRS | 4.2 |
| 2F_06.01 | Aux Electrical Supply-VF.TRS | 3.0 |
| 2F_06.03 | Air Supply-VF.TRS | 3.0 |
| 2F_06.04 | Fire Protection and Extinction-VF.TRS | 3.0 |
| 2F_08.01 | Interior Lighting-VF.TRS | 3.0 |
| 2F_08.02 | Toilet-VF.TRS | 3.0 |
| 2F_08.03 | Catering-VF.TRS | 2.0 |
| 2F_08.04 | Interior Doors-VF.TRS | 3.0 |
| **2F_09.02** | **Train Protection-VF.TRS** | **3.1** |
| **2F_09.03** | **Conventional Train Control-VF.TRS** | **3.1** |
| 2F_09.04 | TCMS – Operational-VF.TRS | 3.0 |
| 2F_10.01 | Diagnostics-VF.TRS | 3.0 |
| 2F_10.03 | Visualisation-VF.TRS | 4.0 |
| 2F_11 | Passenger Information and Entertainment-VF.TRS | 3.1 |
| 2F_12 | Train to Wayside Communication-VF.TRS | 3.0 |
| 2F_13 | CCTV-VF.TRS | 2.0 |
| 2F_14 | Exterior Door and Step Systems-VF.TRS | 3.1 |
| 2F_15.01 | Cab HVAC-VF.TRS | 3.0 |
| 2F_15.02 | Saloon HVAC-VF.TRS | 3.0 |


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